MATHEMATICS

STRAIGHT LINE

CONTENTS

STRAIGHT LINE

JEE EXERCISES

EXERCISE–J-I ..... Page –02

EXERCISE–J-II ..... Page –23

JEE MAIN..... Page –65

JEE ADV. ..... Page –103

CHALLENGER EXERCISES

EXERCISE–C-I ..... Page –107

EXERCISE–C-II ..... Page –125

COORDINATE SYSTEM, DISTANCE, SECTION FORMULAE

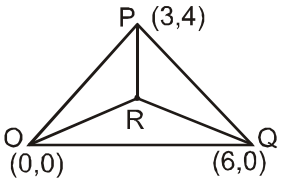
1. Let O(0, 0), P (3, 4), Q(6, 0) be the vertices of the triangle OPQ. The point R inside the triangle OPQ is such that the triangles OPR, PQR, OQR are of equal area. The coordinates of R are :

   माना O(0, 0), P (3, 4) व Q(6, 0) त्रिभुज OPQ के शीर्ष है। त्रिभुज OPQ के अन्दर बिन्दु R इस प्रकार है कि त्रिभुज OPR, PQR व OQR के क्षेत्रफल समान है। तब बिन्दु R के निर्देशांक :

   (A) $\left(\frac{4}{3},3\right)$ (B) $\left(3,\frac{2}{3}\right)$ (C) $\left(3,\frac{4}{3}\right)$ (D) $\left(\frac{4}{3},\frac{2}{3}\right)$

Ans. C

Sol.



R is centroid hence R $\equiv \left(3, \frac{4}{3}\right)$

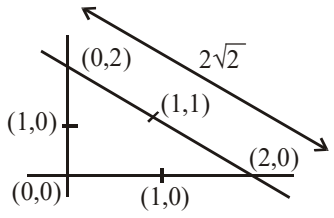
2. The x–coordinate of the incentre of the triangle that has the coordinates of mid point of its sides as (0, 1) (1, 1) and (1, 0) is : **[JEE(Main)2013]**

   एक त्रिभुज जिसकी भुजाओं के मध्य बिन्दुओं के निर्देशांक (0, 1) (1, 1) तथा (1, 0) है, के अन्तः केन्द्र का x निर्देशांक होगा : **[JEE(Main)2013]**

   (A) $2+\sqrt{2}$ (B) $2-\sqrt{2}$ (C) $1+\sqrt{2}$ (D) $1-\sqrt{2}$

Ans. B

Sol.



$$x=\frac{ax_1+bx_2+cx_3}{a+b+c}$$

$$=\frac{2(0)+2\sqrt{2}(0)+2(2)}{2+2+2\sqrt{2}}$$

$$=\frac{4}{4+2\sqrt{2}}=\frac{2}{2+\sqrt{2}}=2-\sqrt{2}$$

3. Let the orthocentre and centroid of a triangle be A(–3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is : **[JEE(Main)2018]**

माना एक त्रिभुज का लम्ब केन्द्र तथा केन्द्रक क्रमशः A(–3, 5) तथा B(3, 3) है। यदि इस त्रिभुज का परिकेन्द्र C है, तो रेखाखण्ड AC को व्यास मान कर बनाए जाने वाने वृत की त्रिज्या है : **[JEE(Main)2018]**

(A) $3\sqrt{\frac{5}{2}}$ (B) $\frac{3\sqrt{5}}{2}$ (C) $\sqrt{10}$ (D) $2\sqrt{10}$

Ans. A

Sol.



B divides AC in 2 : 1

$AC = \frac{3AB}{2}$; $r = \frac{AC}{2} = \frac{3AB}{4}$

$= \frac{3\sqrt{40}}{4}$; $= 3\sqrt{\left(\frac{5}{2}\right)}$

AREA OF TRIANGLE

4. If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P. with the same common ratio, then the points $(x_1\ y_1)$, (x_2, y_2) and (x_3, y_3) :

(A) are vertices of a triangle (B) lie on a straight line

(C) lie on an ellipse (D) lie on a circle

यदि x_1, x_2, x_3 तथा y_1, y_2, y_3 दोनो गुणौत्तर श्रेढ़ी में है तथा सार्व अनुपात समान है तब बिन्दु $(x_1\ y_1)$, (x_2, y_2) तथा (x_3, y_3) होगे:

(A) त्रिभुज के शीर्ष (B) सरल रेखा पर होंगे

(C) दीर्घवृत्त पर होंगे (D) वृत्त पर होंगे

Ans. B

Sol. $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_1r & y_1r & 1 \\ x_1r^2 & y_1r^2 & 1 \end{vmatrix} = \frac{x_1y_1}{2}\begin{vmatrix} 1 & 1 & 1 \\ r & r & 1 \\ r^2 & r^2 & 1 \end{vmatrix} = 0$

5. Let A (h, k),B(1, 1) and C (2, 1) be the vertices of a right angled triangle with AC as its hypotenuse. If the area of the triangle is 1 square unit, then the set of values which 'k' can take is given by :

माना किसी समकोण त्रिभुज के शीर्ष A (h, k),B(1, 1) एवं C (2, 1) है जहां AC कर्ण है यदि इस त्रिभुज का क्षेत्रफल एक वर्ग इकाई है तो 'k' का मान हो सकता है :

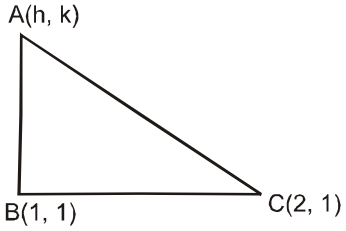
(A) {–1, 3} (B) {–3, – 2) (C) {1, 3} (D) {0, 2}

Ans. A

Sol. $AB = \sqrt{(h-1)^2+(k-1)^2}$

$BC = 1$

$AC = \sqrt{(h-2)^2+(k-1)^2}$



$AB^2 + BC^2 = AC^2 \Rightarrow (h-1)^2 + (k-1)^2 + 1 = (h-2)^2 + (k-1)^2$

$\Rightarrow 2h = 2 \Rightarrow h = 1$

Area of $\Delta ABC = \frac{1}{2}\sqrt{(h-1)^2+(k-1)^2} \times 1 = 1$

$(K-1)^2 = 4 \Rightarrow k - 1 = \pm 2 \Rightarrow k = 3, -1.$

6. Let k be an integer such that the triangle with vertices (k, –3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point : **[JEE(Main)2017]**

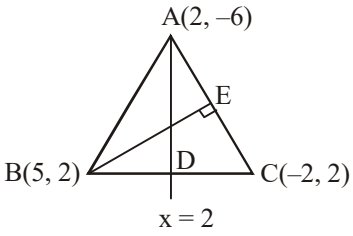
माना k एक ऐसा पूर्णांक है कि त्रिभुज, जिसके शीर्ष (k, –3k), (5, k) तथा (–k, 2) हैं, का क्षेत्रफल 28 वर्ग इकाई है, तो त्रिभुज के लम्ब–केन्द्र जिस बिन्दु पर है, वह है : **[JEE(Main)2017]**

(A) $\left(1, -\frac{3}{4}\right)$ (B) $\left(2, \frac{1}{2}\right)$ (C) $\left(2, -\frac{1}{2}\right)$ (D) $\left(1, \frac{3}{4}\right)$

Ans. B

Sol. $\begin{vmatrix} k & -3k & 1 \\ 5 & k & 1 \\ -k & 2 & 1 \end{vmatrix} = \pm 56$

$\Rightarrow k = 2$ (k is an integer)



$m_{AC} = \frac{-8}{4} = -2$

$BE \Rightarrow (y-2) = \frac{1}{2}(x-5)$

$x - 2y = 1$

$x = 2$

$y = \frac{1}{2}$

orthocenter $\equiv \left(2, \frac{1}{2}\right)$

EQUATION OF LINE AND ANGLE BETWEEN LINES

7. A square of side a lies above the x–axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x–axis. The equation of its diagonal not passing through the origin is :

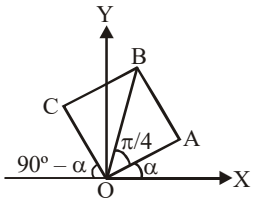
यदि 'a' भुजा का वर्ग x–अक्ष के ऊपर स्थित है उसका एक शीर्ष मूल बिन्दु पर है। मूल बिन्दु से गुजरने वाली भुजा x–अक्ष की धनात्मक दिशा से $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ कोण बनाती है। इसके मूल बिन्दु से न गुजरने वाली विकर्ण की समीकरण है :

(A) $y(\cos\alpha + \sin\alpha) + x(\cos\alpha - \sin\alpha) = a$ (B) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

(C) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$ (D) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

Ans. A

Sol. Co-ordinates of A = $(a\cos\alpha, a\sin\alpha)$



Equation of OB, $y = \tan\left(\frac{\pi}{4} + \alpha\right)x$

$\because$ CA $\perp^r$ to OB

$\therefore$ Slope of CA = $-\cot\left(\frac{\pi}{4} + \alpha\right)$

Equation of CA,

$$y - a\sin\alpha = -\cot\left(\frac{\pi}{4} + \alpha\right)(x - a\cos\alpha)$$

$\Rightarrow y(\sin\alpha + \cos\alpha) + x(\cos\alpha - \sin\alpha) = a$

$\Rightarrow y(\cos\alpha + \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$

M – 2 :

$A(a\cos\alpha, a\sin\alpha)$

$C(-a\cos(90^\circ - \alpha), a\sin(90^\circ - \alpha))$

$C(-a\sin\alpha, a\cos\alpha)$

$$AC \Rightarrow (y - a\cos\alpha) = \left(\frac{\cos\alpha - \sin\alpha}{-\sin\alpha - \cos\alpha}\right)(x + a\sin\alpha)$$

$$(\cos\alpha - \sin\alpha)x + (\cos\alpha + \sin\alpha)y = a$$

8. The equation of the straight line passing through the point (4, 3) and making intercepts on the co–ordinate axes whose sum is –1 is :

(A) $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$ (B) $\frac{x}{2} - \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

(C) $\frac{x}{2} + \frac{y}{3} = 1$ and $\frac{x}{2} + \frac{y}{1} = 1$ (D) $\frac{x}{2} + \frac{y}{3} = -1$ and $\frac{x}{-2} + \frac{y}{1} = -1$

बिन्दु (4, 3) से गुजरने वाली तथा निर्देश अक्षों पर वह अंतःखण्ड बनाने वाली जिनका योग –1 है, रेखा का समीकरण है :

(A) $\frac{x}{2} - \frac{y}{3} = 1$ तथा $\frac{x}{-2} + \frac{y}{1} = 1$ (B) $\frac{x}{2} - \frac{y}{3} = -1$ तथा $\frac{x}{-2} + \frac{y}{1} = -1$

(C) $\frac{x}{2} + \frac{y}{3} = 1$ तथा $\frac{x}{2} + \frac{y}{1} = 1$ (D) $\frac{x}{2} + \frac{y}{3} = -1$ तथा $\frac{x}{-2} + \frac{y}{1} = -1$

Ans. A

Sol. Here a + b = 1. Required line is

$$\frac{x}{a} - \frac{y}{1+a} = 1 \qquad \text{..... (i)}$$

Since line (i) passes through (4, 3)

$$\therefore \quad \frac{4}{a} - \frac{3}{1+a} = 1 \Rightarrow 4 + 4a - 3a = a + a^2$$

$$\Rightarrow a^2 = 4 \Rightarrow a = \pm 2$$

$\therefore$ Required lines are $\frac{x}{2} - \frac{y}{3} = 1$ and $\frac{x}{-2} + \frac{y}{1} = 1$

9. The line parallel to the x–axis and passing through the intersection of the lines ax + 2by + 3b = 0 and bx – 2ay – 3a = 0, where (a, b) ≠ (0, 0) is :

(A) below the x–axis at a distance of $\frac{3}{2}$ from it (B) below the x–axis at a distance of $\frac{2}{3}$ from it

(C) above the x–axis at distance of $\frac{3}{2}$ from it (D) above the x–axis at a distance of $\frac{2}{3}$ from it

x–अक्ष के समांतर तथा रेखाओं ax + 2by + 3b = 0 तथा bx – 2ay – 3a = 0 के प्रतिच्छेद बिन्दु से गुजरने वाली रेखा, जहां (a, b) ≠ (0, 0) है :

(A) x–अक्ष के नीचे इससे $\frac{3}{2}$ इकाई दूरी पर (B) x–अक्ष के ऊपर इससे $\frac{2}{3}$ इकाई दूरी पर

(C) x–अक्ष के ऊपर इससे $\frac{3}{2}$ इकाई दूरी पर (D) x–अक्ष के नीचे इससे $\frac{2}{3}$ इकाई दूरी पर

Ans. A

Sol. M-1 :

Point of intersection of both the lines is $\left(0,-\frac{3}{2}\right)$

M-2 :

The lines passing through the intersection of the lines $ax + 2by + 3b = 0$ and $bx – 2ay – 3a = 0$ is

$ax + 2by + 3b + \lambda(bx – 2ay – 3a) = 0$

$\Rightarrow (a + b\lambda)x + (2b – 2a\lambda)y + 3b – 3\lambda a = 0$ ..... (i)

Line (i) is parallel to x-axis

$\therefore\ a + b\lambda = 0 \Rightarrow \lambda = \frac{-a}{b} = 0$

Put the value of λ in (i)

$ax + 2by + 3b – \frac{a}{b}(bx – 2ay – 3a) = 0$

$y\left(2b+\frac{2a^2}{b}\right) + 3b+\frac{3a^2}{b} = 0,$

$y\left(\frac{2b^2+2a^2}{b}\right) = -\left(\frac{3b^2+3a^2}{b}\right)$

$y = \frac{-3(a^2+b^2)}{2(b^2+a^2)} = \frac{-3}{2},\ y = -\frac{3}{2}$

So, it is 3/2 unit below x-axis

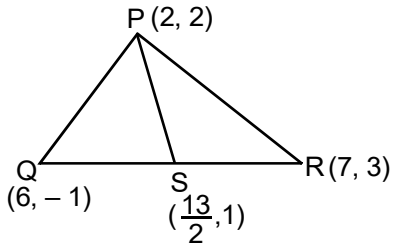
10. Let PS be the median of the triangle with vertices P(2, 2), Q(6, –1) and R(7, 3). The equation of the line passing through (1, –1) and parallel to PS is : **[JEE(Main)2014]**

माना PS एक त्रिभुज की माध्यिका है जिसके शीर्ष P(2, 2), Q(6, –1) तथा R(7, 3) हैं। (1, –1) से होकर जाने वाली रेखा, जो PS के समांतर है, का समीकरण हैं : **[JEE(Main)2014]**

(A) $4x + 7y + 3 = 0$ (B) $2x – 9y – 11 = 0$ (C) $4x – 7y – 11 = 0$ (D) $2x + 9y + 7 = 0$

Ans. D

Sol.



$$\text{Slope of PS} = \frac{2-1}{2-\frac{13}{2}} = \frac{-2}{9}$$

Hence equation of line through $(1, -1)$ & parallel to PS is:

$$(y+1) = \frac{-2}{9}(x-1)$$

$$9y + 2x + 7 = 0$$

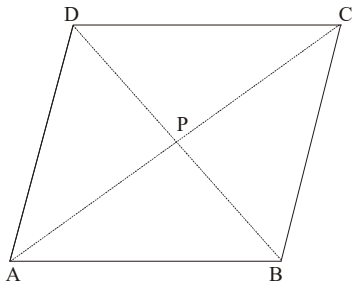
11. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at $(-1, -2)$, then which one of the following is a vertex of this rhombus? **[JEE(Main)2016]**

यदि एक समचतुर्भुज की दो भुजाएँ, रेखाओं $x - y + 1 = 0$ तथा $7x - y - 5 = 0$ की दिशा में हैं तथा इसके विकर्ण बिन्दु $(-1, -2)$ पर प्रतिच्छेद करते हैं, तो इस समचतुर्भुज का निम्न में से कौन–सा शीर्ष है? **[JEE(Main)2016]**

(A) $\left(\frac{1}{3}, -\frac{8}{3}\right)$ (B) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$ (C) $(-3, -9)$ (D) $(-3, -8)$

Ans. A

Sol.



$AB = 7x - y - 5 = 0$

$AD = x - y + 1 = 0$

$A(+1, 2)$

$P(-1, -2)$

$\Rightarrow C(-3, -6)$

$BC = x - y - 3 = 0$

$CD = 7x - y + 15 = 0$

$\Rightarrow B\left(\frac{1}{3},\frac{-8}{3}\right)$

PARAMETRIC FORM OF A LINE

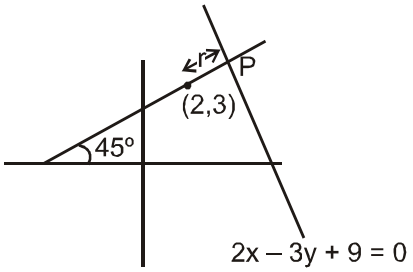
12. The distance of the point (2, 3) from the line $2x - 3y + 9 = 0$ measured along a line $x - y + 1 = 0$ is :

बिन्दु (2, 3) की रेखा $x - y + 1 = 0$ के अनुदिश रेखा $2x - 3y + 9 = 0$ से दूरी है :

(A) $5\sqrt{3}$ (B) $4\sqrt{2}$ (C) $3\sqrt{2}$ (D) $2\sqrt{2}$

Ans. B

Sol. Let coordinates of point P by parametric $P(2 + r\cos 45^\circ, 3 + r\sin 45^\circ)$



It satisfies the line $2x - 3y + 9 = 0$; $2\left(2+\frac{r}{\sqrt{2}}\right) - 3\left(3+\frac{r}{\sqrt{2}}\right) + 9 = 0 \Rightarrow \quad r = 4\sqrt{2}$

13. P lies on the line $y = x$ and Q lies on $y = 2x$. The equation for the locus of the mid point of PQ, if $|PQ| = 4$, is :

यदि बिन्दु P तथा Q क्रमशः रेखा $y = x$ एवं $y = 2x$ पर है। यदि $|PQ| = 4$ तो P एवं Q के मध्य बिन्दु का बिन्दुपथ होगा :

(A) $25x^2 + 36xy + 13y^2 = 4$ (B) $25x^2 - 36xy + 13y^2 = 4$

(C) $25x^2 - 36xy - 13y^2 = 4$ (D) $25x^2 + 36xy - 13y^2 = 4$

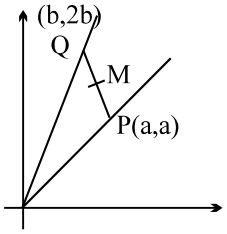
Ans. B

Sol. $2h = a + b$

$2k = a + 2b$

$\therefore \quad b = 2(k - h)$

$a = 2(2h - k)$ now proceed



DISTANCE OF A POINT FROM LINE

14. The line L_1 given by $\frac{x}{5}+\frac{y}{b}=1$ passes through the point M(13, 32). The line L_2 is parallel to L_1 and has the equation $\frac{x}{c}+\frac{y}{3}=1$. Then the distance between L_1 and L_2 is :

सरल रेखा $L_1 : \frac{x}{5}+\frac{y}{b}=1$, जो बिन्दु M(13, 32) से गुजरती है तथा रेखा L_2, सरल रेखा L_1 के समान्तर है तथा उसका समीकरण $\frac{x}{c}+\frac{y}{3}=1$ हो, तो L_1 एवं L_2 के बीच की दूरी होगी :

(A) $\sqrt{17}$ (B) $\frac{17}{\sqrt{15}}$ (C) $\frac{23}{\sqrt{17}}$ (D) $\frac{23}{\sqrt{15}}$

Ans. C

Sol. Slope of line $L_1 = \frac{-b}{5}$ and slope of line $L_2 = \frac{-3}{c}$

Line L_1 is parallel to line $L_2 \Rightarrow \frac{b}{5}=\frac{3}{c} \Rightarrow bc = 15$

As (13, 32) lies on L_1, so $\frac{13}{5}+\frac{32}{b}=1 \Rightarrow \frac{32}{b}=\frac{-8}{5} \Rightarrow b=-20$

So, $c = \frac{-3}{4}$

$\therefore$ Equation of $L_2 : y-4x=3$

Hence distance between L_1 and $L_2 = \frac{|52-32+3|}{\sqrt{17}} = \frac{23}{\sqrt{17}}$

POSITION OF TWO POINTS WITH RESPECT TO A LINE

15. If (a, a^2) falls inside the angled made by the lines $y=\frac{x}{2}$, $x>0$ and $y=3x$, $x>0$, then a along to :

यदि (a, a^2) रेखाओं $y=\frac{x}{2}$, $x>0$ एवं $y=3x$, $x>0$ के मध्य के कोण के अंदर स्थित है, तब 'a' निम्न में स्थित है :

(A) $\left(0,\frac{1}{2}\right)$ (B) $(3, \infty)$ (C) $\left(\frac{1}{2},3\right)$ (D) $\left(-3,-\frac{1}{2}\right)$

Ans. C

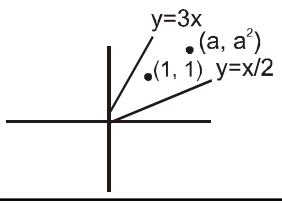
Sol. Let $L_1(x-2y)$ & $L_2(3x-y)$

Since (1, 1) and (a, a^2) Both lies same side with respect to both lines

$L_1(1, 1) \times L_1(a, a^2) > 0 \Rightarrow \quad a-2a^2<0 \quad \Rightarrow \quad 2a^2-a>0$

$\Rightarrow \quad a(2a-1)>0$

$a \in (-\infty, 0) \cup \left(\frac{1}{2}, \infty\right)$



similarly $L_2(1, 1) \times L_2(a, a^2) > 0 \Rightarrow 3a - a^2 > 0 \Rightarrow a^2 - 3a < 0 \Rightarrow a \in (0, 3)$

Hence after taking intersection $a \in \left(\frac{1}{2}, 3\right)$

16. A triangle ABC is formed by the lines $2x-3y-6=0$; $3x-y+3=0$ and $3x+4y-12=0$. If the points $P(\alpha, 0)$ and $Q(0, \beta)$ always lie on or inside the Δ ABC, then :

(A) $\alpha \in [-1, 2]$ and $\beta \in [-2, 3]$ (B) $\alpha \in [-1, 3]$ and $\beta \in [-2, 4]$

(C) $\alpha \in [-2, 4]$ and $\beta \in [-3, 4]$ (D) $\alpha \in [-1, 3]$ and $\beta \in [-2, 3]$

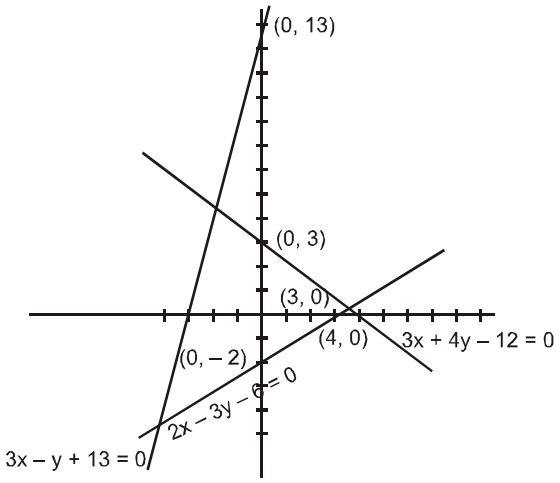
एक त्रिभुज ABC रेखा $2x-3y-6=0$; $3x-y+3=0$ एवं $3x+4y-12=0$ से बन रहा है। यदि बिन्दु $P(\alpha, 0)$ एवं $Q(0, \beta)$ हमेशा त्रिभुज पर या त्रिभुज के भीतर है, तो :

(A) $\alpha \in [-1, 2]$ एवं $\beta \in [-2, 3]$ (B) $\alpha \in [-1, 3]$ एवं $\beta \in [-2, 4]$

(C) $\alpha \in [-2, 4]$ एवं $\beta \in [-3, 4]$ (D) $\alpha \in [-1, 3]$ एवं $\beta \in [-2, 3]$

Ans. D

Sol. $L_1 = 2x - 3y - 6 = 0$



$L_2 = 3x - y + 3 = 0 \quad \Rightarrow \quad L_3 = 3x + 4y - 12 = 0$

Hence अतः $\alpha \in [-1, 3] \quad \Rightarrow \quad \beta \in [-2, 3]$

CONDITION OF CONCURRENCY OF 3 LINES

17. If the lines

$$\left.\begin{aligned} \lambda x + (\sin\alpha)y + \cos\alpha = 0 \\ x + (\cos\alpha)y + \sin\alpha = 0 \\ x - (\sin\alpha)y + \cos\alpha = 0 \end{aligned}\right]$$ pass through the same point where $\alpha \in R$ then λ lies in the interval :

यदि रेखाऐं

$\left.\begin{array}{l}\lambda x+(\sin\alpha)y+\cos\alpha=0\\ x+(\cos\alpha)y+\sin\alpha=0\\ x-(\sin\alpha)y+\cos\alpha=0\end{array}\right]$ एक बिन्दु से गुजरती है। जहाँ $\alpha \in R$ तो λ इनमें से किस अन्तराल में होगा :

(A) $[-1, 1]$ (B) $\left[-\sqrt{2},\sqrt{2}\right]$ (C) $[-2, 2]$ (D) $(-\infty, \infty)$

Ans. B

Sol. $\begin{vmatrix}\lambda & \sin\alpha & \cos\alpha\\ 1 & \cos\alpha & \sin\alpha\\ 1 & -\sin\alpha & \cos\alpha\end{vmatrix}=0$

$\Rightarrow \lambda(1)-1(2\sin\alpha\cos\alpha)+1(\sin^2\alpha-\cos^2\alpha)=0$

$\Rightarrow \lambda=\sin 2\alpha+\cos 2\alpha$

$\left[-\sqrt{2},\sqrt{2}\right]$

FAMILY OF LINES

18. The lines $ax + by + c = 0$, where $3a + 2b + 4c = 0$ are concurrent at the point :

यदि $3a + 2b + 4c = 0$ हो, तो $ax + by + c = 0$ से प्रदर्शित सरल रेखाएँ निम्न में से किस बिन्दु पर संगामी हैं :

(A) $\left(\frac{1}{2},\frac{3}{4}\right)$ (B) $(1, 3)$ (C) $(3, 1)$ (D) $\left(\frac{3}{4},\frac{1}{2}\right)$

Ans. D

Sol. $ax + by + c = 0 \quad \Rightarrow \quad \frac{3a}{4}+\frac{b}{2}+c=0$

compare both $(x, y) \equiv \left(\frac{3}{4},\frac{1}{2}\right)$. Hence given family passes through $\left(\frac{3}{4},\frac{1}{2}\right)$

19. Given the family of lines, $a(2x + y + 4) + b(x - 2y - 3) = 0$. Among the lines of the family, the number of lines situated at a distance of $\sqrt{10}$ from the point $M(2, -3)$ is :

रेखाओं के समूह $a(2x + y + 4) + b(x - 2y - 3) = 0$ में से कितनी रेखाऐं ऐसी है कि जो बिन्दु $M(2, -3)$ से $\sqrt{10}$ की दूरी पर होगी :

(A) 0 (B) 1 (C) 2 (D) ∞

Ans. B

Sol. $(2x + y + 4) + \lambda(x - 2y - 3) = 0$

$(2 + \lambda)x + 1(1 - 2\lambda)y + (4 - 3\lambda) = 0$

$m(2, -3)$

$$\frac{4 + 2\lambda + 6\lambda - 3 + 4 - 3\lambda}{\sqrt{(2 \times \lambda)^2 + (1 - 2\lambda)^2}} = \pm\sqrt{10}$$

$$\Rightarrow \frac{(\lambda + 1)^2}{(\lambda^2 + 1)} = 2$$

$$\Rightarrow \lambda^2 + 1 + 2\lambda = 2\lambda^2 + 2$$

$$\Rightarrow \lambda^2 + 1 - 2\lambda = (\lambda - 1)^2 = 0$$

$\lambda = 1$

CONCEPT OF IMAGE

20. The equation of image of pair of lines $y = |x - 1|$ with respect to y–axis is :

रेखा युग्म $y = |x - 1|$ का y–अक्ष के सापेक्ष प्रतिबिम्ब का समीकरण होगा :

(A) $x^2 - y^2 - 2x + 1 = 0$ (B) $x^2 - y^2 - 4x + 4 = 0$

(C) $4x^2 - 4x - y^2 + 1 = 0$ (D) $x^2 - y^2 + 2x + 1 = 0$

Ans. D

Sol. $y = (x - 1)$

$y = x - 1$ $y = 1 - x$

y axis $\rightarrow x \rightarrow -x$

$(x + y + 1) \times (x - y + 1) = 0$

$(x + 1)^2 - y^2 = 0$

$x^2 - y^2 + 2x + 1 = 0$

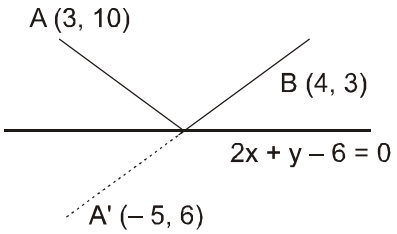
21. A light beam emanating from the point A(3, 10) reflects from the straight line $2x + y - 6 = 0$ and then passes through the point B(4, 3). The equation of the reflected beam is :

बिन्दु A(3, 10) से उत्सर्जित होने वाली एक प्रकाश किरण, सरल रेखा $2x + y - 6 = 0$ से परावर्तित होकर बिन्दु B(4, 3) से गुजरती है, तो परावर्तित प्रकाश किरण का समीकरण होगा :

(A) $3x - y + 1 = 0$ (B) $x + 3y - 13 = 0$ (C) $3x + y - 15 = 0$ (D) $x - 3y + 5 = 0$

Ans. B

Sol. Image of A(3, 10) in $2x + y - 6 = 0$



$$\frac{x-3}{2} = \frac{y-10}{1} = -2\left(\frac{6+10-6}{2^2+1^2}\right)$$

$$\frac{x-3}{2} = \frac{y-10}{1} = -4$$

A' = (– 5, 6)

Equation of A'B is $y - 3 = \left(\frac{6-3}{-5-4}\right)(x-4) \Rightarrow y - 3 = -\frac{1}{3}(x-4)$

$3y - 9 = -x + 4 \Rightarrow x + 3y - 13 = 0$

22. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x–axis, the equation of the reflected ray is :

[JEE(Main)2013]

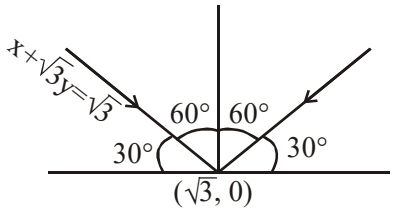
$x + \sqrt{3}y = \sqrt{3}$ की दिशा में जाती है हुई एक प्रकाश की किरण x–अक्ष पर पहुँच कर परावर्तित हो जाती है। इस परावर्तित किरण का समीकरण है :

[JEE(Main)2013]

(A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$ (C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

Ans. B

Sol.



$$y = \frac{1}{\sqrt{3}}(x - \sqrt{3})$$

$\sqrt{3}y = x - \sqrt{3}$

ANGLE BISECTORS

23. Find the equation of the bisector of the acute angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$:

रेखाओं $3x - 4y + 7 = 0$ एवं $12x + 5y - 2 = 0$ के मध्य न्यून कोण के अर्द्धक का समीकरण ज्ञात कीजिए :

(A) $11x - 3y + 9 = 0$ (B) $3x + 11y - 13 = 0$ (C) $3x + 11y - 3 = 0$ (D) $11x - 3y + 2 = 0$

Ans. A

Sol. $3x - 4y + 7 = 0$ $\quad a_1a_2 + b_1b_2 = -36 + 20$

$-12x - 5x + 2 = 0$ $\quad = -16 < 0$

$$\frac{(3x-4y+7)}{5} = +\frac{(-12x-5y+2)}{13}$$

$39x - 52y + 91 = -60x - 25y + 10$

$99x - 27y + 81 = 0$

$11x - 3y + 9 = 0$

24. The equation of the bisector of the angle between two lines $3x - 4y + 12 = 0$ and $12x - 5y + 7 = 0$ which contains the points $(-1, 4)$ is :

दो रेखाओं $3x - 4y + 12 = 0$ एवं $12x - 5y + 7 = 0$ के बीच बिन्दु $(-1, 4)$ को समाहित करने वाले कोण के अर्द्धक का समीकरण है :

(A) $21x + 27y - 121 = 0$ (B) $21x - 27y + 121 = 0$

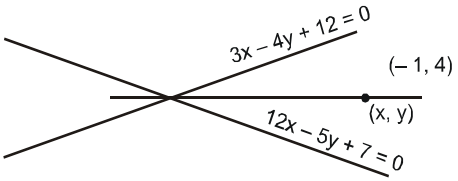
(C) $21x + 27y + 191 = 0$ (D) $\frac{-3x + 4y - 12}{5} = \frac{12x - 5y + 7}{13}$

Ans. A

Sol. at $(-1, 4)$

$3x - 4y + 12 < 0$ and $12x - 5y + 7 < 0$

$\Rightarrow \quad \frac{3x-4y+12}{12x-5y+12} > 0$ at $(-1, 4)$



So we have to take the bisector with + sign

$$\frac{3x-4y+12}{5} = \frac{12x-5y+7}{13} \quad \Rightarrow \quad 21x + 27y - 121 = 0$$

25. The equation of bisectors of two lines L_1 and L_2 are $2x - 16y - 5 = 0$ and $64x + 8y + 35 = 0$. If the line L_1 passes through $(-11, 4)$, the equation of acute angle bisector of L_1 and L_2 is :

(A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) data insufficient (D) none of these

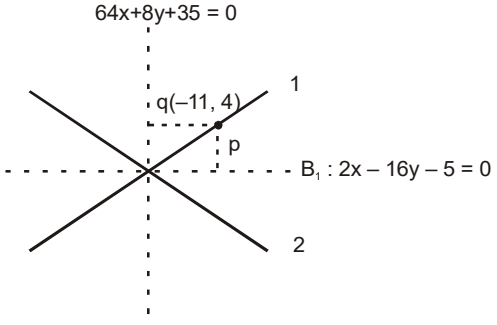
दो रेखाओं L_1 एवं L_2 के समद्विभाजकों की समीकरण $2x - 16y - 5 = 0$ एवं $64x + 8y + 35 = 0$ हैं। यदि रेखा L_1 बिन्दु $(-11, 4)$ से गुजरती है, तो L_1 एवं L_2 के मध्य न्यून कोण अर्द्धक की समीकरण है :

(A) $2x - 16y - 5 = 0$ (B) $64x + 8y + 35 = 0$ (C) आँकड़े अपर्याप्त हैं (D) इनमें से कोई नहीं

Ans. A

Sol. $p = \left|\frac{-22-64-5}{\sqrt{2^2+(-16)^2}}\right| = \frac{91}{\sqrt{260}}$

$q = \left|\frac{-64\times11+8\times4+35}{\sqrt{64^2+8^2}}\right| = \frac{637}{\sqrt{4160}} = \frac{159.25}{\sqrt{260}}$



$p < q$ Hence $2x - 16y - 5 =$ - is a cute angle besectory

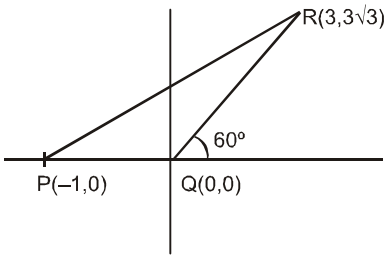
26. Let P = (–1, 0), Q = (0, 0) and R $(3, 3\sqrt{3})$ be three point. The equation of the bisector of the angle PQR is :

माना P = (–1, 0), Q = (0, 0) एवं R $(3, 3\sqrt{3})$ तीन बिन्दु है ∠PQR के समद्विभाजक की समीकरण है :

(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3y} = 0$ (C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

Ans. C

Sol.



The line segment QR makes an angle 60º with the positive direction of x-axis.
hence bisector of angle PQR will make 120º with +ve direction of x-axis.
$\therefore$ Its equation

$y - 0 = \tan 120^\circ (x - 0) \quad \Rightarrow y = -\sqrt{3}x \Rightarrow x\sqrt{3} + y = 0$

IMPORTANT CONCEPT RELATED TO OPTIMIZING

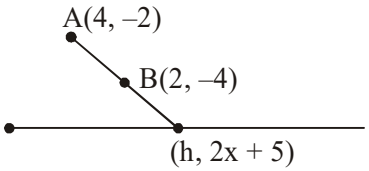
27. The co–ordinates of a point P on the line $2x - y + 5 = 0$ such that $|PA - PB|$ is maximum where A is $(4, -2)$ and B is $(2, -4)$ will be :

यदि रेखा $2x - y + 5 = 0$ पर मौजूद बिन्दु P इस प्रकार से है कि $|PA - PB|$ अधिकतम है जहाँ A $(4, -2)$ एवं B $(2, -4)$ तो P का निर्देशांक होगा :

(A) $(11, 27)$ (B) $(-11, -17)$ (C) $(-11, 17)$ (D) $(0, 5)$

Ans. B

Sol.



$L_A, L_B > 0$

$$\begin{vmatrix} h & 4 & 2 & h \\ 2h+5 & -2 & -4 & 2h+5 \end{vmatrix} = 0$$

$-2h - 20 - 16 + 4 + 10 = 0$

$-2h = 22$

$h = -11$

$2h + 5 = -22 + 5 = -17$

GENERAL LOCUS PROBLEMS

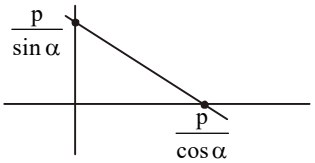
28. Locus of mid point of the portion between the axes of $x\cos\alpha + y\sin\alpha = p$ whre p is constant is :

रेखा $x\cos\alpha + y\sin\alpha = p$ के अक्षों के मध्य काटे गये अन्त खण्ड के मध्य बिन्दु का बिन्दुपथ होगा जहां p अचर है :

(A) $x^2 + y^2 = \frac{4}{p^2}$ (B) $x^2 + y^2 = 4p^2$ (C) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{2}{p^2}$ (D) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$

Ans. D

Sol.



$x\cos\alpha + y\sin\alpha = p$

$x = \frac{p}{2\sin\alpha} \quad \Rightarrow \cos\alpha = \frac{p}{2x}$

$y = \frac{p}{2\sin\alpha} \quad \Rightarrow \sin\alpha = \frac{p}{2y}$

$\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$

29. Locus of centroid of the triangle whose vertices are (a cos t, a sin t), (b sin t, – b cos t) and (1, 0) where t is a parameter, is :

एक त्रिभुज जिसके शीर्ष (a cos t, a sin t), (b sin t, – b cos t) तथा (1, 0) है तब त्रिभुज के केन्द्रक का बिन्दुपथ होगा। जहां t परिमाप है :

(A) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$ (B) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$ (D) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

Ans. C

Sol. (a cot t, a sin t), (b sin t, – b cos t), (1, 0)

$x = \frac{a\cos t + b\sin t + 1}{3}$, $y = \frac{a\sin t - b\cos t}{3}$

$(3x - 1) = a\cos t + b\sin t$ $3y = a\sin t - b\cos t$

$(3x - 1)^2 + (3y)^2 = a^2 + b^2$

30. Let P be the point (1, 0) and Q a point on the locus $y^2 = 8x$. The locus of mid point of PQ is :

माना P बिन्दु (1, 0) तथा Q बिन्दुपथ $y^2 = 8x$ पर स्थित एक बिन्दु है। PQ के मध्य बिन्दु का बिन्दुपथ है :

(A) $y^2 - 4x + 2 = 0$ (B) $y^2 + 4x + 2 = 0$ (C) $x^2 + 4y + 2 = 0$ (D) $x^2 - 4y + 2 = 0$

Ans. A

Sol. P(1, 0); Q $y^2 = 8x$

$\left(\frac{\alpha^2}{8}, \alpha\right)$ $x = \frac{\alpha^2}{8} + 1$, $y = \frac{\alpha}{2}$

$\alpha = 2y$

$2x = \frac{y^2}{2} + 1$

$4x = y^2 + 2$

$y^2 - 4x + 2 = 0$

31. The equatons $x = t^3 + 9$ and $y = \frac{3t^3}{4} + 6$ represents a straight line where t is a parameter. Then y–intercept of the line is :

यदि समीकरण $x = t^3 + 9$ तथा $y = \frac{3t^3}{4} + 6$ एक सरल रेखा को निरूपित करता है, जहाँ t एक प्राचल है। तो रेखा का y–काट होगा:

(A) $-\frac{3}{4}$ (B) 9 (C) 6 (D) 1

Ans. A

Sol. $x = t^3 + 9$

$y = \frac{3t^3}{4} \times 6$

$4y = 3t^3 \times 24$

$\Rightarrow 4y = 3(x - 9) + 24$

$\Rightarrow 4y = 3x - 3$

$\Rightarrow y = \frac{3x}{4} - \frac{3}{4}$

$c = -\frac{3}{4}$

32. Locus of the image of the point (2, 3) in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0$ $k \in R$ is a : **[JEE(Main)-2015]**

(A) straight line parallel to x–axis (B) straight line parallel to y–axis

(C) circle of radius $\sqrt{2}$ (D) circle of radius $\sqrt{3}$

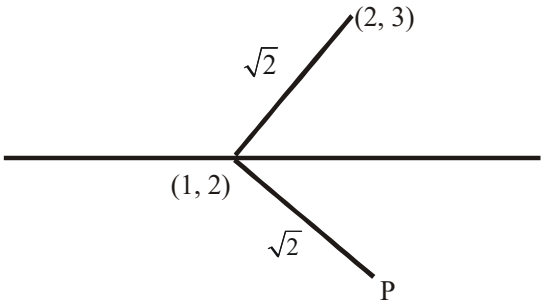
बिन्दु (2, 3) के रेखा $(2x - 3y + 4) + k(x - 2y + 3) = 0$ $k \in R$ से प्रतिबिंब का बिंदुपथ एक : **[JEE(Main)2015]**

(A) x–अक्ष के समांतर रेखा है (B) y–अक्ष के समांतर अक्ष है।

(C) $\sqrt{2}$ त्रिज्या का वृत है (D) $\sqrt{3}$ त्रिज्या का वृत है

Ans. C

Sol.



$2x - 3y + 4 = 0$

$x - 2y + 3 = 0$

Line passing through $(1, 2)$

P will at distance of $\sqrt{2}$ from $(1, 2)$

So locus of P will circle of radius $\sqrt{2}$

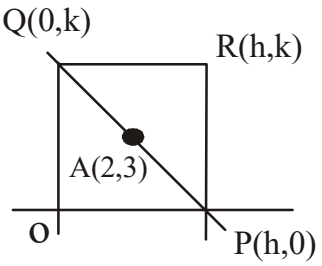
33. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is : **[JEE(Main)2018]**

एक सरल रेखा, जो एक अचर बिन्दु (2, 3) से होकर जाती है, निर्देशांक अक्षों को दो विभिन्न बिन्दुओं P तथा Q पर प्रतिच्छेद करती है। यदि O मूल बिन्दु है तथा आयत OPRQ को पूरा किया जाता है तो R का बिन्दुपथ है : **[JEE(Main)2018]**

(A) $3x + 2y = xy$ (B) $3x + 2y = 6xy$ (C) $3x + 2y = 6$ (D) $2x + 3y = xy$

Ans. A

Sol.



P, A, Q will be collinear

$$\begin{vmatrix} h & 0 & 1 \\ 2 & 3 & 1 \\ 0 & k & 1 \end{vmatrix} = 0$$

$3h + 2k = hk$

locus of R (h, k) will be

$3x + 2y = xy$

PAIR OF STRAIGHT LINES

34. If the slope of one line of the pair of lines represented by $ax^2 + 10xy + y^2 = 0$ is four times, the slope of the other line, then a equal to :

यदि समीकरण $ax^2 + 10xy + y^2 = 0$ द्वारा निरूपित रेखाओं के युग्म में से एक रेखा की प्रवणता दूसरी रेखा की प्रवणता से चार गुना है, तो a का मान है :

(A) 1 (B) 2 (C) 4 (D) 16

Ans. D

Sol. $m_1 + m_2 = -10 \quad \Rightarrow \quad m_1 m_2 = \frac{a}{1}$

given $m_1 = 4m_2 \quad \Rightarrow \quad m_2 = -2,\ m_1 = -8,\ a = 16$

35. The pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are perpendicular to each other for :

(A) two values of a (B) $\forall\ a \in R$ (C) for one value of a (D) for no value of a

निम्न में से किसके लिए रेखा युग्म $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ परस्पर लम्बवत् है :

(A) a के दो मान (B) $\forall\ a \in R$ (C) a के एक मान के लिए (D) a के किसी मान के लिए नही

Ans. A

Sol. $x^2 - 2pxy - y^2 = 0, \quad x^2 - 2axy - y^2 = 0$

$ca^2 = 2m^2 + 5m + 3a = 0$

$$m_1m_2 = -1 = \frac{3a}{a^2-2} \quad \Rightarrow \quad 2 - a^2 = 3a$$

$$a^2 + 3a - 2 = 0$$

36. If one of the lines of $my^2 + (1 - m^2)\,xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is :

यदि $my^2 + (1 - m^2)\,xy - mx^2 = 0$ में से सरल रेखाओं $xy = 0$ के मध्य के कोण का समद्विभाजक है, तब m है :

(A) 1 (B) 2 (C) $-\frac{1}{2}$ (D) –2

Ans. A

Sol. Bisector of $x = 0$ and $y = 0$ is either $y = x$ or $y = -x$
If $y = x$ is Bisector, then
$mx^2 + (1 - m^2)x^2 - mx^2 = 0 \Rightarrow m + 1 - m^2 - m = 0 \Rightarrow m^2 = 1 \Rightarrow m = \pm 1$.

HOMOGENISATION

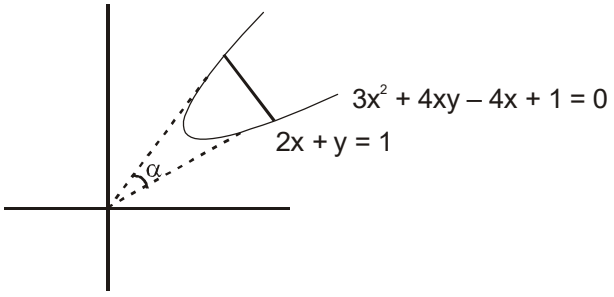
37. The straight lines joining the origin to the points of intersection of the line $2x + y = 1$ and curve $3x^2 + 4xy - 4x + 1 = 0$ include an angle :

सरल रेखा $2x + y = 1$ एवं वक्र $3x^2 + 4xy - 4x + 1 = 0$ के प्रतिच्छेद बिन्दुओं को मूल बिन्दु से मिलाने वाली सरल रेखाओं के मध्य कोण है :

(A) $\frac{\pi}{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{6}$

Ans. A

Sol. Homogenize given curve with given line
$3x^2 + 4xy - 4x(2x + y) + 1(2x + y)^2 = 0$
$3x^2 + 4xy - 8x^2 - 4xy + 4x^2 + y^2 + 4xy = 0$
$-x^2 + 4xy + y^2 =$
coeff. x^2 + coeff. $y^2 = 0$
Hence angle is 90º



38. If the straight lines joining the origin and the points of intersection of the curve $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$ are equally inclined to the co–ordinate axes then the value of k :

(A) is equal to 1 (B) is equal to -1
(C) is equal to 2 (D) does not exist in the set of real numbers

यदि वक्र $5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ तथा $x + ky - 1 = 0$ के प्रतिच्छेद बिन्दु को (0, 0) से मिलाने पर जो रेखा प्राप्त होती है वह दोनों अक्षों से बराबर झुकाव रखती हो तो k का मान होगा :

(A) 1 के बराबर (B) -1 के बराबर

(C) 2 के बराबर (D) वास्तविक संख्याओं के समुच्चय के लिये अस्तित्व में नहीं

Ans. B

Sol. Homogenise and put co-efficient of $xy = 0$

39. The angles between the straight lines joining the origin to the points common to $7x^2 + 8y^2 - 4xy + 2x - 4y - 8 = 0$ and $3x - y = 2$ is :

वक्र $7x^2 + 8y^2 - 4xy + 2x - 4y - 8 = 0$ एवं रेखा $3x - y = 2$ के उभयनिष्ट बिन्दु को $(0, 0)$ से मिला दिया जाये तो सरल रेखाओं के बीच का कोण होगा :

(A) $\tan^{-1}\sqrt{2}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Ans. D

Sol. $7x^2 + 8y^2 - 4xy + x(3x - y) - 2y(3x - y) - 2(3x - y^2) = 0$

$\Rightarrow -8x^2 + 8y^2 + 11xy = 0$

$\Rightarrow a + b = 0 \quad Q = \frac{\pi}{2}$

Exercise J-II

SUBJECTIVE / INTEGER TYPE QUESTIONS

COORDINATE SYSTEM, DISTANCE, SECTION FORMULAE

1. Determine the ratio in which the point P(3 , 5) divides the join of A(1, 3) & B(7, 9). Find the harmonic conjugate of P w.r.t. A & B.

वह अनुपात ज्ञात करो जिस अनुपात में बिन्दु P(3 , 5) बिन्दु A(1, 3) एवं B(7, 9) को मिलाने वाले रेखाखण्ड को विभाजित करता है एवं बिन्दु P का A एवं B के सापेक्ष हरात्मक संयुग्मी ज्ञात करो।

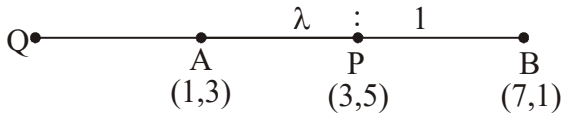
Ans. 1 : 2 ; Q(–5, –3)

Sol. $\frac{7\lambda+1}{\lambda+1}=3$

$7\lambda + = 3\lambda + 3$

$4\lambda = 2$

$1 : 2$



$x = \frac{2-7}{2-1} = -5$

$y = \frac{6-9}{2-1} = y - 3$

$Q(-5, -3)$

2. Line $\frac{x}{6}+\frac{y}{8}=1$ intersects the x and y axes at M and N respectively. If the coordinates of the point P lying inside the triangle OMN (where 'O' is origin) are (a, b) such that the areas of the triangle POM, PON and PMN are equal. Find

(a) the coordinates of the point P and

(b) the radius of the circle escribed opposite to the angle N.

सरल रेखा $\frac{x}{6}+\frac{y}{8}=1$ अक्षों x एवं y को क्रमशः M तथा N बिन्दुओं पर काटती है। यदि बिन्दु P के निर्देशांक (a, b) त्रिभुज OMN के अन्दर इस तरह स्थित हो (जहाँ 'O' मूल बिन्दु है) कि त्रिभुजों POM, PON एवं PMN का क्षेत्रफल बराबर है, ज्ञात कीजिये।

(a) बिन्दु P के निर्देशांक

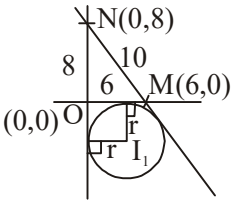
(b) कोण N के विपरीत बाह्य वृत की त्रिज्या

Ans. (a) $\left(2, \frac{8}{3}\right)$; (b) 4

Sol. $\frac{x}{6}+\frac{y}{8}=1$

(a) $P\left(2,\frac{8}{3}\right)$

(b) $x=\frac{0\times10+6\times8-6\times0}{8-6+10} \quad \Rightarrow \frac{-48}{12}$



$y=-2$

$r=4$

M-2 :

$r_1=\frac{\Delta}{(s-a)}$

3. In an acute triangle ABC, the base BC has the equation $4x-3y+3=0$. If the coordinates of the orthocentre (H) and circumcentre (P) of the triangle are $(1, 2)$ and $(2, 3)$ respectively, then the radius of the circle circumscribing the triangle is $\frac{\sqrt{m}}{n}$ where m and n are relatively prime. Find the value of $(m+n)$.

(You may use the fact that the distance between orthocentre and circumcentre of the triangle is given by $R\sqrt{1-8\cos A\cos B\cos C}$).

एक न्यूकोणीय त्रिभुज ABC में, आधार BC का समीकरण $4x - 3y + 3 = 0$ है। यदि त्रिभुज का लम्ब केन्द्र (H) तथा परिकेन्द्र (P) क्रमशः $(1, 2)$ व $(2, 3)$ है, तो त्रिभुज की परित्रिज्या $\frac{\sqrt{m}}{n}$ से व्यक्त की जायेगी। जहाँ m और n का महत्तम समापवर्तक 1 है, तो $(m + n)$ का मान ज्ञात करों।

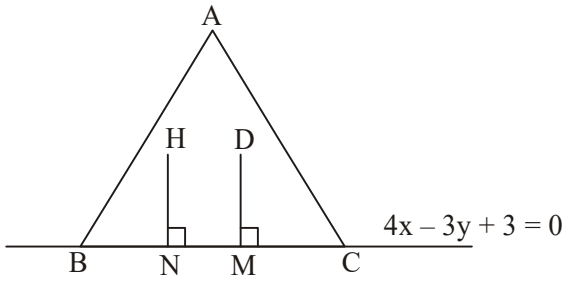
(आप इस तथ्य का इस्तेमाल कर सकते है कि त्रिभुज के लम्ब केन्द्र तथा परिकेन्द्र के बीच की दूरी $R\sqrt{1-8\cos A\cos B\cos C}$ होगी)

Ans. 63

Sol. $R=\frac{\sqrt{m}}{n}$

$HP=R\sqrt{1-8\cos A\cos B\cos C}$

$R=\frac{\sqrt{2}}{\sqrt{1-8\cos A\cos B\cos C}}$



$HN = 2r \cos B \cos C$

$DM = R \cos A$

$$\cos A \cos B \cos C = \frac{PM \times HN}{2R^2}$$

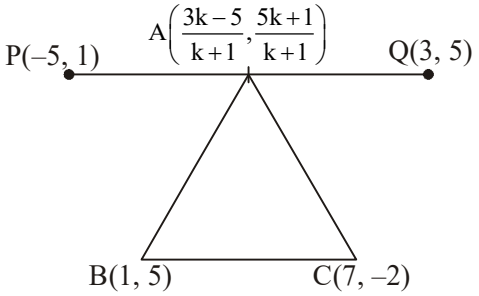
AREA OF TRIANGLE

4. The point A divides the join of $P(-5, 1)$ & $Q(3, 5)$ in the ratio $K : 1$. Find the two values of K for which the area of triangle ABC, where B is (1, 5) & C is (7, –2), is equal to 2 units in magnitude.

यदि बिन्दु A, $P(-5, 1)$ एवं $Q(3, 5)$ को मिलाने वाले रेखा खण्ड को $K : 1$ अनुपात में विभाजित करता है, तो K के वह दो मान ज्ञात करे, जिसके लिए त्रिभुज ABC के क्षेत्रफल का परिमाण 2 इकाई है। जहाँ बिन्दु B एवं बिन्दु C के निर्देशांक क्रमशः (1, 5) एवं (7, –2) है।

Ans. $K = 7$ or $\frac{31}{9}$

Sol.



$$\begin{vmatrix} 1 & 7 & \frac{3k-5}{k+1} & 1 \\ 5 & -2 & \frac{5k+1}{k+1} & 5 \end{vmatrix} = \pm 4$$

$$\Rightarrow -2 - 35 + \frac{35k+7}{k+1} - \frac{6k-10}{k+1} + \frac{15k-25}{k+1} - \frac{5k+1}{k-1} = \pm 4$$

$$\Rightarrow -37+\frac{51k-29}{k+1}=\pm 4$$

$$\begin{cases} 10k=70 & k=7 \\ 18k=62 & k=\dfrac{62}{18}=\dfrac{31}{9} \end{cases}$$

5. If $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$ $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ and $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$ then

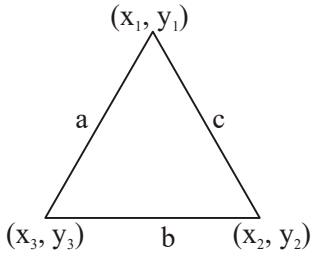
$$\lambda\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c).$$ Find the value of λ.

यदि $(x_1 - x_2)^2 + (y_1 - y_2)^2 = a^2$, $(x_2 - x_3)^2 + (y_2 - y_3)^2 = b^2$ एवं $(x_3 - x_1)^2 + (y_3 - y_1)^2 = c^2$ तो

$$\lambda\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}^2 = (a+b+c)(b+c-a)(c+a-b)(a+b-c).$$ तो λ का मान ज्ञात कीजिये।

Ans. 4

Sol. $A(x_1, y_2)$ $B(x_2, y_2)$, $C(x_3, y_3)$



$$\Delta=\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$\Delta=\sqrt{s(s-a)(s-b)(s-c)} \quad \Rightarrow \sqrt{\frac{(a+b+c)}{2}\frac{(b+c-a)}{2}\frac{(a+c-b)}{2}\frac{(a+b-c)}{2}}$$

$$\Delta^2 \Rightarrow \frac{1}{16}(a+b+c)$$

$$(a+b+c)(a+b-c)(a+c-b)(b+c-a)=160^2$$

$\Rightarrow 16 \times \frac{1}{4}$

$\lambda = 4$

6. The area of a triangle is 5. Two of its vertices are (2, 1) & (3, –2). The third vertex lies on y = x + 3. Find the third vertex.

यदि किसी त्रिभुज का क्षेत्रफल 5 ईकाई हो, एवं शीर्ष बिन्दुओं के निर्देशांक (2, 1) एवं (3, –2) हो, तथा तीसरा शीर्ष बिन्दु रेखा y = x + 3 पर स्थित हो तो तीसरे शीर्ष बिन्दु के निर्देशांक ज्ञात करों।

Ans. $\left(\frac{7}{2},\frac{13}{2}\right)$ or $\left(-\frac{3}{2},\frac{3}{2}\right)$

Sol. $\Delta = 5$ (2, 1) and (3, –2) and (h, h + 3)

$$\begin{vmatrix} h & 2 & 3 & h \\ h+3 & 1 & -2 & h+3 \end{vmatrix} = \pm 10$$

$h - 2n - 6 - 4 - 3 + 3h + 9 + 2h = \pm 10$

$$4n - 4 = \pm 10 \begin{cases} 4h = 14 & \Rightarrow h = \frac{7}{2} \\ 4n = -6 & h = \frac{-3}{2} \end{cases}$$

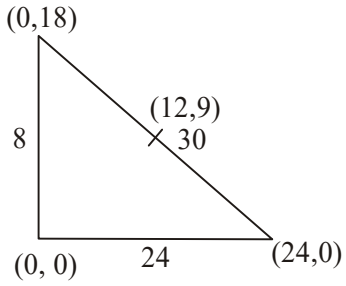
$\left(\frac{7}{2},\frac{13}{2}\right),\left(\frac{-3}{2},\frac{3}{2}\right)$

7. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle.

एक त्रिभुज की भुजाओं की लम्बाई 18, 24 एवं 30 है, उस त्रिभुज का क्षेत्रफल ज्ञात कीजिये जिसके शीर्ष बिन्दु दिये गये त्रिभुज के अंतः केन्द्र, परिकेन्द्र एवं केन्द्रक है।

Ans. 3 Units

Sol.



C(12,9), G(8, 6)

$$I \equiv \left(\frac{0+18\times 24+0}{72}, \frac{0+0+18\times 24}{72} \right)$$

I (6, 6)

$$\Delta CGI = \frac{1}{2}\begin{vmatrix} 12 & 8 & 6 & 2 \\ 9 & 6 & 6 & 9 \end{vmatrix}$$

$$\Rightarrow \frac{1}{2}|72-72+48-36+54-72|$$

$$= \frac{1}{2}\times(6)$$

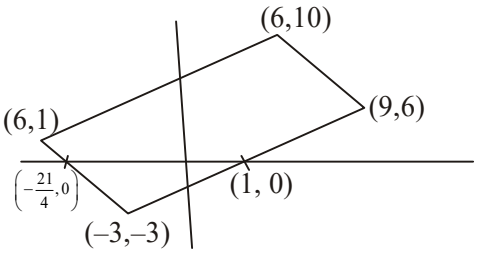
= 3

8. The points (–6, 1), (6, 10), (9, 6) and (–3, –3) are the vertices of a rectangle. If the area of the portion of this rectangle that lies above the x axis is a/b, find the value of (a + b), given a and b are coprime.

बिन्दु (–6, 1), (6, 10), (9, 6) एवं (–3, –3) किसी आयत के शीर्ष बिन्दु हो, तथा x अक्ष के उपर स्थित भाग का क्षेत्रफल a/b है तो (a + b) का मान ज्ञात करो यदि a एवं b परस्पर अभाज्य (coprime) है।

Ans. 533

Sol.



(–6, 1), (6, 10), (9,6) and (–3, –3)

$$\text{Area of rectangle} = \left(\sqrt{81+144}\right)\left(\sqrt{16+9}\right) = 75$$

$$\text{area below x–axis} = \frac{1}{2}\times\frac{25}{4}\times 3 = \frac{75}{8}$$

$$\text{Area above x–axis} = \frac{525}{8}$$

a + b = 533

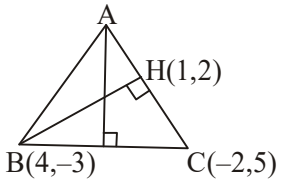
EQUATION OF LINE AND ANGLE BETWEEN LINES

9. Two vertices of a triangle are (4, –3) & (–2, 5). If the orthocentre of the triangle is at (1, 2), find the coordinates of the third vertex.

यदि त्रिभुज के दो शीर्ष बिन्दुओं के निर्देशांक (4, –3) एवं (–2, 5) है, एवं लम्ब केन्द्र बिन्दु (1, 2) पर स्थित हो, तो तीसरे शीर्ष बिन्दु के निर्देशांक ज्ञात कीजिये।

Ans. (33, 26)

Sol. orthocentre of ΔBCH will be A. Hence



$$AH \Rightarrow (y-2) = \frac{6}{8}(x-1)$$

$$AC \Rightarrow (y-5) = \frac{3}{5}(x+2)$$

A(33, 26)

10. A line is such that its segment between the straight lines $5x - y - 4 = 0$ and $3x + 4y - 4 = 0$ is bisected at the point (1, 5). Obtain the equation.

उस रेखा का समीकरण ज्ञात करे। जिसका सरल रेखाओं $5x - y - 4 = 0$ एवं $3x + 4y - 4 = 0$ के मध्य काटा गया रेखाखण्ड, बिन्दु (1, 5) पर समद्विभाजित होता है।

Ans. $83x - 35y + 92 = 0$

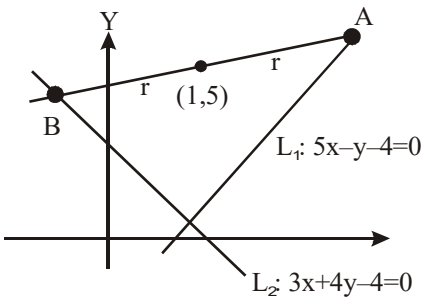
Sol. Coordinate of A which is at a distance 'r' from (1, 5) is $(1 + r\cos\theta, 5 + r\sin\theta)$

which satisfies $L_1 = 0$

$5(1 + r\cos\theta) - (5 + r\sin\theta) - 4 = 0$

$r(5\cos\theta - \sin\theta) = 4$

$$r = \frac{4}{5\cos\theta - \sin\theta}$$



Similarly coordinate of point B $(1 - r\cos\theta, 5 - r\sin\theta)$ which lies on $L_2 = 0$

$\therefore\ 3(1 - r\cos\theta) + 4(5 - r\sin\theta) - 4 = 0$

$$r = \frac{19}{3\cos\theta + 4\sin\theta}$$

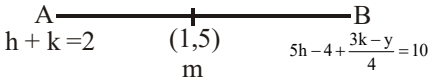
Equating both values of r & get tan θ (= m) and equation of the line passing through (1, 5).

$$m = \tan\theta = \frac{83}{35}$$

Alternate solution

$5x - y - 1 = 0$ $\qquad$ $3x + 4y - 4 = 0$

$(4, 5h, -4)$ $\qquad$ $\left(k, \frac{3x-4}{4}\right)$



$20h + 3k = 40 + 20 = 60$

$3h + 3k = 6$

$17h = 54$

$$h = \frac{54}{17}$$

$$k = 2 - \frac{54}{17}$$

$$k = \frac{-20}{17}$$

$$\left(\frac{54}{17}, \frac{270-68}{17}\right)$$

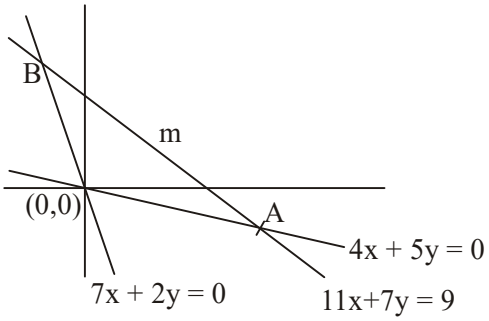
$$\left(\frac{54}{17}, \frac{202}{17}\right)$$

11. Two consecutive sides of a parallelogram are $4x + 5y = 0$ and $7x + 2y = 0$. If the equation to one diagonal is $11x + 7y = 9$, find the equation to the other diagonal.

यदि सामान्तर चतुर्भुज की दो संलग्न भुजाओं के समीकरण $4x + 5y = 0$ एवं $7x + 2y = 0$ हो, एवं एक विकर्ण का समीकरण $11x + 7y = 9$, तो दूसरे विकर्ण का समीकरण ज्ञात करो।

Ans. $x - y = 0$

Sol.



$A\left(\frac{5}{3},-\frac{4}{3}\right)$

$B\left(-\frac{2}{3},\frac{7}{3}\right)$

$M\left(\frac{1}{2},\frac{1}{2}\right)$

Equation of $OM \rightarrow y = x$

12. The line $3x + 2y = 24$ meets the y–axis at A & the x–axis at B. The perpendicular bisector of AB meets the line through $(0, -1)$ parallel to x–axis at C. Find the area of the triangle ABC.

यदि रेखा $3x + 2y = 24$ y–अक्ष को बिन्दु A एवं x अक्ष को बिन्दु B पर मिलती है, एवं रेखाखंड AB का लम्ब समद्विभाजक, रेखा जो कि x अक्ष के समातर एवं बिन्दु $(0, -1)$ से गुजरती है, को बिन्दु C पर प्रतिच्छेद करता है, तो त्रिभुज ABC का क्षेत्रफल ज्ञात करें।

Ans. 91 sq.units

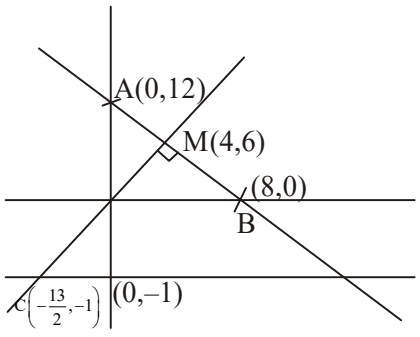
Sol. $3x + 2y = 24$

$CM \Rightarrow (y-6) = \frac{2}{3}(x-4)$

$y = -1$ $\qquad C\left(-\frac{13}{2},-1\right)$

$\frac{-7\times 3}{2} + 4 = x \Rightarrow \frac{-21+8}{2} = \frac{-13}{2}$

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} 0 & 8 & -\frac{13}{2} & 0 \\ 12 & 0 & -1 & 12 \end{vmatrix} = \frac{1}{2}[-96-8-78] = \frac{182}{2} = 91$$



13. Find the area of the triangle formed by the straight lines whose equations are $x + 2y - 5 = 0$; $2x + y - 7 = 0$ and $x - y + 1 = 0$. Also compute the tangent of the interior angles of the triangle and hence comment upon the nature of triangle.

रेखाओं, जिनका समीकरण $x + 2y - 5 = 0; 2x + y - 7 = 0$ एवं $x - y + 1 = 0$ है, से बनने वाले त्रिभुज का क्षेत्रफल ज्ञात करो। साथ ही आंतरिक कोणो की स्पर्शज्याऐं ज्ञात कीजिये एवं त्रिभुज की प्रकृति पर टिप्पणी करे।

Ans. $\frac{3}{2}$ sq. units, $\left(3, 3, \frac{3}{4}\right)$, isosceles

Sol. $x + 2y - 5 = 0$ $\quad m_2 = -\frac{1}{2}$

$2x + y - 7 = 0$ $\quad m_3 = -2$ $\quad m_1 > m_2 > m_3$

$x - y + 1 = 0$ $\quad m_1 = 1$ $\quad 1 \quad -\frac{1}{2} \quad -2$

$$\tan I = \left|\frac{1+\frac{1}{2}}{1+\frac{1}{2}}\right| = 3$$

$$\tan II = \left|\frac{-\frac{1}{2}+2}{1+1}\right| \Rightarrow \frac{\frac{3}{2}}{2} = \frac{3}{4}$$

$$\tan III = \left|\frac{-2-1}{1+2}\right| \Rightarrow 3 \quad \Rightarrow III \Rightarrow \text{Isosceles}$$

$L_2L_3\,(2, 3)$ $\quad 3x - 6 = 0$

$x = 2 \quad y = 3$

$L_3L_1\,(1,2)$ $\quad x + 2y - 5 = 0$

$2x - 2y + 2 = 0$

$3x - 3 = 0$

$x = 1, y = 2$

$L_1L_2\,(3,1)$ $\quad 4x + 2y - 14 = 0$

$x + 2y - 5 = 0$

$3x = 9$

$x = 3, y = 1$

$$\Delta = \frac{1}{2}\begin{vmatrix} 2 & 1 & 3 & 2 \\ 3 & 2 & 1 & 3 \end{vmatrix}$$

$$= \frac{1}{2}[4 - 3 + 1 - 6 + 9 - 2]$$

$$\Rightarrow \frac{1}{2}(3) = \frac{3}{2} \text{ sq. units}$$

14. The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$. Find c & the remaining vertices.

बिन्दु (1, 3) एवं (5, 1) किसी आयत के विपरीत शीर्षो के निर्देशांक है एवं अन्य दो शीर्ष बिन्दु रेखा $y = 2x + c$ पर स्थित है, तो c का मान एवं अन्य शीर्ष बिन्दु के निर्देशांक ज्ञात करो।

Ans. $c = -4$; B(2 , 0); D(4 , 4)

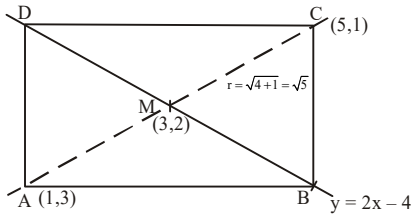
Sol. A (1,3) & C(5,1)

M (3,2)

$y = 2x + c \Rightarrow y = 2x - 4$

$2 = 6 + c$

$c = -4$



$B_1D(3 \pm r\cos\theta, 2 \pm r\sin\theta)$

$$B\left(3 + \sqrt{5} \times \frac{1}{\sqrt{5}}, 2 + \sqrt{5} \times \frac{2}{\sqrt{5}}\right) = (4,4)$$

D (3 – 1, 2 – 2) = (2, 0)

15. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

एक सरल रेखा L जो कि रेखा $5x - y = 1$ के लम्बवत् है, यदि रेखा L एवं अक्षो के साथ मिलकर बनने वाले त्रिभुज का क्षेत्रफल 5 हो, तो रेखा का समीकरण ज्ञात करों।

Ans. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

Sol. $5x - y = 1$

$L : x + 5y = \lambda \qquad a = \lambda, \; b = \frac{\lambda}{5}$

$$\frac{1}{2}\times\lambda\times\frac{\lambda}{5}=5$$

$\lambda^2 = 50$

$\lambda = \pm 5\sqrt{2}$

$x + 5y = 5\sqrt{2}$

$x + 5y - 5\sqrt{2} = 0$

16. The equations of the perpendicular bisectors of the sides AB and AC of a triangle ABC are $x - y + 5 = 0$ and $x + 2y = 0$, respectively. If the point A is $(1, -2)$ find the equation of the line BC.

किसी त्रिभुज ABC की भुजाओं AB एवं AC के लम्ब समद्विभाजको समीकरण क्रमशः $x - y + 5 = 0$ एवं $x + 2y = 0$ हो, एवं बिन्दु A के निर्देशांक $(1, -2)$ हो तो रेखा BC का समीकरण ज्ञात करो।

Ans. $14x + 23y = 40$

Sol.

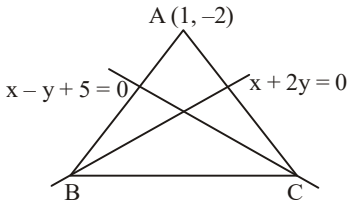


Image of A

B

$$\frac{x-1}{1}=\frac{y+2}{-1}=\frac{-2(1+2+5)}{2}$$

$x = 1 - 8 = -7$ B $(-7, 6)$

$y = -2 + 8 = 6$

C

$$\frac{x-1}{1}=\frac{y+2}{2}=\frac{-2(1-4)}{5}$$

$$x = 1+\frac{6}{5}=\frac{11}{5} \qquad C\left(\frac{11}{5},\frac{2}{5}\right)$$

$$y = -2+\frac{12}{5}=\frac{2}{5}$$

BC :→

$$(y-6)=\left(\frac{6-\frac{2}{5}}{-7-\frac{11}{5}}\right)(x+7)$$

$$(y-6)=\frac{28}{-46}(x+7)$$

$-23y + 138 = 14x + 98$

$14x + 23y - 40 = 0$

17. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the Δ ABC are G, H and S respectively. $L = 0$ denotes the equation of side BC.

(a) If $L_1 : 2x - y = 0$ and $L_2 : x + y = 3$ and G (2, 3) then find the slope of the line $L = 0$.

(b) If $L_1 : 2x + y = 0$ and $L_2 : x - y + 2 = 0$ and H (2, 3) then find the y-intercept of $L = 0$.

(c) If $L_1 : x + y - 1 = 0$ and $L_2 : 2x - y + 4 = 0$ and S(2, 1) then find the x-intercept of the line $L = 0$.

किसी त्रिभुज ABC जिसकी भुजाओं AB एवं AC के समीकरण यदि $L_1 = 0$ एवं $L_2 = 0$ हो, एवं केन्द्रक, लम्बकेन्द्र एवं परिकेन्द्र क्रमशः G, H एवं S हो, एवं $L = 0$ रेखा BC का समीकरण है।

(a) यदि $L_1 : 2x - y = 0$ एवं $L_2 : x + y = 3$ एवं G (2, 3) हो तो रेखा $L = 0$ की प्रवणता ज्ञात करो।

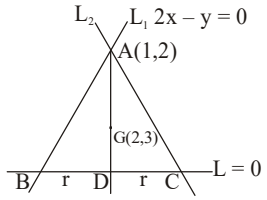
(b) यदि $L_1 : 2x + y = 0$ एवं $L_2 : x - y + 2 = 0$ एवं H (2, 3) तो रेखा $L = 0$ द्वारा y अक्ष पर काटे गये अंतखण्ड की लम्बाई ज्ञात करो।

(c) यदि $L_1 : x + y - 1 = 0$ एवं $L_2 : 2x - y + 4 = 0$ तथा S(2, 1) तो $L = 0$ द्वारा x अक्ष पर काटे गये अंतः खण्ड की लम्बाई ज्ञात करो।

Ans. (a) 5 (b) 2 (c) 3/2

Sol. (a) ΔABC, AB & AC $L_1 = 0$ & $L_2 = 0$

BC $\Rightarrow L - 0$ G → Centroid, H → Ortho, S → Circum



D Externally 3,1

$$\left(\frac{6-1}{2},\frac{9-2}{2}\right)\equiv\left(\frac{5}{2},\frac{7}{2}\right)$$

Let $B\left(\frac{5}{2}+r\cos\theta,\frac{7}{2}+r\sin\theta\right)$

$2x - y = 0$

$$5+2r\cos\theta-\frac{7}{2}-r\sin\theta=0$$

$$r\left(2\cos\theta-\sin\theta\right)=-\frac{3}{2}$$

$$r=\frac{3}{2\left(\sin\theta-2\cos\theta\right)}$$

$2\sin\theta - 4\cos\theta = \cos\theta + \sin\theta$

$\sin\theta = 5\cos\theta \Rightarrow \tan\theta = 5$

(b) $CH \Rightarrow x - 2y + 4 = 0$

$AC \Rightarrow x - y + 2 = 0$

C(0,2)

$BH \Rightarrow x + y - 5 = 0$

$AB \Rightarrow 2x + y = 0$

B (–5,10)

$BC \Rightarrow 8x + 5y = 10$

y-intercept = 2

(c) A (–1,2)

$C \rightarrow$ Image of A w.r.t. to SE

$SE \rightarrow x + 2y - 4 = 0$

$$C\left(\frac{13}{5},\frac{46}{5}\right)$$

$B \rightarrow$ Image of A w.r.t. SF

$SF \rightarrow x - y - 1 = 0$

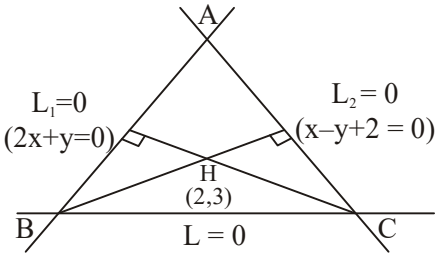
B(3, –2)

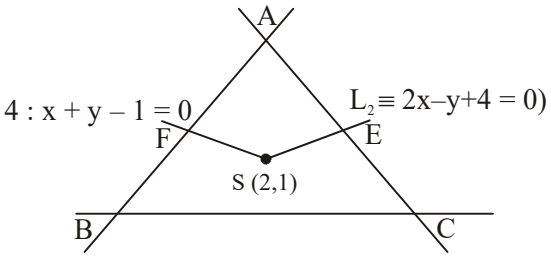
$C\left(\frac{5}{2}-r\cos\theta,\frac{7}{2}-r\sin\theta\right)$

$x + y = 3$

$$\frac{5}{2}+\frac{7}{2}-3=r\left(\cos\theta-\sin\theta\right)$$

$$r=\frac{3}{\cos\theta+\sin\theta}$$



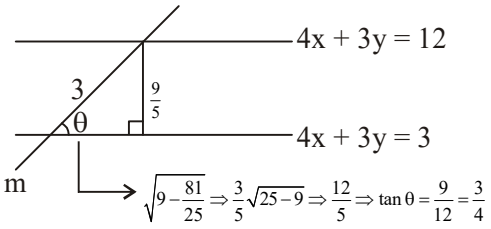


18. Find the equation of the straight lines passing through (–2, –7) & having an intercept of length 3 between the straight lines $4x + 3y = 12$, $4x + 3y = 3$.

बिन्दु (–2, –7) से गुजरने वाली रेखा का समीकरण ज्ञात करो, जो कि रेखाओं $4x + 3y = 12$, $4x + 3y = 3$ के मध्य 3 इकाई का अंतखण्ड काटती है।

Ans. $7x + 24y + 182 = 0$ or $x = -2$

Sol.



$$\tan\theta = \left|\frac{m + \frac{4}{3}}{1 - \frac{4}{3}m}\right| \Rightarrow \frac{3m+4}{3-4m} = \pm\frac{3}{4}$$

$\frac{3m+4}{3-4m} = \frac{3}{4}$

$12m + 16 = 9 - 12m$

$24m = -7$

$m = -\frac{7}{24}$

$\frac{3m+4}{3-4m} = -\frac{3}{4}$

$12m + 16 = -9 + 12m$

$m = \infty$

$(y + 7) = m(x + 2)$

$24y + 168 = -7x - 14$

$7x + 24y + 182 = 0$

$x = -2$

19. Consider a Δ ABC whose sides AB, BC and CA are represented by the straight lines $2x + y = 0$, $x + py = q$ and $x - y = 3$ respectively. The point P is (2, 3).

(a) If P is the centroid, then find the value of $(p + q)$.

(b) If P is the orthocentre, then find the value of $(p + q)$.

(c) If P is the circumcentre, then find the value of $(p + q)$.

किसी त्रिभुज ABC , जिसकी भुजाऐं AB, BC एवं CA क्रमशः रेखाओं के समीकरण $2x + y = 0$, $x + py = q$ तथा $x - y = 3$ से प्रदर्शित है। यदि P के निर्देशांक (2, 3)

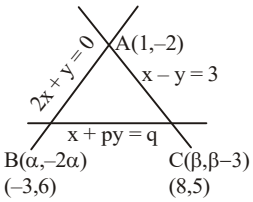
(a) यदि P केन्द्रक है, तो $(p + q)$ का मान ज्ञात करो

(b) यदि P लम्बकेन्द्र है, तो $(p + q)$ का मान ज्ञात करो

(c) यदि P परिकेन्द्र है, तो $(p + q)$ का मान ज्ञात करो

Ans. (a) 74 (b) 50 (c) 47

Sol.



(a) centroid (2, 3)

$1 + \alpha + \beta = 6$ ...(1)

$-2 - 2\alpha + \beta - 3 = 9$ ...(2)

$\alpha + \beta = 5$

$2\alpha - \beta = -14$

$3\alpha = -9$

$\alpha = -3,\ \beta = 8$

$x + py = q$ $\quad 8 + 5p = q$

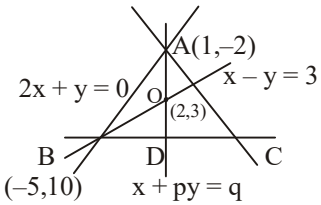
$-3 + p6 = q$

$-3 + 6p = 8 + 5p \Rightarrow p = 11$ & $q = 63$

$x + 11y = 63$

$p + q = 11 + 63 = 74$

(b) orthocentre (2, 3)



$$m_{AD} = \frac{3+2}{2-1} = 5$$

$$m_{BC} = -\frac{1}{5} = -\frac{1}{p}$$

$p = 5$

$BE \rightarrow x + y = \lambda$

(2, 3)

$x + y = 5$

$2x + y = 0$

$-x = 5$

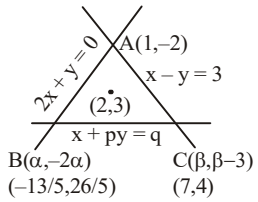
$x = -5$

$y = 10$

$-5 + 50 = q$

$q = 45$

$p + q = 45 + 5 = 50$

(C) 

(2,3) circumcentre

$R = \sqrt{1+25} = \sqrt{26}$

$(\alpha-2)^2 + (3+2\alpha)^2 = 26$

$\Rightarrow 5\alpha^2 - 4\alpha + 4 + 12\alpha + 9 = 26$

$\Rightarrow 5\alpha^2 + 8\alpha - 13 = 0 \rightarrow 1, \alpha$

$\alpha = -\frac{13}{5}$

$\alpha(\beta-2)^2 + (\beta-6)^2 = 26$

$\Rightarrow 2\beta^2 - 16\beta + 40 - 26 = 0$

$\Rightarrow 2\beta^2 - 16\beta + 14 = 0 \rightarrow 1, \beta$

$\beta = 7$

$7 + 4p = q$

$-13 + 26\,p = 5q$

$\underline{35 + 20\,p = 5q}$

$-48 + 6p = 0$

$p = 8$

$q = 7 + 32 = 39$

$\Rightarrow p + q = 8 + 39 = 47$

PARAMETRIC FORM OF A LINE

20. A line through the point P(2, –3) meets the lines $x - 2y + 7 = 0$ and $x + 3y - 3 = 0$ at the points A and B respectively. If P divides AB externally in the ratio 3 : 2 then find the equation of the line AB.

यदि बिन्दु P(2, –3) से गुजरने वाली रेखा, सरल रेखाओं $x - 2y + 7 = 0$ एवं $x + 3y - 3 = 0$ को क्रमशः बिन्दुओं A एवं B पर मिलती है। यदि बिन्दु P रेखाखण्ड AB को अनुपात 3 : 2 में बाह्यविभाजित करे तो AB का समीकरण ज्ञात कीजिये।

Ans. $2x + y - 1 = 0$

Sol. P(2, –3) $x - 2yb + 7 = 0$ & $x + 3y - 3 = 0$

$2 + 3k\cos\theta + 6 - 6k\sin\theta + 7 = 0$

$6k\sin\theta - 3k\cos\theta = 15$

$2k\sin\theta - k\cos\theta = 5$ ...(1)

$2 + 2k\cos\theta - 9 + 6k\sin\theta - 3 = 0$

$2k\cos\theta + 6k\sin\theta = 10$

$k\cos\theta + 3k\sin\theta = 5$ ...(2)

(1)/(2)

$$\frac{2s - c}{c + 3s} = 1$$

$$\Rightarrow \frac{2t - 1}{3t +} = 1$$

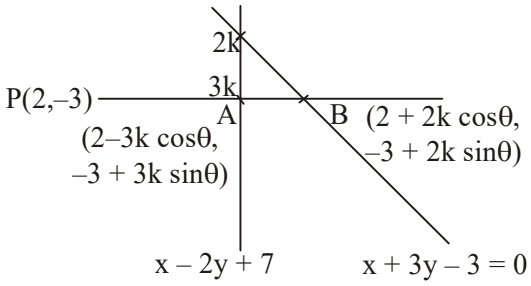
$\Rightarrow 2t - 1 = 3t + 1$

$\Rightarrow t = -2$

$(y + 3) = -2(x - 2)$

$y + 3 = -2x + y$

$2x + y = 1$



21. If the straight line drawn through the point $P(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ.

बिन्दु $P(\sqrt{3}, 2)$ से गुजरने वाली एवं x अक्ष से $\frac{\pi}{6}$ कोण बनाने वाली रेखा, किसी अन्य रेखा $\sqrt{3}x - 4y + 8 = 0$ को बिन्दु Q पर मिलती है, तो रेखाखण्ड PQ की लम्बाई ज्ञात करो।

Ans. 6 units

Sol. $P\left(\sqrt{3},2\right)$ $\quad Q\left(\sqrt{3}+\frac{r\sqrt{3}}{2},2+\frac{r}{2}\right)$

$\sqrt{3}x-4y+8=0$

$\Rightarrow 3+\frac{3r}{2}-8-2r+8=0$

$\Rightarrow \frac{r}{2}=3\Rightarrow r=6$

FAMILY OF LINES

22. Consider the family of lines $(x-y-6)+\lambda(2x+y+3)=0$ and $(x+2y-4)+\mu(3x-2y-4)=0$. If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.

रेखाओं का निकाय $(x-y-6)+\lambda(2x+y+3)=0$ एवं $(x+2y-4)+\mu(3x-2y-4)=0$. यदि दोनों निकायो की रेखायें एक दूसरे से समकोण बनाती है तो इनके प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात करो।

Ans. $x^2+y^2-3x+4y-3=0$

Sol. $(x-y-6)+1(2x+y+3)=0$

$2x+y+3=0$

$3x=3$

$x=1 \quad y=-5 \qquad (1,-5)$

$x+2y-4=0 \qquad 2+2y-4=0$

$3x-2y-4=0 \qquad 1+y-2=0$

$4x=8 \qquad y=1 \qquad (2,1)$

$x=2$

$(x-1)(x-2)+(y+5)(y-1)=0$

$x^2+y^2-3x+4y-3=0$

CONCEPT OF IMAGE

23. Let P be the point (3, 2). Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.

बिन्दु P (3, 2), जहाँ बिन्दु Q, बिन्दु P का, x अक्ष के सापेक्ष एवं बिन्दु R, बिन्दु Q का, रेखा $y = -x$ के सापेक्ष प्रतिबिम्ब बिन्दु है। R का मूल बिन्दु के सापेक्ष प्रतिबिम्ब S है तथा PQRS एक उत्तल चतुर्भुज है तो PQRS का क्षेत्रफल ज्ञात करो।

Ans. 15

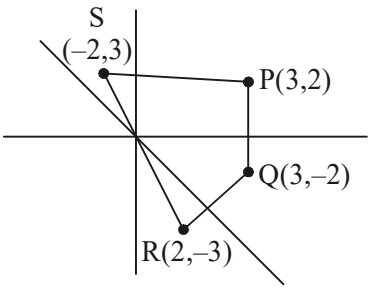
Sol. P (3, 2)

Q (3, –2)

$$\Delta = \frac{1}{2}\begin{vmatrix} 3 & 3 & 2 & -2 & 3 \\ 2 & -2 & -3 & 3 & 2 \end{vmatrix}$$

$$= \frac{1}{2}[-6-6-9+4+6-6-4-9]$$

$$= \frac{1}{2}[30] = 15$$



ANGLE BISECTORS

24. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ & its third side passes through the point (1, –10). Determine the equation of the third side.

यदि समद्विबाहु त्रिभुज की दो समान भुजाओं के समीकरण $7x - y + 3 = 0$ एवं $x + y - 3 = 0$ हो, एवं तीसरी भुजा बिन्दु (1, –10) से गुजरती हो, तो तीसरी भुजा का समीकरण ज्ञात करो।

Ans. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$

Sol. $7x - y + 3 = 0$ $\qquad \frac{7x - y + 3}{x\sqrt{5}} = \infty$ $\qquad \pm\frac{x + y - 3}{\sqrt{2}}$

$x + y - 3 = 0$

$7x - y + 3 = 5x + 5y - 15$ $\qquad$ $7x - y + 3 = -5x - 5y + 15$

$2x - 6y + 18 = 0$ $\qquad$ $12x + 4y - 12 = 0$

$x - 3y + 9 = 0$ $\qquad$ $3x + y - 3 = 0$

$m = \frac{1}{3}$ $\qquad$ $m = -3$

$(y+10)=\frac{1}{3}(x-1)$ $\qquad$ $(y+10)=-3(x-1)$

$3y+30=x-1$ $\qquad$ $y+10=-3x+3$

$x-3y-31=0$ $\qquad$ $3x+y+7=0$

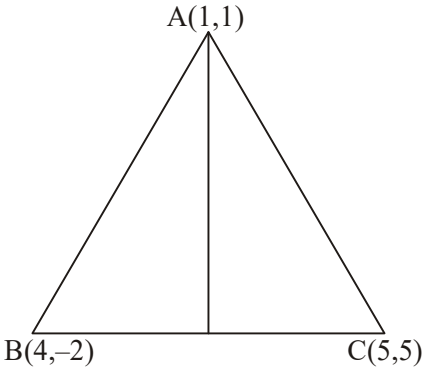
25. Given vertices A(1, 1), B(4, –2) and C(5, 5) of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

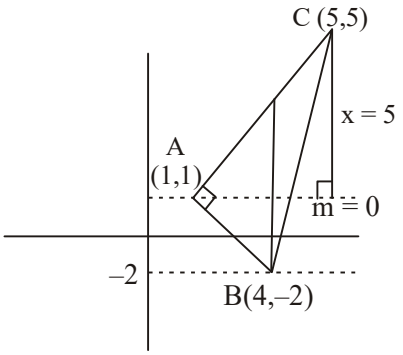
A(1, 1), B(4, –2) एवं C(5, 5) त्रिभुज के शीर्ष बिन्दु हो, तो बिन्दु C से कोण A के आन्तरिक कोणार्धक पर डाले गये लम्ब का समीकरण ज्ञात करो।

Ans. $x-5=0$

Sol. A(1,1), B(4, –2) & C(5, 5)

$$\left.\begin{matrix} m_{AC}=1 \\ m_{AB}=-1 \end{matrix}\right] \text{bisector} \qquad m=0$$





GENERAL LOCUS PROBLEMS

26. A variable line, drawn through the point of intersection of the straight lines $\frac{x}{a}+\frac{y}{b}=1$ & $\frac{x}{b}+\frac{y}{a}=1$, meets the coordinate axes in A and B. Find the locus of the mid point of AB.

रेखाओं $\frac{x}{a}+\frac{y}{b}=1$ एवं $\frac{x}{b}+\frac{y}{a}=1$ के प्रतिच्छेद बिन्दुओं से गुजरने वाली चर रेखा, निर्देशांक अक्षों को क्रमशः A एवं B पर मिलती है, तो रेखाखंड AB के मध्य बिन्दु के बिन्दुपथ (locus) का समीकरण ज्ञात करो।

Ans. $2xy(a+b)=ab(x+y)$

Sol. By family of lines $\left(\frac{x}{a}+\frac{y}{b}-1\right)+\lambda\left(\frac{x}{b}+\frac{y}{a}-1\right)=0$ it cuts x-axis at $A\left(\frac{\lambda+1}{\frac{1}{a}+\frac{\lambda}{b}},\ 0\right)$

y - axis at $B\left(0,\ \frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}\right)$. Let mid point of AB is P(h, k) $\Rightarrow$ $2h=\frac{\lambda+1}{\frac{1}{a}+\frac{\lambda}{b}},\ 2k=\frac{\lambda+1}{\frac{\lambda}{a}+\frac{1}{b}}$

eliminate λ from both equations we get $2hk(a+b)=ab(h+k)$

Hence locus of P(h, k) is $2xy(a+b)=ab(x+y)$

27. In the xy plane, the line 'l_1' passes through the point (1, 1) and the line 'l_2' passes through the point (–1, 1). If the difference of the slopes of the lines is 2. Find the locus of the point of intersection of the lines l_1 and l_2

xy समतल में, यदि सरल रेखा line 'l_1' बिन्दु (1, 1) एवं सरल रेखा 'l_2' बिन्दु (–1, 1) से गुजरती है, एवं उसकी प्रवणताओं का अंतर 2 है तो सरल रेखाओं l_1 एवं l_2 के प्रतिच्छेद बिन्दु का बिन्दुपथ ज्ञात करो।

Ans. $y=x^2$ and $y=2-x^2$

Sol.

$m_1-m_2=2$	or	$m_1-m_2=-2$
$\frac{k-1}{h-1}-\frac{k-1}{h+1}=2$		$\frac{k-1}{h-1}-\frac{k-1}{h+1}=-2$
.....		.....
$x^2=y$		$y=2-x^2$

28. The area of the triangle formed by the intersection of a line parallel to x–axis and passing through P (h, k) with the lines $y=x$ and $x+y=2$ is $4h^2$. Find the locus of the point P.

x अक्ष के समान्तर एवं P (h, k) से गुजरने वाली रेखा तथा रेखाओं $y=x$ एवं $x+y=2$ द्वारा निर्मित त्रिभुज का क्षेत्रफल $4h^2$ है तो बिन्दु P का बिन्दु पथ ज्ञात कीजिये।

Ans. $y=2x+1,\ y=-2x+1$

Sol. A line passing through P(h, k) and parallel to x-axis is $y=k$. .....(i)

The other lines given are $y=x$ .....(ii)

and $y+x=2$ .....(iii)

Let ABC be the Δ formed by the points of intersection of the lines (i), (ii) and (iii).

$\therefore$ A(k, k) , B(1, 1), C (2–k, k)

$$\therefore \qquad \text{Area of } \Delta ABC = \frac{1}{2}\begin{vmatrix} k & k & 1 \\ 1 & 1 & 1 \\ 2-k & k & 1 \end{vmatrix} = 4h^2$$

$$C_1 \to C_1 - C_2 \quad \frac{1}{2}\begin{vmatrix} 0 & k & 1 \\ 0 & 1 & 1 \\ 2-2k & k & 1 \end{vmatrix} = 4h^2$$

$$\Rightarrow \qquad \frac{1}{2}\,|(2-2k)(k-1)| = 4h^2 \qquad \Rightarrow \qquad (k-1)^2 = 4h^2$$

$$\Rightarrow \qquad k-1 = 2h, \qquad k-1 = -2h \qquad \Rightarrow \qquad k = 2h+1 \qquad k = -2h+1$$

$\therefore$ locus of (h, k) is $y = 2x + 1$ $y = -2x + 1$

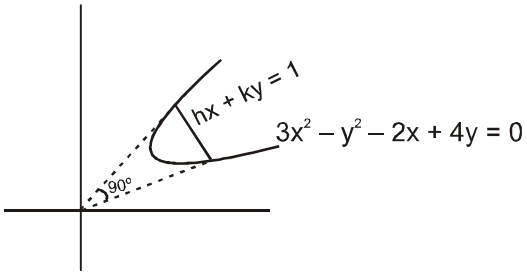
HOMOGENISATION

29. Show that all the chords of the curve $3x^2 - y^2 - 2x + 4y = 0$ which subtend a right angle at the origin are concurrent. Does this result also hold for the curve, $3x^2 + 3y^2 - 2x + 4y = 0$? If yes, what is the point of concurrency & if not, give reasons.

दशाओं कि किसी वक्र $3x^2 - y^2 - 2x + 4y = 0$ की सभी जीवाऐं, जो कि मूल बिन्दु पर समकोण बनाती है, संगामी होती है। क्या उपरोक्त कथन वक्र $3x^2 + 3y^2 - 2x + 4y = 0$ के लिये भी सत्य है, यदि हाँ, तो संगमन बिन्दु ज्ञात करो यदि नही, तो कारण बताओं।

Ans. $(1, -2)$, yes $\left(\frac{1}{3}, -\frac{2}{3}\right)$

Sol. Let equation of chords $hx + ky = 1$



By homogenisation

$3x^2 - y^2 - 2x(hx + ky) + 4y(hx + ky) = 0$

$\therefore$ it makes 90°. Hence

coeff. x^2 + coeff. $y^2 = 0$

$3 - 2h - 1 + 4k = 0 \Rightarrow h - 2k = 1$

Hence all chords are concurrent at $(1, -2)$

$3x^2 + 3y^2 - 2x + 4y = 0$

$$x^2 + y^2 - \frac{2}{3}x + \frac{y}{3}y = 0$$

circle centre at $\left(\frac{1}{3},-\frac{2}{3}\right)$

Passing through origin chords subtend 90º at origin will be diameters P.O. concurrents

$\left(\frac{1}{3},-\frac{2}{3}\right)$

30. A straight line is drawn from the point (1, 0) to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

यदि बिन्दु (1, 0) से वक्र $x^2 + y^2 + 6x - 10y + 1 = 0$ पर रेखा इस तरह खीची जाए कि रेखा द्वारा वक्र पर काटे जाने वाला अन्तः खण्ड मूल बिन्दु पर समकोण बनाये तो रेखा का समीकरण ज्ञात करो।

Ans. $x + y = 1;\ x + 9y = 1$

Sol. (1,0) $\qquad x^2 + y^2 + 6x - 10y + 1 = 0$

$x + py = 1 \qquad x^2 + y^2 + 6x(x + py) - 10y(x + py) + (x + py)^2 = 0$

$\Rightarrow x^2 + y^2 + 6x^2 + 6pxy - 10xy - 10py^2 + x^2 + p^2y^2 + 2pxy = 0$

$\Rightarrow 8x^2 + (1 - 10p)y^2 + (8p - 10)xy = 0$

Perpendicular if $a + b = 0 \Rightarrow p^2 - 10p + 9 = 0$

$\Rightarrow p = 9, 1$

$x + y = 1$

$x + 9y = 1$

31. If the straight line joining the origin to the points of intersection of $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0$ and $2x + 3y = k$ are at right angles, then find the value of $5k - 6k^2$.

यदि वक्र $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0$ तथा रेखा $2x + 3y = k$ की उभयनिष्ट निर्देशांको को मूल बिन्दु से मिलाकर रेखा बनायी जाये तथा वो दोनों रेखा के बीच 90° का कोण बनता है, तो $5k - 6k^2$ का मान बताओ।

Ans. 52

Sol. Curve : $3x^2 - xy + 3y^2 + 2x - 3y + 4 = 0 \qquad .....(1)$

Line : $2x + 3y = k \quad \Rightarrow \quad \frac{2x + 3y}{k} = 1 \quad ....(2)$

homogenise equation (1) with the help of equation (2), combined equation of line pair is

$$(3x^2 - xy + 3y^2) + (2x - 3y)\left(\frac{2x + 3y}{k}\right) + 4\left(\frac{2x + 3y}{k}\right)^2 = 0$$

given angle between the lines is 90° $\quad \Rightarrow \quad$ coefficient of x^2 + coefficient of $y^2 = 0$

$$\left(3+\frac{4}{k}+\frac{16}{k^2}\right)+\left(3-\frac{9}{k}+\frac{66}{k^2}\right)=0$$

$$6-\frac{5}{k}+\frac{52}{k^2}=0 \quad \Rightarrow \quad 6k^2-5k+52=0$$

$\therefore \quad 5k-6k^2=52$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 32 to 34

Let ABC be an acute angled triangle and AD, BE and CF are its medians, where E and F are the points E(3, 4) and F(1, 2) respectively and centroid of Δ ABC is G(3, 2), then answer the following questions:

मानाकि ABC एक न्यूनकोण त्रिभुज है तथा AD, BE तथा CF इसकी माध्यिकाएँ है जहाँ E तथा F क्रमशः बिन्दु E(3, 4) और F(1, 2) है तथा त्रिभुज ABC का केन्द्रक G(3, 2) है, तो निम्न प्रश्नों के उत्तर दीजिए।

32. The equation of side AB is :

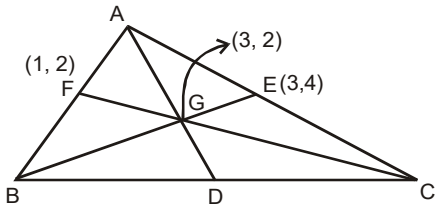
(A) $2x+y=4$ (B) $x+y-3=0$ (C) $4x-2y=0$ (D) none of these

भुजा AB का समीकरण है :

(A) $2x+y=4$ (B) $x+y-3=0$ (C) $4x-2y=0$ (D) इनमें से कोई नहीं

Ans. A

Sol. Taking $A(x_1, y_1)$, $B(x_2, y_2)$ and $C(x_3, y_3)$



then by $\frac{x_1+x_2}{2}=1$, $\frac{x_2+x_3}{2}=5$, $\frac{x_3+x_1}{2}=3$ and $\frac{y_1+y_2}{2}=2$, $\frac{y_2+y_3}{2}=0$, $\frac{y_1+y_3}{2}=4$

we get A(–1, 6) , B(3, –2) , C(7, 2) equation of AB is $2x+y=4$

33. Coordinate of D are :

D के निर्देशांक है :

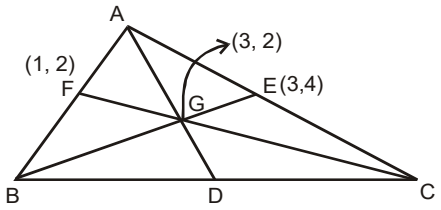
(A) (7, –4) (B) (5, 0) (C) (7, 4) (D) (– 3, 0)

Ans. B

Sol. Let the co-ordinates of D(α, β) then $\frac{\alpha+1+3}{3}=3 \quad \Rightarrow \quad \alpha = 5$

and $\frac{\beta+2+4}{3}=2 \quad \Rightarrow \quad \beta = 0$

$\therefore \quad$ D(5, 0)



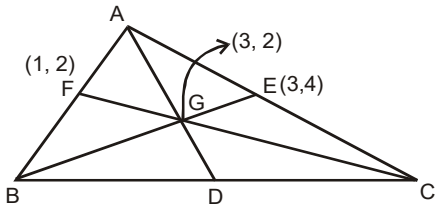
34. Height of altitude (in units) drawn from point A is :

बिन्दु A से डाले गये शीर्षाभिलम्ब की ऊँचाई (इकाई में) है :

(A) $4\sqrt{2}$ (B) $3\sqrt{2}$ (C) $6\sqrt{2}$ (D) $2\sqrt{3}$

Ans. C

Sol.



Height of altitude from A is $= \frac{2 \times \text{area}\,(\Delta ABC)}{BC} = 6\sqrt{2}$

Paragraph for Question Nos. 35 to 36

Consider a triangle ABC with vertex A(2, –4). The internal bisectors of the angles B and C are x + y = 2 and x – 3y = 6 respectively. Let the two bisectors meet at I.

माना एक त्रिभुज ABC जिसका शीर्ष A(2, –4) है। कोण B तथा C के अन्तःकोण समद्विभाजकों का समीकरण क्रमशः x + y = 2 तथा x – 3y = 6 है। माना दो समद्विभाजक I पर मिलते है।

35. If (a, b) is incentre of the triangle ABC then (a + b) has the value equal to :

यदि (a, b) त्रिभुज ABC का अन्तः केन्द्र है, तो (a + b) का मान बराबर होगा :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. x – 3y = 6

$x + y = 2$

$I(3, -1) \equiv (a, b)$

$a + b = 2$

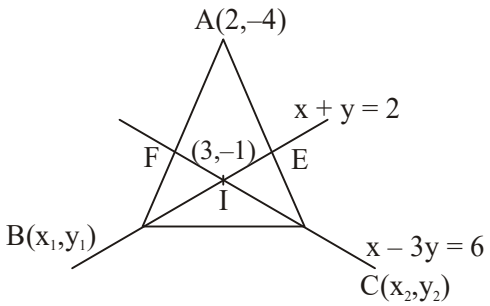
36. If (x_1, y_1) and (x_2, y_2) are the co–ordinates of the point B and C respectively, then the value of $(x_1 x_2 + y_1 y_2)$ is equal to :

यदि (x_1, y_1) तथा (x_2, y_2) क्रमशः बिन्दु B तथा C के निर्देशांक है, तो $(x_1 x_2 + y_1 y_2)$ का मान होगा :

(A) 4 (B) 5 (C) 6 (D) 8

Ans. D

Sol. P = image of A (2, – 4) in $x - 3y = 6$



$P\left(\frac{2}{5}, \frac{4}{5}\right)$

Q = Image of A (2,–4) in $x + y = 2$ Q (6,0)

Equation of BC = Equation of PQ $\Rightarrow$

BC $\Rightarrow x + 7y = 6$

$C \rightarrow x + 7y = 6$ $B \rightarrow x + 7y = 6$

$x + y = 2$ $x - 3y = 6$

$C\left(\frac{4}{3}, \frac{2}{3}\right)$ B (6,0)

$x_1x_2 + y_1y_2 = 8$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

37. The area of triangle ABC is 20 cm^2 . The co–ordinates of vertex A are (–5, 0) and B are (3, 0). The vertex C lies on the line, $x - y = 2$. The co–ordinates of C are :

Δ ABC का क्षेत्रफल 20 cm^2 है। शीर्ष A (–5, 0) तथा B (3, 0) है। शीर्ष C रेखा $x - y = 2$ पर स्थित है। C के निर्देशांक है :

(A) (5, 3) (B) (–3, –5) (C) (–5, –7) (D) (7, 5)

Ans. BD

Sol. Let any point on the line x – y = 2 be C ≡ (h, h – 2). Given area of ΔABC is

$$\left|\frac{1}{2}\begin{vmatrix} h & h-2 & 1 \\ -5 & 0 & 1 \\ 3 & 0 & 1 \end{vmatrix}\right| = 20$$

$\Rightarrow |8(h-2)| = 40$

$\Rightarrow h - 2 = \pm 5$

$\hbar = 7, -3$

Hence, the points are (7,5) and (–3,–5)

38. Three vertices of a triangle are A(4, 3) ; B(1, – 1) and C(7, k) . Value(s) of k for which centroid, orthocentre, incentre and circumcentre of the Δ ABC lie on the same straight line is/are :

(A) 7 (B) – 1 (C) – 19/8 (D) none

त्रिभुज के तीन शीर्ष A(4, 3) ; B(1, – 1) तथा C(7, k) है। k के मान जिनके लिये केन्द्रक, लम्बकेन्द्र, अन्तःकेन्द्र तथा परिकेन्द्र एक ही सरल रेखा पर है :

(A) 7 (B) – 1 (C) – 19/8 (D) कोई नहीं

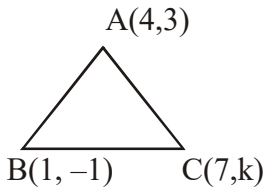
Ans. BC

Sol. AB = AC

$\Rightarrow 9 + 16 = 9 + (k-3)^2$

$\Rightarrow k = 3 \pm 4 \Rightarrow k = 7, -1$

for k = 7 C(7,7), B(1,–1), A(4,3) are collinear.



so, $k \neq 7$

AB = BC

$\Rightarrow 9 + 16 = 36 + (k+1)^2$

Not possible

AC = BC

$\Rightarrow 36 + (k+1)^2 = 9 + (k-3)^2$

$27 + k^2 + +2k + 1 = k^2 - 6k + 9$

$\Rightarrow 8k = -19 \Rightarrow k = -\frac{19}{8}$

39. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines $ax + by + c = 0$ are concurrent at :

यदि $a^2 + 9b^2 - 4c^2 = 6ab$ तो रेखाओं का समूह $ax + by + c = 0$ किस बिन्दु पर संगामी होगा :

(A) (1/2, 3/2) (B) (–1/2, –3/2) (C) (–1/2, 3/2) (D) (1/2, –3/2)

Ans. CD

Sol. Given $a^2 + 9b^2 - 4c^2 = 6ab$

$\Rightarrow (a-3b)^2 = (2c)^2$

$\Rightarrow (a-3b) = \pm c$

$\therefore a - 3b + 2c = 0$ $\quad a - 3b = 2c$

$c = \frac{3b-a}{2}$ $\quad c = \frac{a-3b}{2}$

family of lines is $ax + by + c = 0$

$ax + by + \frac{3b-a}{2} = 0$ $\quad ax + by + \frac{a-3b}{2} = 0$

if these two lines are concurrent then

$3b - a = 0$

now, $ax + by + \frac{3b-a}{2} = 0$

$3bx + by = 0$ $\quad y = 3x$

$\therefore$ The points will be $\left(-\frac{1}{2}, \frac{3}{2}\right)$ and $\left(\frac{1}{2}, \frac{-3}{2}\right)$.

40. The straight lines $x + y = 0$, $3x + y - 4 = 0$ and $x + 3y - 4 = 0$ form a triangle which is :

(A) isosceles (B) right angled (C) obtuse angled (D) equilateral

रेखाओं $x + y = 0$, $3x + y - 4 = 0$ एवं $x + 3y - 4 = 0$ से बनने का वाला त्रिभुज होगा :

(A) समद्विबाहु (B) समकोणीय (C) अधिक कोणीय (D) समबाहु

Ans. AC

Sol. $x + y = 0$ _____(i)

$3x + y - 4 = 0$ _____(ii)

$x + 3y - 4 = 0$ _____(iii)

Solving lines (i) and (ii), we get

$-x = -3x + 4 \Rightarrow x = 2 \Rightarrow y = -2$

$\therefore$ (i) and (ii) intersect at $A = (2, -2)$

solving lines (ii) and (iii), we get

$$-3x + 4 = \frac{4 - x}{3} \Rightarrow -9x + 12 = 4 - x \Rightarrow x = 1 \Rightarrow y = 1$$

$\therefore$ (ii) and (iii) intersect at $B = (1,1)$

Solving lines (i) and (iii), we get

$$-x = \frac{4 - x}{3} \Rightarrow -3x = 4 - x \Rightarrow x = -2 \Rightarrow y = 2$$

$\therefore$ (i) and (iii) intersect at $C = (-2,2)$

So, AB, BC, AC form a triangle ABC

Now, $AB = \sqrt{(1-2)^2 + (1+2)^2} = \sqrt{10}$

$$BC = \sqrt{(-2-1)^2 + (2-1)^2} = \sqrt{10}$$

$$AC = \sqrt{(-2-2)^2 + (2+2)^2} = \sqrt{32}$$

$\therefore AB = BC$

Since, two sides of a triangle are equal then the triangle formed by A, B, C is isosceles triangle.

41. Two equal sides of an isosceles triangle are given by the equations $7x - y + 3 = 0$ and $x + y - 3 = 0$ and its third side passes through the point $(1, -10)$. The equation of the third side can be :

किसी समद्विबाहु त्रिभुज की दो समान भुजा क्रमशः $7x - y + 3 = 0$ एवं $x + y - 3 = 0$ है तथा इसकी तीसरी भुजा यदि $(1, -10)$ से गुजरती हो, तो तीसरी भुजा का समीकरण होगा :

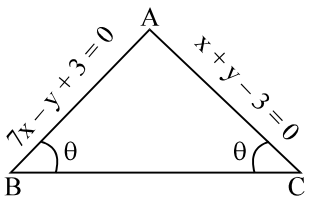
(A) $x + 3y + 29 = 0$ (B) $x - 3y = 31$ (C) $3x - y = 13$ (D) $3x + y + 7 = 0$

Ans. BD

Sol. $m_{AB} = 7$ and $m_{AC} = -1$

Let slope of BC (third side) be m.

Now $\left|\frac{7-m}{1+7m}\right| = \left|\frac{-1-m}{1-m}\right| \Rightarrow \left(\frac{7-m}{1+7m}\right) = \pm\left(\frac{-1-m}{1-m}\right)$



$\therefore$ Taking '+' sign, we get

$m^2 + 1 = 0$ (no real values of m possible)

And taking '–' sign, we get

$3m^2 + 8m - 3 = 0 \Rightarrow (3m - 1)(m + 3) = 0 \Rightarrow m = \frac{1}{3}, -3$

$\therefore$ Required equation of third side can be $x - 3y - 31 = 0$ or $3x = y + 7 = 0$

42. Straight lines $2x + y = 5$ and $x - 2y = 3$ intersect at the point A. Points B and C are chosen on these two lines such that $AB = AC$. Then the equation of a line BC passing through the point $(2, 3)$ is :

रेखा $2x + y = 5$ एवं $x - 2y = 3$ एक दूसरे को बिन्दु A पर प्रतिच्छेदित करती है। बिन्दु B एवं C क्रमशः इन दोनों रेखाओं पर इसप्रकार से है कि $AB = AC$। यदि BC रेखा बिन्दु $(2, 3)$ से गुजरती हो, तो BC का समीकरण होगा :

(A) $3x - y - 3 = 0$ (B) $x + 3y - 11 = 0$ (C) $3x + y - 9 = 0$ (D) $x - 3y + 7 = 0$

Ans. AB

Sol. Note that the lines are perpendicular. Find the equation of the lines through $(2, 3)$ and parallel to the bisectors of the given lines, the slopes of the bisectors being $-1/3$ and 3

43. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angles between them is θ. Which of the following relations hold good?

(A) $m_1 + m_2 = 5/4$

(B) $m_1 m_2 = 3/8$

(C) acute angle between L_1 and L_2 is $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$

(D) sum of the abscissa and ordinate of the point P is -1

रेखा युग्म L_1 एवं L_2 समीकरण $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ से व्यक्त होता हैं तथा P पर प्रतिच्छेदित करता है तथा जिनकी प्रवणता m_1 एवं m_2 है। यदि उनके बीच का न्यून कोण θ हो, तो इनमें से कौनसा सम्बन्ध सत्य होगा?

(A) $m_1 + m_2 = 5/4$

(B) $m_1 m_2 = 3/8$

(C) L_1 एवं L_2 का बीच का न्यून कोण $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$ होगा

(D) बिन्दु P की भुजा एवं कोटि का योग -1 होगा

Ans. BCD

Sol. $f(x, y) = (x + 2y + 3)(3x + 4y + 5)$

$m_1 + m_2 = -\frac{2h}{b} = -\frac{10}{8} = -\frac{5}{4}$

$m_1 m_2 = \frac{a}{b} = \frac{3}{8} \Rightarrow$ (B) is correct

also L_1 and L_2 intersect at $(1, -2)$

$\therefore$ sum of abscissa and ordinate $= -1 \Rightarrow$ (D) is correct

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a+b} = \frac{2\sqrt{25-24}}{11} = \frac{2}{11}$$

$$\sin\theta = \frac{2}{5\sqrt{5}} \Rightarrow \theta = \sin^{-1}\left(\frac{2}{5\sqrt{5}}\right) \Rightarrow \text{(C) is correct}$$

44. Let B $(1, -3)$ and D $(0, 4)$ represent two vertices of rhombus ABCD in (x, y) plane, then coordinates of vertex A, if $\angle BAD = 60°$, can be equal to :

यदि B $(1, -3)$ एवं D $(0, 4)$ किसी समचतुर्भुज के शीर्ष है यदि $\angle BAD = 60°$ तो शीर्ष A का निर्देशांक होगा :

(A) $\left(\frac{1-7\sqrt{3}}{2}, \frac{1-\sqrt{3}}{2}\right)$ (B) $\left(\frac{1+7\sqrt{3}}{2}, \frac{1+\sqrt{3}}{2}\right)$

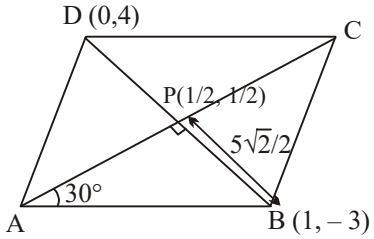
(C) $\left(\frac{1-14\sqrt{3}}{2}, \frac{1-2\sqrt{3}}{2}\right)$ (D) $\left(\frac{1+14\sqrt{3}}{2}, \frac{1+2\sqrt{3}}{2}\right)$

Ans. AB

Sol. Let inclination of AC is θ

$\tan\theta = \frac{1}{7}$ $\sin\theta = \frac{1}{5\sqrt{2}}$, $\cos\theta = \frac{7}{5\sqrt{2}}$

$AP = \frac{5}{2}\sqrt{2}\cot 30° = \frac{5\sqrt{2}}{2}\sqrt{3}$



$$A = \left(\frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\frac{7}{5\sqrt{2}}, \frac{1}{2} - \frac{5\sqrt{2}\sqrt{3}}{2}\frac{1}{5\sqrt{2}}\right)$$

or $$\left(\frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\frac{7}{5\sqrt{2}}, \frac{1}{2} + \frac{5\sqrt{2}\sqrt{3}}{2}\frac{1}{5\sqrt{2}}\right)$$

$\Rightarrow \quad A = \left(\frac{1}{2} - \frac{7\sqrt{3}}{2}, \frac{1}{2} - \frac{\sqrt{3}}{2}\right)$ or $\left(\frac{1}{2} + \frac{7\sqrt{3}}{2}\right), \left(\frac{1}{2} + \frac{\sqrt{3}}{2}\right)$

45. The equation of the bisectors of the angle between the two intersecting lines :

$\frac{x-3}{\cos\theta} = \frac{y+5}{\sin\theta}$ and $\frac{x-3}{\cos\varphi} = \frac{y+5}{\sin\varphi}$ are $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$ and $\frac{x-3}{\beta} = \frac{y+5}{\gamma}$ then :

दो प्रतिच्छेदी रेखाओं $\frac{x-3}{\cos\theta} = \frac{y+5}{\sin\theta}$ एवं $\frac{x-3}{\cos\varphi} = \frac{y+5}{\sin\varphi}$ के मध्य कोणों के अर्द्धकों की समीकरण $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$ एवं $\frac{x-3}{\beta} = \frac{y+5}{\gamma}$ हो, तो :

(A) $\alpha = \frac{\theta+\phi}{2}$ (B) $\beta = -\sin\alpha$ (C) $\gamma = \cos\alpha$ (D) $\beta = \sin\alpha$

Ans. ABC

Sol. one bisector makes an angle $\left(\frac{\theta+\phi}{2}\right)$ with x-axis, then other bisector makes an angle $90^\circ + \left(\frac{\theta+\phi}{2}\right)$ with x-axis

Equation of bisector are

$$\frac{x-3}{\cos\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\sin\left(\frac{\theta+\phi}{2}\right)} \qquad ...(1)$$

and $\frac{x-3}{\cos\left(\frac{\pi}{2}+\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\sin\left(\frac{\pi}{2}+\frac{\theta+\phi}{2}\right)}$

$$\frac{x-3}{-\sin\left(\frac{\theta+\phi}{2}\right)} = \frac{y+5}{\cos\left(\frac{\theta+\phi}{2}\right)} \qquad ...(2)$$

But given bisector are $\frac{x-3}{\cos\alpha} = \frac{y+5}{\sin\alpha}$

$\therefore \alpha = \frac{\theta+\phi}{2}$

and $\frac{x-3}{\beta}=\frac{y+5}{\gamma}$

$\therefore \beta=-\sin\left(\frac{\theta+\phi}{2}\right)=-\sin\alpha$

and $y=\cos\left(\frac{\theta+\phi}{2}\right)=\cos\alpha$

Hence, option (A), (B), (C) is correct answer.

46. Equation of a straight line passing through the point (4, 5) and equally inclined to the lines,$3x = 4y + 7$ and $5y = 12x + 6$ is :

बिन्दु (4, 5) से गुजरने वाली एवं रेखाओं $3x = 4y + 7$ तथा $5y = 12x + 6$ से समान कोण पर झुकी रेखा की समीकरण है :

(A) $9x - 7y = 1$ (B) $9x + 7y = 71$ (C) $7x + 9y = 73$ (D) $7x - 9y + 17 = 0$

Ans. AC

Sol. The lines will pass through (4, 5) & parallel to the bisectors between them $\frac{3x-4y-7}{5}=\pm\frac{12x-5y+6}{13}$

by taking + sign, we get $21x + 27y + 121 = 0$. Now by taking – sign, we get $99x - 77y - 61 = 0$

so slopes of bisectors are $-\frac{7}{9}, \frac{9}{7}$

Equation of lines are $y-5=\frac{-7}{9}(x-4)$ and $y-5=\frac{9}{7}(x-4)$

$\Rightarrow$ $7x + 9y = 73$ and $9x - 7y = 1$

47. One side of a rectangle lies along the line $4x + 7y + 5 = 0$. Two of its vertices are (–3, 1) and (1, 1). Then the equations of other sides are :

एक आयत की एक भुजा रेखा $4x + 7y + 5 = 0$ के अनुदिश है। इसके दो शीर्ष (–3, 1) एवं (1, 1) हैं। तो इसकी अन्य भुजाओं के समीकरण है :

(A) $7x - 4y + 25 = 0$ (B) $7x + 4y + 25 = 0$ (C) $7x - 4y - 3 = 0$ (D) $4x + 7y = 11$

Ans. ACD

Sol. Line $\perp$ to $4x + 7y + 5 = 0$ is $7x - 4y + \lambda = 0$

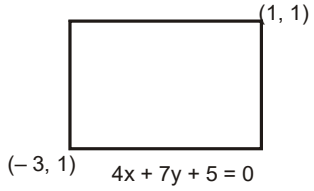
It passes through (– 3, 1) and (1, 1)

$-21 - 4 + \lambda = 0 \quad \Rightarrow \lambda = 25$

$7 - 4 + \lambda = 0 \quad \Rightarrow \lambda = -3$

Hence lines are $7x - 4y + 25 = 0$, $7x - 4y - 3 = 0$

line || to $4x + 7y + 5 = 0$ passing through (1, 1) is $4x + 7y + \lambda = 0$



48. Suppose ABCD is a quadrilateral such that the coordinates of A, B and C are (1, 3), (–2, 6) and (5, – 8) respectively. For what choice of coordinates of D will make ABCD a trapezium :

माना ABCD एक चतुर्भुज है, तथा बिन्दु A, B तथा C के निर्देशांक क्रमशः (1, 3), (–2, 6) तथा (5, – 8) है। ABCD एक समलम्ब चतुर्भुज होने के लिए दिये गये विकल्पों में से D के निर्देशांक हो सकते है :

(A) (3, – 6) (B) (6, –9) (C) (0, 5) (D) (3, –1)

Ans. ABCD

Sol. One pair of parallel line should have

49. The points A(0, 0), B(cosα, sinα) and C(cosβ, sinβ) are the vertices of a right angled triangle if :

बिन्दु A(0, 0), B(cosα, sinα) तथा C(cosβ, sinβ) एक समकोण त्रिभुज के शीर्ष है, यदि :

(A) $\sin\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (B) $\cos\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$

(C) $\cos\left(\frac{\alpha-\beta}{2}\right)=\frac{1}{\sqrt{2}}$ (D) $\sin\left(\frac{\alpha-\beta}{2}\right)=-\frac{1}{\sqrt{2}}$

Ans. ABCD

Sol. OA = 1 and OB = 1 $\therefore AB=\sqrt{2}$

$\therefore (\cos\beta-\cos\alpha)^2+(\sin\beta-\sin\alpha)^2=2$

$\Rightarrow \cos^2\alpha+\sin^2\alpha+\cos^2\beta+\sin^2\beta-2\sin\beta\sin\alpha-2\cos\beta\cos\alpha=2$

$\Rightarrow 2\sin\beta\sin\alpha+2\cos\beta\cos\alpha=0$

$\therefore \cos(\alpha-\beta)=0$

$\Rightarrow \sin\left(\frac{\alpha-\beta}{2}\right)=\pm\frac{1}{\sqrt{2}}$

$\rightarrow \cos\left(\frac{\alpha-\beta}{2}\right)=\pm\frac{1}{\sqrt{2}}$

50. Two vertices of the Δ ABC are at the points A(– 1, – 1) and B(4, 5) and the third vertex lines on the straight line y = 5(x – 3) . If the area of the Δ is 19/2 then the possible co–ordinates of the vertex C are :

ΔABC का शीर्ष A(– 1, – 1) तथा B(4, 5) है। यदि तीसरा शीर्ष रेखा y = 5(x – 3) पर है तथा त्रिभुज का क्षेत्रफल 19/2 है, तो बिन्दु C का समीकरण निर्देशांक होगा :

(A) (5, 10) (B) (3, 0) (C) (2, –5) (D) (5, 4)

Ans. AB

Sol. A(–1,–1), B(4,5), C(α, $5\alpha - 15$)

$$\begin{vmatrix} -1 & -1 & 1 \\ 4 & 5 & 1 \\ \alpha & 5\alpha-15 & 1 \end{vmatrix} = \pm 19 \Rightarrow \alpha = 5, 3$$

51. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x - \sqrt{3}\, y = 0$ then the co–ordinates of the third vertex are :

यदि a भुजा की लम्बाई वाले समबाहु त्रिभुज की एक शीर्ष (0, 0) पर है तथा दूसरी रेखा $x - \sqrt{3}\, y = 0$ पर हो, तो तीसरे शीर्ष का निर्देशांक होगा :

(A) (0, a) (B) $\left(\frac{\sqrt{3}a}{2}, -\frac{a}{2}\right)$ (C) (0, – a) (D) $\left(-\frac{\sqrt{3}a}{2}, \frac{a}{2}\right)$

Ans. ABCD

Sol. Make a figure and interpret

52. The sides of a triangle are the straight lines $x + y = 1$; $7y = x$ and $\sqrt{3}\, y + x = 0$. Then which of the following is an interior point of the triangle?

(A) circumcentre (B) centroid (C) incentre (D) orthocentre

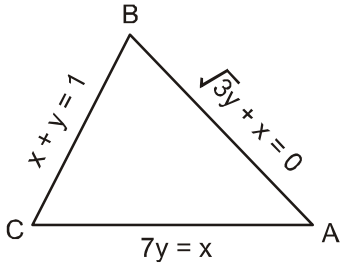
यदि त्रिभुज ABC की भुजाओं का समीकरण $x + y = 1$; $7y = x$ और $\sqrt{3}\, y + x = 0$ हैं, तो इनमें से कौनसा बिन्दु त्रिभुज के भीतर होगा?

(A) परिकेन्द्र (B) केन्द्रक (C) अन्तः केन्द्र (D) लम्बकेन्द्र

Ans. BC

Sol. Let slope of given lines $m_1 = \frac{1}{7}, m_2 = \frac{-1}{\sqrt{3}}, m_3 = -1$

Hence interior angle of triangle



$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{\frac{1}{7} + \frac{1}{\sqrt{3}}}{1 - \frac{1}{7\sqrt{3}}} = \frac{\sqrt{3} + 7}{7\sqrt{3} - 1} > 0$$

$$\tan B = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{-\frac{1}{\sqrt{3}} + 1}{1 + \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} > 0 \quad \Rightarrow \quad \tan C = \frac{m_3 - m_1}{1 + m_2 m_1} = \frac{-1 - \frac{1}{7}}{1 - \frac{1}{7}} = \frac{-8}{6} < 0$$

Hence angle C is obtuse therefore circumcentre and orthocentre lies outside the triangle.

MATCH THE COLUMN TYPE QUESTION(S)

53. Match the following column :

Column –I	**Column – II**
(A) Slope of line bisecting co–ordinate axis, is	(P) 3
(B) Area of Δ formed by line $3x + 4y + 12 = 0$, with co–ordinate axis is	(Q) 1
(C) If the equation $2x^2 - 2xy - y^2 - 6x + 6y + c = 0$ represents a pair of lines then 'C' is	(R) 6
(D) If distance between the pair of parallel lines $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ is $25\sqrt{2}$, then 'a/5 is equal to	(S) – 1

निम्न कॉलम को सुमेलित कीजिए :

Column –I	**Column – II**
(A) निर्देशांक अक्षों के मध्य के कोण को समद्विभाजित करने वाली रेखा की प्रवणता होगी	(P) 3
(B) रेखा $3x + 4y + 12 = 0$ तथा निर्देशांक अक्षों से बने त्रिभुज का क्षेत्रफल होगा	(Q) 1
(C) समीकरण $2x^2 - 2xy - y^2 - 6x + 6y + c = 0$ एक रेखा युग्म को निरूपित करती है तो 'C' का मान होगा	(R) 6
(D) यदि समान्तर रेखाओं $x^2 + 2xy + y^2 - 8ax - 8ay - 9a^2 = 0$ के मध्य की दूरी $25\sqrt{2}$, है तो 'a/5 का मान होगा	(S) – 1

Ans. A – QS, B – R, C – P, D – QS

Sol. (A) Equation of bisector

$y = mx$

$m = \tan 45^\circ = 1$

or $m = \tan 135^\circ = -1$

$\therefore\ y = \pm x$

slope $= m = \pm 1$

(B) Equation of x-axis is $y = 0$ _____(1)

Equation of y-axis is $x = 0$ _____(2)

Let given equation $3x + 4y + 12 = 0$ meets x-axis at point A and y-axis at B.

At A we have $y = 0 \Rightarrow 3x = -12$

$\Rightarrow x = -4$

$\therefore$ A is $(-4,0)$

$\Rightarrow OA = \sqrt{(-4)^2 + 0} = 4$ units

At B we have $x = 0 \Rightarrow 4y = -12$

$\Rightarrow y = -3$

$\therefore$ B is $(0, -3)$

$\Rightarrow OB = \sqrt{0 + (-3)^2} = 3$ *units*

$$\text{Area of triangle} = \frac{1}{2} \times OA \times OB = \frac{1}{2} \times 4 \times 3 = 6 \text{ sq units}$$

(C) $\Delta = 0 \Rightarrow C = 3$

$abc + 2fgh - af^2 - bg^2 - ch^2 = 0$

$\Rightarrow -2c + 18 - 18 + 9 - c = 0$

$\Rightarrow -3c = -9 \Rightarrow c = 3$

(D) $(x + y)^2 - 8a(x + y) - 9a^2 = 0$

$(x + y - 9a)(x + y + a) = 0$

$$25\sqrt{2} = \left|\frac{10a}{\sqrt{2}}\right|$$

$|a| = 5 \Rightarrow a = \pm 5$

54. Match the following column :

Column–I	**Column–II**
(A) If a, b, c are in A. P., then lines $ax + by + c = 0$ are concurent at	(P) $(-4, -7)$
(B) A point on the line $x + y = 4$ which lies at a unit distance from the line $4x + 3y = 10$ is	(Q) $(-7, 11)$
(C) Orthocentre of trianlge made by lines $x + y = 1$, $x - y + 3 = 0$, $2x + y = 7$ is	(R) $(1, -2)$
(D) Two vertice of a triangle are $(5, -1)$ and $(-2, 3)$. If orthcentre is the origin then coordinated of the third vertex are	(S) $(-1, 2)$ (T) $(0, 0)$

निम्न कॉलम को सुमेलित कीजिए :

Column–I	**Column–II**
(A) यदि a, b, c समान्तर श्रेणी में है, तो रेखाऐं $ax+by+c= 0$ किस बिन्दु पर संगामी है	(P) $(-4, -7)$
(B) एक बिन्दु रेखा $x + y = 4$ पर स्थित है, रेखा $4x + 3y = 10$ से इकाई दूरी पर स्थित है, के निर्देशांक होंगे	(Q) $(-7, 11)$
(C) रेखाओं $x + y = 1, x - y + 3 = 0, 2x + y = 7$ से बने त्रिभुज का लम्बकेन्द्र होगा	(R) $(1, -2)$
(D) यदि त्रिभुज के दो शीर्ष $(5, -1)$ तथा $(-2, 3)$ है। यदि इसका लम्बकेन्द्र मूल बिन्दु पर स्थित है तो त्रिभुज का तीसरा शीर्ष होगा	(S) $(-1, 2)$

Ans. A – R, B – Q, C – S, D – P

Sol. (A) $a + c = 2b \Rightarrow a + c - 2b = 0$

$\Rightarrow ax + by + c = 0$ satisfies $(1, -2)$

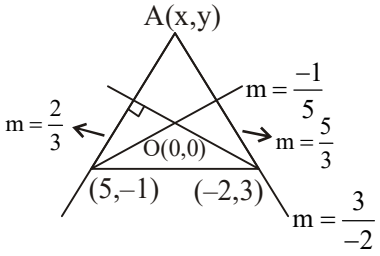
(B) Perpendicular distance of $P(\lambda, 4 - \lambda)$ from $4x + 3y = 10$

$$\left|\frac{4\lambda + 3(4-\lambda) - 10}{5}\right| = 1$$

$$\Rightarrow |\lambda + 2| = 5 \Rightarrow \lambda - 2 \pm 5 = 3, -7$$

P(3,1), (–7,+11)

(C) Point B = (–1,2)



(D) $\frac{y}{x} = \frac{7}{4}$

$$\frac{4+1}{y-5} = \frac{2}{3}, \frac{21}{4}x = 2x - 13$$

$x = -4$, $y = -7$

55. Consider the lines given by

$L_1 = x + 3y - 5 = 0$

$L_2 = 3x - ky - 1 = 0$

$L_3 = 5x + 2y - 12 = 0$

Match the statements / Expression in Column–I with the statements / Expressions in Column–II and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in OMR :

Column–I	**Column–II**
(A) L_1, L_2, L_3 are concurrent, if	(P) $k = -9$
(B) One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(Q) $k = -\frac{6}{5}$
(C) L_1, L_2, L_3 form a triangle, if	(R) $k = \frac{5}{6}$
(D) L_1, L_2, L_3 do not form a triangle, if	(S) $k = 5$

माना रेखाओं के समीकरण

$L_1 = x + 3y - 5 = 0$

$L_2 = 3x - ky - 1 = 0$

$L_3 = 5x + 2y - 12 = 0$

कॉलम–I से कॉलम–II का मिलान कीजिये तथा अपने उत्तर को 4 × 4 के आव्यूह के रूप में उत्तर पुस्तिका पर अंकित कीजिये :

Column–I	**Column–II**
(A) L_1, L_2, L_3 संगामी है, यदि	(P) $k = -9$
(B) L_1, L_2, L_3 मे से कोई एक शेष दो में से एक के समान्तर है, यदि	(Q) $k = -\frac{6}{5}$
(C) L_1, L_2, L_3 एक त्रिभुज बनाती है, यदि	(R) $k = \frac{5}{6}$
(D) L_1, L_2, L_3 एक त्रिभुज नही बनाती है, यदि	(S) $k = 5$

Ans. (A) S (B) P,Q (C) R (D) P,Q,S

Sol. (A) L_1, L_2 and L_3, intersect at (2,1).

L_1, L_2, L_3 are concurrent if L_2, passes through (2,1) that is if $k = 5$.

(B) As L_1 and L_3 intersect at (2,1) they are not parallel.

So L_2 is parallel to L_1 if $\frac{3}{k} = -3 \Rightarrow k = -9$

and L_2 is parallel to L_3 if $\frac{3}{k} = -\frac{5}{2} \Rightarrow k = -\frac{6}{5}$.

(C) The lines L_1, L_2, L_3 will form a triangle, fi no two of them are parallel and they are not concurrent. Thsu $k = \frac{5}{6}$.

(D) Lines L_1, L_2, L_3 will not form a triangle, if two of them are parallel or all of them are concurrent.

Alter :

$x + 3y - 5 = 0$, $3x - xy - 1 = 0$, $5x + 2y - 12 = 0$

(A) For concurrency

$$\begin{vmatrix} 1 & 3 & -5 \\ 3 & -k & -1 \\ 5 & 2 & -12 \end{vmatrix} = 0$$

$\Rightarrow$ $(12k + 2) - 3(-36 + 5) - 5(6 + 5k) = 0$

$\therefore$ $k = 5$

(B) For parallel

either $-\frac{1}{3} = \frac{3}{k}$ $\therefore k = -9$

or $-\frac{5}{2} = \frac{3}{k}$ $\therefore k = \frac{-6}{5}$

(C) They will form triangle

when $k \neq 5, -9, \frac{-6}{5}$

(D) They will not form triangle

when $k = 5, -9, \frac{-6}{5}$

56. Let ABC be a triangle such that the coordinates of A are (– 3, 1). Equation of the median through B is $2x + y - 3 = 0$ and equation of the angular bisector of C is $7x - 4y - 1 = 0$. Then match the entries of column–I with their corresponding correct entries of column–II :

Column–I	**Column–II**
(A) Equation of the line AB is	(P) $2x + y - 3 = 0$
(B) Equation of the line BC is	(Q) $2x - 3y + 9 = 0$
(C) Equation of CA is	(R) $4x + 7y + 5 = 0$
	(S) $18x - y - 49 = 0$

यदि त्रिभुज ABC जिसके बिन्दु A के निर्देशांक (– 3, 1) है, बिन्दु B से गुजरने वाली माध्यिका का समीकरण $2x + y - 3 = 0$ एवं कोण C के कोर्णाधक का समीकरण $7x - 4y - 1 = 0$ है तो कॉलम–I एवं कॉलम–II का मिलान कीजिए :

Column–I	**Column–II**
(A) रेखा AB का समीकरण	(P) $2x + y - 3 = 0$
(B) रेखा BC का समीकरण	(Q) $2x - 3y + 9 = 0$
(C) रेखा CA का समीकरण	(R) $4x + 7y + 5 = 0$
	(S) $18x - y - 49 = 0$

Ans. (A) R; (B) S ; (C) Q

Sol. $D\left(\frac{\alpha-3}{2}, \frac{7\alpha+3}{8}\right)$ In $2x + y - 3 = 0$

$$\alpha - 3 + \frac{7\alpha+3}{8} - 3 = 0$$

$$\Rightarrow 8\alpha - 24 + 7\alpha + 3 - 24 = 0$$

$$\Rightarrow 15\alpha = 45$$

$$(\alpha - 3)$$

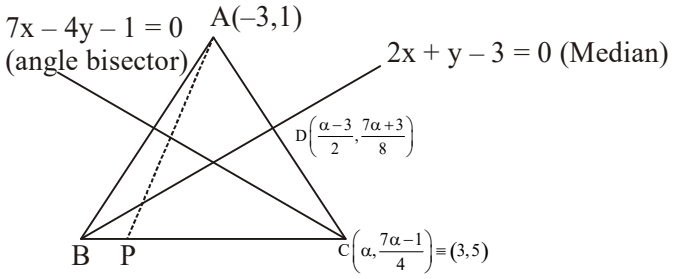
P is image of A(–3,1) in $7x - 4y - 1 = 0$

$$\frac{x+3}{7} = \frac{y-1}{-4} = -\frac{2(-21-4-1)}{49+16} = \frac{2\times 26}{65} = \frac{4}{5}$$

$$x = \frac{28}{5} - 3 = \frac{13}{5},\ y = -\frac{16}{5} + 1 = -\frac{11}{5}$$

$$P\left(\frac{13}{5}, -\frac{11}{5}\right)$$

$$\text{Equation of BC} = (y-1) = \frac{-\frac{11}{5}-1}{\frac{13}{5}+3}(x+3)$$



$$(y-1) - \frac{16}{28}(x+3)$$

$$\Rightarrow 7y - 4 = -4x - 12$$

$$\Rightarrow 4x + 7y + 8 = 0 \qquad \text{BC}$$

$$2x + y - 3 = 0 \qquad \text{Median}$$

$$B\left(\frac{13}{5}, -\frac{11}{5}\right)$$

(A) AB $\Rightarrow 4x + 7y + 5 = 0$

(B) BC $\Rightarrow 18x - y - 49 = 0$

(C) CA $\Rightarrow 2x - 3y + 9 = 0$

Exercise J-M

1. The equations of the sides AB and AC of a triangle ABC are $(\lambda + 1)x + \lambda y = 4$ and $\lambda x + (1 - \lambda)\, y + \lambda = 0$ respectively. Its vertex A is on the y-axis and its orthocentre is (1, 2). The length of the tangent from the point C to the part of the parabola $y^2 = 6x$ in the first quadrant is : **[JEE(Main)Jan.2023]**

एक त्रिभुज ABC की भुजाओं AB और AC के समीकरण क्रमशः $(\lambda + 1)x + \lambda y = 4$ और $\lambda x + (1 - \lambda)\, y + \lambda = 0$ है। इसका शीर्ष A, y-अक्ष पर है और इसका लंबकेन्द्र (1, 2) है। बिंदु C से परवलय $y^2 = 6x$ के प्रथम चतुर्थांश के भाग पर खींची गई स्पर्श रेखा की लंबाई है : **[JEE(Main)Jan.2023]**

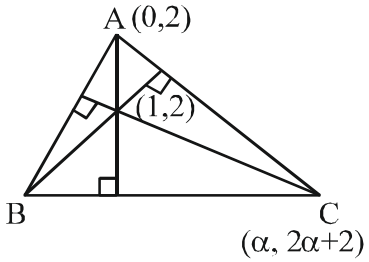
(A) $2\sqrt{2}$ (B) 2 (C) 4 (D) $\sqrt{6}$

Ans. A

Sol. $AB : (\lambda+1)x + \lambda y = 4$

$AC : \lambda x + (1-\lambda)y + \lambda = 0$

Vertex A is on y-axis



$\Rightarrow x = 0$

$y = \frac{4}{\lambda}, y = \frac{\lambda}{\lambda - 1}$

$\Rightarrow \frac{4}{\lambda} = \frac{\lambda}{\lambda - 1}$

$\Rightarrow \lambda = 2$

$AB : 3x + 2y = 4$

$AC : 2x - y + 2 = 0$

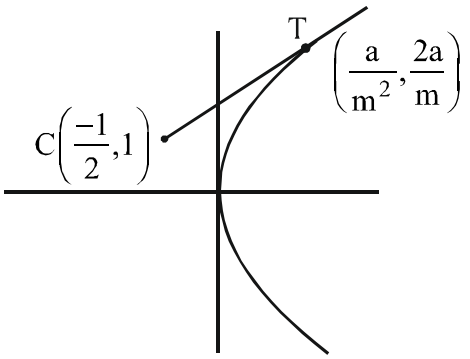
$\Rightarrow A(0, 2)$

Let $C(\alpha, 2\alpha + 2)$

Now (Slope of Altitude through C) $\left(-\frac{3}{2}\right) = -1$

$\left(\frac{2\alpha}{\alpha - 1}\right)\left(-\frac{3}{2}\right) = -1 \Rightarrow \alpha = -\frac{1}{2}$

So $C\left(-\frac{1}{2}, 1\right)$



Let Equation of tangent be $y = mx + \frac{3}{2m}$

$m^2 + 2m - 3 = 0$

$\Rightarrow m = 1, -3$

So tangent which touches in first quadrant at T is

$T \equiv \left(\frac{a}{m^2},\frac{2a}{m}\right)$

$\equiv \left(\frac{3}{2},3\right)$

$\Rightarrow CT = \sqrt{4+4} = 2\sqrt{2}$

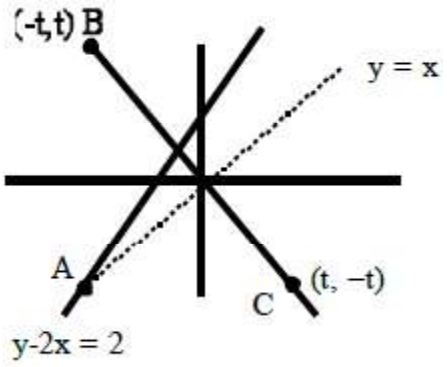
2. Let B and C be the two points on the line $y + x = 0$ such that B and C are symmetric with respect to the origin. Suppose A is a point on $y - 2x = 2$ such that ΔABC is an equilateral triangle. Then, the area of the ΔABC is : **[JEE(Main)Jan.2023]**

माना रेखा $y + x = 0$ पर दो बिंदु B व C मूल बिंदु के सममित है। माना $y - 2x = 2$ पर एक बिंदु A इस प्रकार है कि ΔABC एक समबाहु त्रिभुज है। तब ΔABC का क्षेत्रफल है : **[JEE(Main)Jan.2023]**

(A) $3\sqrt{3}$ (B) $2\sqrt{3}$ (C) $\frac{8}{\sqrt{3}}$ (D) $\frac{10}{\sqrt{3}}$

Ans. C

Sol.



At A $x = y$

$Y - 2x = 2$

$(-2, -2)$

Height from line $x + y = 0$

$$h = \frac{4}{\sqrt{2}}$$

$$\text{Area of } \Delta = \frac{\sqrt{3}}{4}\frac{h^2}{\sin^2 60} = \frac{8}{\sqrt{3}}$$

3. A light ray emits from the origin making an angle 30° with the positive x-axis. After getting reflected by the line $x + y = 1$, if this ray intersects x-axis at Q, then the abscissa of Q is : **[JEE(Main)Jan.2023]**

X-अक्ष की धनात्मक दिशा से 30° का कोण बनाती हुई मूल बिंदु से एक प्रकाश किरण उत्सर्जित होती है। रेखा $x + y = 1$ से परावर्तित होने के बाद, यदि यह किरण x-अक्ष को Q पर मिलती है, तब Q का भुज है : **[JEE(Main)Jan.2023]**

(A) $\frac{2}{3-\sqrt{3}}$ (B) $\frac{\sqrt{3}}{2\left(\sqrt{3}+1\right)}$ (C) $\frac{2}{\left(\sqrt{3}-1\right)}$ (D) $\frac{2}{3+\sqrt{3}}$

Ans. D

Sol. Image of O (0, 0) in line $x + y - 1 = 0$ lies on reflected ray.

$$\frac{x-0}{1} = \frac{y-0}{1} = \frac{-2(0+0-1)}{2} \Rightarrow B\ (1, 1).$$

Also, upon solving $y = \frac{x}{\sqrt{3}}$ and $x + y - 1 = 0$ we get $P \equiv \left(\frac{3-\sqrt{3}}{2}, \frac{\sqrt{3}-1}{2}\right)$

equation of reflected ray is same as line passing through BP.

$$\text{slope} = \frac{\frac{\sqrt{3}}{2} - \frac{1}{2} - 1}{\frac{3}{2} - \frac{\sqrt{3}}{2} - 1} = \frac{\sqrt{3}-3}{1-\sqrt{3}} = \sqrt{3}$$

Equation of line BP is

$y - 1 = \sqrt{3}(x - 1)$

Put $y = 0 \Rightarrow -\frac{1}{\sqrt{3}} = x - 1$

Required point $\left(1 - \frac{1}{\sqrt{3}}, 0\right)$

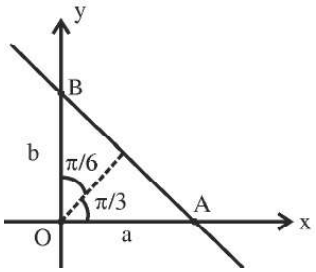
4. A straight line cuts off the intercepts OA = a and OB = b on the positive directions of *x*-axis and *y*-axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of *y*-axis and the area of ΔOAB is $\frac{98}{3}\sqrt{3}$, then $a^2 - b^2$ is equal to : **[JEE(Main)Jan.2023]**

एक सरल रेखा x-अक्ष तथा y-अक्ष की घनात्मक दिशाओं पर क्रमशः OA = a तथा OB = b अंतःखंड काटती है। यदि मूलबिंदु से इस रेखा पर अभिलंब y-अक्ष की घनात्मक दिशा से $\frac{\pi}{6}$ का कोण बनाता है तथा ΔOAB का क्षेत्रफल $\frac{98}{3}\sqrt{3}$ है, तो $a^2 - b^2$ बराबर है : **[JEE(Main)Jan.2023]**

(A) 98 (B) $\frac{196}{3}$ (C) 196 (D) $\frac{392}{3}$

Ans. D

Sol.



Equation of straight line : $\frac{x}{a} + \frac{y}{b} = 1$

Or $x\cos\frac{\pi}{3} + y\sin\frac{\pi}{3} = p$

$\frac{x}{2} + \frac{y\sqrt{3}}{2} = p$

$\frac{x}{3p} + \frac{y}{2p} = 1$

Comparing both : $a = 2p,\ b = \frac{2p}{\sqrt{3}}$

Now area of $\Delta OAB = \frac{1}{2}\cdot ab = \frac{98}{3}\cdot\sqrt{3}$

$\frac{1}{2}\cdot 2p\cdot\frac{2p}{\sqrt{3}} = \frac{98}{3}\cdot\sqrt{3}$

$p^2 = 49$

$$a^2 - b^2 = 4p^2 - \frac{4p^2}{3} = \frac{2}{3}4p^2$$

$$= \frac{8}{3} \cdot 49 = \frac{392}{3}$$

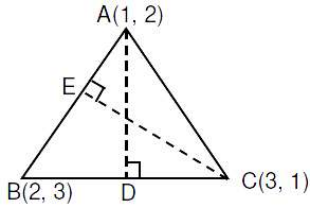
5. If the orthocenter of the triangle, whose vertices are (1, 2), (2, 3) and (3, 1) is (α, β), then the quadratic equation whose roots are $\alpha + 4\beta$ and $4\alpha + \beta$, is : **[JEE(Main)Jan.2023]**

एक त्रिभुज के शीर्ष (1, 2), (2, 3) और (3, 1) है। यदि इसका लंबकेन्द्र (α, β) है, तब $\alpha + 4\beta$ और $4\alpha + \beta$ किस समीकरण के मूल है : **[JEE(Main)Jan.2023]**

(A) $x^2 - 19x + 90 = 0$ (B) $x^2 - 20x + 99 = 0$ (C) $x^2 - 18x + 80 = 0$ (D) $x^2 - 22x + 120 = 0$

Ans. B

Sol.



$$\text{Slope (AB)} = m_{AB} = \frac{2-3}{1-2} = 1$$

$\therefore$ Slope (CE) $= -1$

$\therefore$ Equation of CE is : $y - 1 = -1(x - 3)$

$\Rightarrow y - 1 = -x + 3$

$\Rightarrow x + y = 4$ ...(i)

$$\text{Similarly } m_{BC} = \frac{2}{-1} = -2$$

$$\text{Equation of AD is : } y - 2 = \frac{1}{2}(x - 1)$$

$2y - 4 = x - 1$

$\therefore x - 2y + 3 = 0$ ...(ii)

Solving equation (i) & (ii)

$$\text{We have } x = \frac{5}{3},\ y = \frac{7}{3}$$

$$\therefore \alpha = \frac{5}{3}, \beta = \frac{7}{3}$$

$$\therefore \alpha + 4\beta = \frac{5}{3} + \frac{28}{3} = \frac{33}{3} = 11$$

$4\alpha+\beta=4\times\frac{5}{3}+\frac{7}{3}=\frac{27}{3}=9$

$\therefore$ Quadratic Equation is

$x^2-(\text{sum of roots})x+(\text{products of roots})=0$

$x^2-20x+99=0$

6. The combined equation of the two lines $ax+by+c=0$ and $a'x+b'y+c'=0$ can be written as $(ax+by+c)\left(a'x+b'y+c'\right)=0$. The equation of the angle bisectors of the lines represented by the equation $2x^2+xy-3y^2=0$ is : **[JEE(Main)Jan.2023]**

दो रेखाओं $ax+by+c=0$ तथा $a'x+b'y+c'=0$ का संयुक्त समीकरण $(ax+by+c)\left(a'x+b'y+c'\right)=0$ लिखा जा सकता है। समीकरण $2x^2+xy-3y^2=0$ से प्राप्त दो रेखाओं के कोण समद्विभाजकों का समीकरण है। **[JEE(Main)Jan.2023]**

(A) $x^2-y^2-10xy=0$ (B) $3x^2+5xy+2y^2=0$

(C) $x^2-y^2+10xy=0$ (D) $3x^2+xy-2y^2=0$

Ans. A

Sol. Equation of angle bisectors is $\frac{x^2-y^2}{2+3}=\frac{xy}{\frac{1}{2}}$

$\Rightarrow\frac{x^2-y^2}{5}=2xy$

$\Rightarrow x^2-y^2=10xy$

$\Rightarrow x^2-y^2-10xy=0$

7. The equations of the sides AB, BC and CA of a triangle ABC are : $2x+y=0$, $x+py=21a$, $(a\neq0)$ and $x-y=3$ respectively. Let P(2, a) be the centroid of ΔABC. Then $(BC)^2$ is equal to _____. **[JEE(Main)Jan.2023]**

एक त्रिभुज ABC की भुजाओं AB, BC तथा CA के समीकरण क्रमशः $2x+y=0$, $x+py=21a$, $(a\neq0)$ तथा $x-y=3$ है। माना ΔABC का केन्द्रक P(2, a) है। तब $(BC)^2$ बराबर है। **[JEE(Main)Jan.2023]**

Ans. 122

Sol. Now $\frac{1+\alpha+\beta}{3}=2$

$\alpha+\beta=5\ldots(1)$

and $\frac{-2-2\beta+\alpha-3}{3}=a\Rightarrow\alpha-2\beta=3a+5$ _____(2)

from (1) and (2) $\alpha = a + 5, \beta = -a$

B(– a, 2 a), C(a + 5, a + 2)

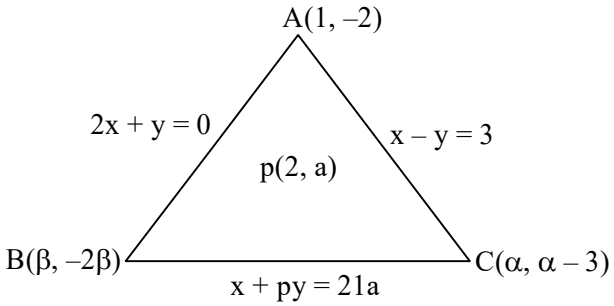
B and C lies on x + py = 21a

–a + p(2a) = 21a $\Rightarrow$ p = 11

and a + 5 + 11(a + 2) = 21a

$\Rightarrow$ a = 3

B(–3, 6), C(8, 5), $BC^2 = 122$

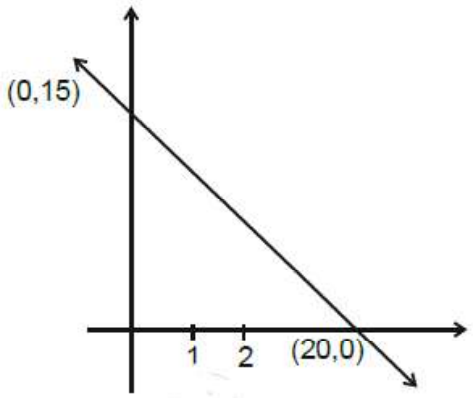


8. A triangle is formed by X-axis, Y-axis and the line $3x + 4y = 60$. Then the number of points P(a,b) which lie strictly inside the triangle, where a is an integer and b is a multiple of a, is _____. **[JEE(Main)Jan.2023]**

X-अक्ष, Y-अक्ष तथा रेखा $3x + 4y = 60$ एक त्रिभुज बनाते है। तो ऐसे बिंदुओं P(a,b) जहाँ a एक पूर्णांक है तथा b, a का एक गुणज है, जो त्रिभुज के अंदर है, की संख्या है। **[JEE(Main)Jan.2023]**

Ans. 31

Sol. If $x = 1, y = \frac{57}{4} = 14.25$



(1, 1) (1, 2) – (1, 14) $\Rightarrow$ 14 pts.

If $x = 2,\ y = \frac{27}{2} = 13.5$

(2, 2) (2, 4) ¡K. (2, 12) $\Rightarrow$ 6 pts.

If $x = 3,\ y = \frac{51}{4} = 12.75$

(3, 3) (3, 6) – (3, 12) $\Rightarrow$ 4 pts.

If x = 4, y = 12

(4, 4) (4, 8) $\Rightarrow$ pts.

If $x = 5.\ y = \frac{45}{4} = 11.25$

(5, 5), (5, 10) $\Rightarrow$ 2 pts.

If $x = 6,\ y = \frac{21}{2} = 10.5$

(6, 6) $\Rightarrow$ 1 pt.

If $x = 7,\ y = \frac{39}{4} = 9.75$

(7, 7) $\Rightarrow$ 1 pt.

If $x = 8,\ y = 9$

(8, 8) $\Rightarrow$ 1 pt.

If $x = 9\ \ y = \frac{33}{4} = 8.25$ $\Rightarrow$ no pt.

Total = 31 pts.

9. The straight lines l_1 and l_2 pass through the origin and trisect the line segment of the line $L : 9x + 5y = 45$ between the axes. If m_1 and m_2 are the slopes of the lines l_1 and l_2 then the point of intersection of the line $y = (m_1 + m_2)x$ with L lies on : **[JEE(Main)April,2023]**

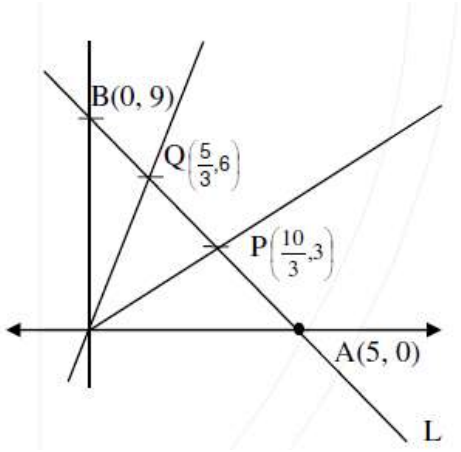
सरल रेखाएँ l_1 व l_2 मूल बिंदु से होकर जाती है और रेखा $L : 9x + 5y = 45$ के अक्षों के बीच रेखाखंड को तीन बराबर भागों में बाँटती है। यदि रेखाओं l_1 व l_2 की प्रवणताएँ m_1 व m_2 हैं, तब रेखाओं $y = (m_1 + m_2)x$ और L का प्रतिच्छेदन बिंदु किस रेखा पर है :

[JEE(Main)April,2023]

(A) $6x + y = 10$ (B) $y - 2x = 5$ (C) $y - x = 5$ (D) $6x - y = 15$

Ans. C

Sol.



$$m_1 = \frac{3}{(10/3)} = \frac{9}{10}$$

$$m_2 = \frac{6}{(5/3)} = \frac{18}{5}$$

$$m_1 + m_2 = \frac{9+36}{10} = \frac{45}{10} = \frac{9}{2}$$

So, $y = (m_1 + m_2)x$ is $y = \frac{9}{2}x$

Now solving $y = \frac{9}{2}x$ and $9x + 5y = 45$

$$\Rightarrow 2y + 5y = 45 \Rightarrow y = \frac{45}{7}, x = \frac{10}{7}$$

$$y - x = \frac{35}{7} = 5$$

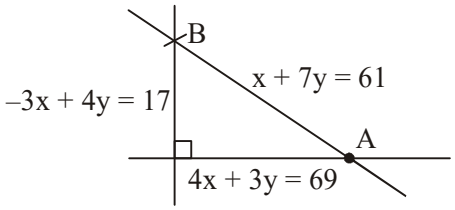
10. Let $C(\alpha, \beta)$ be the circumcenter of the triangle formed by the lines $4x + 3y = 69$, $4y - 3x = 17$, and $x + 7y = 61$. Then $(\alpha - \beta)^2 + \alpha + \beta$ is equal to : **[JEE(Main)April,2023]**

माना रेखाओ $4x + 3y = 69$, $4y - 3x = 17$, तथा $x + 7y = 61$ द्वारा निर्मित त्रिभुज का परिकेन्द्र $C(\alpha, \beta)$ है। तो $(\alpha - \beta)^2 + \alpha + \beta$ बराबर है : **[JEE(Main)April,2023]**

(A) 18 (B) 17 (C) 16 (D) 15

Ans. B

Sol.



$4x + 28y = 244$

$4x + 3y = 69$

$25y = 175$

$y = 7, x = 12$

$A(12, 7)$

$-3x + 4y = 17$

$3x + 21y = 183$

$25y = 200$

$y = 8, x = 5$

$B(5, 8)$

$\therefore$ Circumcenter

$$\alpha = \frac{17}{2} \beta = \frac{15}{2}$$

$\left(\frac{17}{2}, \frac{15}{2}\right)$

$(\alpha-\beta)^2+\alpha+\beta$

$1 + 16 = 17$

11. Let R be a rectangle given by the lines x = 0, x = 2, y = 0 and y = 5. Let A(α, 0) and B(0, β), α ∈ [0, 2] and β ∈ [0, 5], be such that the line segment AB divides the area of the rectangle R in the ratio 4 : 1. Then, the mid-point of AB lies on a : **[JEE(Main)April,2023]**

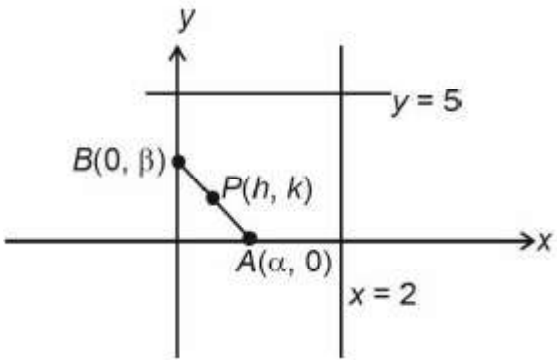
(A) circle (B) straight line (C) hyberbola (D) parabola

माना रेखाओं x = 0, x = 2, y = 0 तथा y = 5 द्वारा निर्मित आयत R है। माना A(α, 0) तथा B(0, β), α ∈ [0, 2], β ∈ [0, 5], इस प्रकार हैं कि रेखाखंड AB आयत R क्षेत्रफल को 4 : 1 के अनुपात में बाँटती है। तो AB का मध्य बिंदु एक **[JEE(Main)April,2023]**

(A) वृत्त (B) सरल रेखा (C) अतिपरवलय (D) परवलय

Ans. C

Sol.



$$\frac{10-\frac{1}{2}\alpha\beta}{\frac{1}{2}\alpha\beta}=\frac{4}{1} \Rightarrow 20-\alpha\beta=4\alpha\beta$$

$\Rightarrow \alpha\beta = 4$

Let $h=\frac{\alpha}{2}, \beta=\frac{k}{2}$

$\therefore \quad 4hk = 4$

$\therefore xy = 1$

12. If the point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on the curve traced by the mid-points of the line segments of the lines

$x\cos\theta + y\sin\theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ between the co-ordinates axes, then α is equal to :

[JEE(Main)April,2023]

यदि रेखाओं $x\cos\theta + y\sin\theta = 7, \theta \in \left(0, \frac{\pi}{2}\right)$ के निर्देशांक अक्षों के बीच रेखाखंडों के मध्य बिंदुओं द्वारा बने वक्र पर एक बिंदु $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ है, तो α बराबर है : **[JEE(Main)April,2023]**

(A) $7\sqrt{3}$ (B) $-7\sqrt{3}$ (C) -7 (D) 7

Ans. D

Sol. $x\cos\theta + y\sin\theta = 7; \quad \theta \in \left(0, \frac{\pi}{2}\right)$

Mid point of line segment between coordinate axes

$$(h,k) \equiv \left(\frac{7}{2\cos\theta}, \frac{7}{2\sin\theta}\right)$$

$$\Rightarrow \text{Locus of (h, k) is } \frac{49}{y^2} + \frac{49}{x^2} = 4 \Rightarrow \frac{1}{x^2} + \frac{1}{y^2} = \frac{4}{49}$$

Point $\left(\alpha, \frac{7\sqrt{3}}{3}\right)$ lies on it

$$\Rightarrow \frac{1}{\alpha^2} + \frac{3}{49} = \frac{4}{49}$$

$$\Rightarrow \frac{1}{\alpha^2} = \frac{1}{49}$$

$$\Rightarrow \alpha = 7$$

13. Let (α, β) be the centroid of the triangle formed by the lines $15x - y = 82$, $6x - 5y = -4$ and $9x + 4y = 17$. Then $\alpha + 2\beta$ and $2\alpha - \beta$ are the roots of the equation : **[JEE(Main)April,2023]**

माना रेखाओं $15x - y = 82$, $6x - 5y = -4$ तथा $9x + 4y = 17$ द्वारा बने त्रिभुज का केन्द्रक (α, β) है। तो $\alpha + 2\beta$ तथा $2\alpha - \beta$ किस समीकरण के मूल हैं : **[JEE(Main)April,2023]**

(A) $x^2 - 7x + 12 = 0$ (B) $x^2 - 13x + 42 = 0$ (C) $x^2 - 10x + 25 = 0$ (D) $x^2 - 14x + 48 = 0$

Ans. B

Sol. $15x - y = 82$

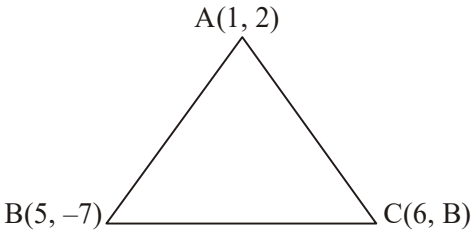
$6x - 5y = -4$

$9x + 4y = 17$

$(\alpha,\beta) \equiv \left(\frac{1+5+6}{3}, \frac{2-7+8}{3}\right) \equiv (4,1)$

$\alpha + 2\beta = 6$ and $2\alpha - \beta = 7$

Equation $x^2 - 13x + 42 = 0$



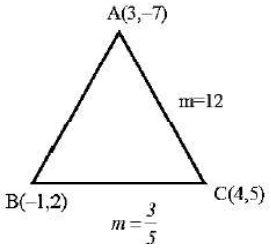
14. If (α, β) is the orthocenter of the triangle ABC with vertices A(3, –7), B(–1, 2) and C(4, 5), then $9\alpha - 6\beta + 60$ is equal to : **[JEE(Main)April,2023]**

यदि त्रिभुज ABC का लम्ब केन्द्र (α, β) है जहाँ त्रिभुज ABC के शीर्ष क्रमशः A(3, –7), B(–1, 2) तथा C(4, 5) हो, तो $9\alpha - 6\beta + 60$ का मान बराबर होगा : **[JEE(Main)April,2023]**

(A) 30 (B) 35 (C) 40 (D) 25

Ans. D

Sol.



Altitude of BC : $y + 7 = \frac{-5}{3}(x - 3)$

$3y + 21 = -5x + 15$

$5x + 3y + 6 = 0$

Altitude of AC : $y - 2 = \frac{-1}{12}(x + 1)$

$12y - 24 = -x - 1$

$x + 12y = 23$

$\alpha = \frac{-47}{19}, \beta = \frac{121}{57}$

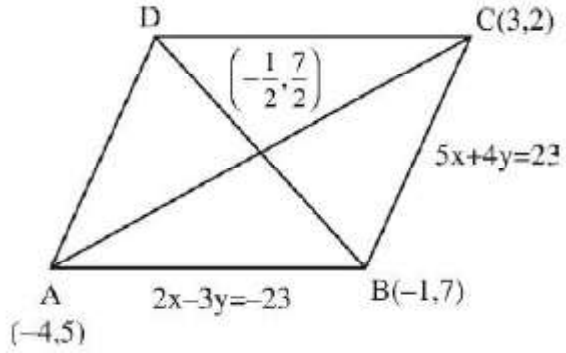
$9\alpha - 6\beta + 60 = 25$

15. Let the equations of two adjacent sides of a parallelogram ABCD be $2x - 3y = -23$ and $5x + 4y = 23$. If the equation of its one diagonal AC is $3x + 7y = 23$ and the distance of A from the other diagonal is d, then $50d^2$ is equal to _____. **[JEE(Main)April,2023]**

माना एक समांतर चतुर्भुज की दो संलग्न भुजाओं के समीकरण $2x - 3y = -23$ तथा $5x + 4y = 23$ हैं। यदि इसके एक विकर्ण AC का समीकरण $3x + 7y = 23$ है तथा A की दूसरे विकर्ण से दूरी d है, तो $50d^2$ बराबर है : **[JEE(Main)April,2023]**

Ans. 529

Sol.



A & C point will be (–4, 5) & (3, 2)

mid point of AC will be $\left(-\frac{1}{2},\frac{7}{2}\right)$

equation of diagonal BD is

$$y-\frac{7}{2}=\frac{\frac{7}{2}}{-\frac{1}{2}}\left(x+\frac{1}{2}\right)$$

$\Rightarrow 7x+y=0$

Distance of A from diagonal BD

$$=d=\frac{23}{\sqrt{50}}$$

$\Rightarrow 50d^2=(23)^2$

$50d^2 = 529$

16. If the line $l_1 : 3y - 2x = 3$ is the angular bisector of the lines $l_2 : x - y + 1 = 0$ and $l_3 : \alpha x + \beta y + 17 = 0$, then $\alpha^2 + \beta^2 - \alpha - \beta$ is equal to _____. **[JEE(Main)April,2023]**

यदि रेखा $l_1 : 3y - 2x = 3$ रेखाओं $l_2 : x - y + 1 = 0$ तथा $l_3 : \alpha x + \beta y + 17 = 0$ की कोण समद्विभाजक है, तो $\alpha^2 + \beta^2 - \alpha - \beta$ बराबर है _____ **[JEE(Main)April,2023]**

Ans. 348

Sol. Point of intersection of $\ell_1 : 3y - 2x = 3$

$\ell_2 : x - y + 1 = 0$ is $P \equiv (0,1)$

Which lies on $\ell_3 : \alpha x + \beta y + 17 = 0$

$\Rightarrow \beta = -17$

Consider a random point $Q \equiv (-1,0)$

on $\ell_2 : x - y + 1 = 0$, image of Q about

$\ell_2 : x - y + 1 = 0$, is $Q' \equiv \left(\frac{-17}{13}, \frac{6}{13}\right)$ which is calculated by formulae

$$\frac{x-(-1)}{2} = \frac{y-0}{-3} = 2\left(\frac{-2+3}{13}\right)$$

Now, Q' lies in $\ell_3 : \alpha x + \beta y + 17 = 0$

$\Rightarrow \alpha = 7$

Now, $\alpha^2 + \beta^2 - \alpha - \beta = 348$

17. Consider the triangle with vertices A(2, 1), B(0, 0) and C(t, 4), $t \in [0,4]$. If the maximum and the minimum perimeters of such triangles are obtained at $t = \alpha$ and $t = \beta$ respectively, then $6\alpha + 21\beta$ is equal to _____.
[JEE(Main)April,2023]

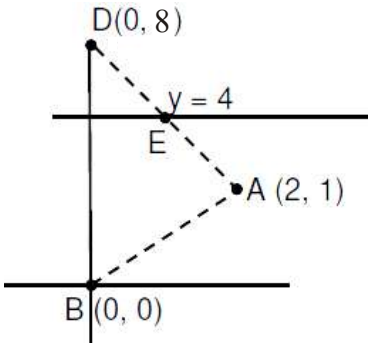
एक त्रिभुज इस प्रकार है जिसके शीर्ष A(2, 1), B(0, 0) तथा C(t, 4), $t \in [0,4]$ है। इस प्रकार के त्रिभुजों का अधिकतम तथा न्यूनतम परिमाप क्रमशः $t = \alpha$ तथा $t = \beta$ पर प्राप्त होता है तो $6\alpha + 21\beta$ का मान होगा _____ **[JEE(Main)April,2023]**

Ans. 48

Sol. A (2, 1), B(0, 0), C (t, 4), $t \in [0, 4]$

Image of B in $y = 4$

D(0, 8)



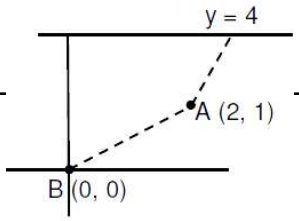
Equation of AD :

$$y - 1 = \frac{7}{-2}(x - 2)$$

$7x + 2y = 16$

$$E\left(\frac{8}{7}, 4\right) \Rightarrow \beta = \frac{8}{7}$$

For maximum perimeter



Maximum perimeter at C (4, 4)

$\therefore t \in [0,4]$

Now, $6\alpha + 21\beta = 6\times 4 + 21\left(\frac{8}{7}\right) = 48$

18. The portion of the line $4x + 5y = 20$ in the first quadrant is trisected by the lines L_1 and L_2 passing through the origin. The tangent of an angle between the lines L_1 and L_2 is : **[JEE(Main)Jan.2024]**

रेखा $4x + 5y = 20$ के प्रथम चतुर्थांश में भाग को मूल बिन्दु से होकर जाने वाली रेखाएँ L_1 तथा L_2 समत्रिभाजित करती हैं। L_1 तथा L_2 के बीच एक कोण की स्पर्शज्या है : **[JEE(Main)Jan.2024]**

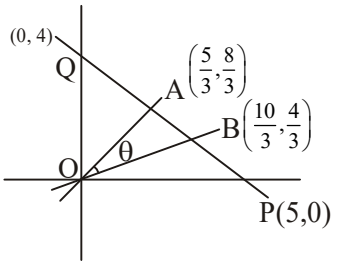
(A) $\frac{8}{5}$ (B) $\frac{2}{5}$ (C) $\frac{30}{41}$ (D) $\frac{25}{41}$

Ans. C

Sol. $A\left(\frac{5}{3},\frac{8}{3}\right)$, $B\left(\frac{10}{3},\frac{4}{3}\right)$

$m_{OA} = m_2 = \frac{8}{5}$

$m_{0B} = m_1 = \frac{2}{5}$



$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right| = \left|\frac{\frac{8}{5} - \frac{2}{5}}{1 + \frac{8}{5}\times\frac{2}{5}}\right|$$

$$= \frac{\frac{6}{5}}{1 + \frac{16}{25}} = \frac{6\times 5}{25 + 16} = \frac{30}{41}$$

19. Let R be the interior region between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin. The set of all values of a, for which the points $\left(a^2, a+1\right)$ lie in R, is : **[JEE(Main)Jan.2024]**

माना रेखाओं $3x - y + 1 = 0$ तथा $x + 2y - 5 = 0$ के अंदर का क्षेत्र, जिसमें मूलबिन्दु है, R है। a के सभी मानों, जिनके लिए $\left(a^2, a+1\right)$ क्षेत्र R में है, का समुच्चय है : **[JEE(Main)Jan.2024]**

(A) $(-3,-1)\cup\left(-\frac{1}{3},1\right)$ (B) $(-3,-1)\cup\left(\frac{1}{3},1\right)$

(C) $(-3,0)\cup\left(\frac{1}{3},1\right)$ (D) $(-3,0)\cup\left(\frac{2}{3},1\right)$

Ans. C

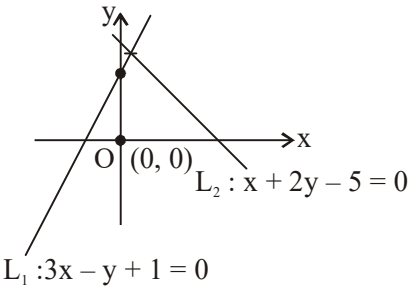
Sol. $P(a^2, a + 1)$

$L_1 = 3x - y + 1 = 0$

Origin and P lies same side w.r.t. L_1

$\Rightarrow L_1(0) \cdot L_1(P) > 0$

$\therefore 3(a^2) - (a + 1) + 1 > 0$



$\Rightarrow 3a^2 - a > 0$

$a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$ _____(1)

Let $L_2 : x + 2y - 5 = 0$

Originand P lies same side w.r.t. L_2

$\Rightarrow L_2(0) \cdot L_2(P) > 0$

$\Rightarrow a^2 + 2(a + 1) - 5 < 0$

$\Rightarrow a^2 + 2a - 3 < 0$

$\Rightarrow (a + 3)(a - 1) < 0$

$\therefore a \in (-3, 1)$ _____(2)

Intersection of (1) and (2)

$a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

20. Let $\left(5, \frac{a}{4}\right)$ be the circumcenter of a triangle with vertices A(a, –2), B(a, 6) and $C\left(\frac{a}{4}, -2\right)$. Let α denote the circumradius, β denote the area and γ denote the perimeter of the triangle. Then $\alpha + \beta + \gamma$ is :

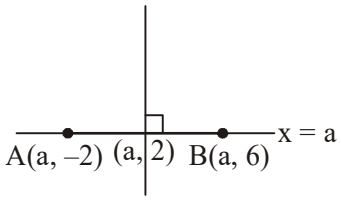
[JEE(Main)Jan.2024]

माना A(a, –2), B(a, 6) तथा $C\left(\frac{a}{4}, -2\right)$ के एक त्रिभुज का परिकेन्द्र $\left(5, \frac{a}{4}\right)$ है। यदि इस त्रिभुज के परिवृत्त की त्रिज्या α, त्रिभुज का क्षेत्रफल β तथा त्रिभुज का परिमाप γ है, तो $\alpha + \beta + \gamma$ is बराबर है : **[JEE(Main)Jan.2024]**

(A) 53 (B) 60 (C) 62 (D) 30

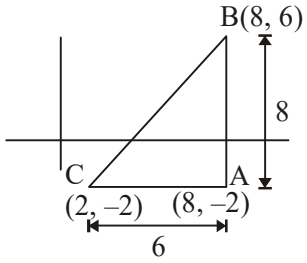
Ans. A

Sol.



Circumcentre lie on perpendicular bisector of AB $\Rightarrow y = 2$

$\Rightarrow \frac{a}{4} = 2$ $\Rightarrow a = 8$



$\Rightarrow$ Perimeter = 6 + 8 + 10 = 24

Area = $\frac{6\times 8}{2} = 24$

Circumradius = half of hypotenuse

$= \frac{10}{2} = 5$

$\alpha + \beta + \gamma = 24 + 24 + 5 = 53$

21. In a ΔABC, suppose $y = x$ is the equation of the bisector of the angle B and the equation of the side AC is $2x - y = 2$. If $2AB = BC$ and the points A and B are respectively (4, 6) and (α, β), then $\alpha + 2\beta$ is equal to :

[JEE(Main)Jan.2024]

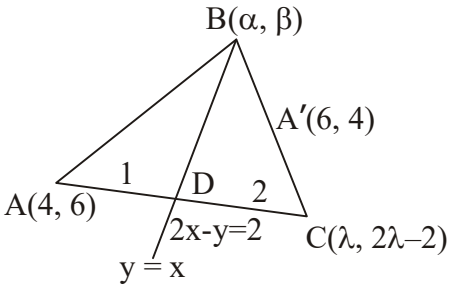
एक ΔABC में माना कोण B के समद्विभाजक का समीकरण $y = x$ है तथा भुजा AC का समीकरण $2x - y = 2$ है। यदि (4, 6) और (α, β) क्रमशः बिन्दु A और B तथा $2AB = BC$ है, तो $\alpha + 2\beta$ बराबर है :

[JEE(Main)Jan.2024]

(A) 48 (B) 39 (C) 45 (D) 42

Ans. D

Sol.



$$\frac{AB}{BC}=\frac{AD}{DC}=\frac{1}{2}$$

$$D\left(\frac{\lambda+8}{3},\frac{2\lambda-2+12}{3}\right) \text{ lies on } y=x$$

$$\frac{\lambda+8}{3}=\frac{2\lambda+10}{3}$$

$$\Rightarrow \lambda=-2 \Rightarrow c(-2,-6)$$

Image of A w.r.t. y = x lies on BC

m(BC) = m(A'C)

$$\Rightarrow \frac{\alpha+6}{\alpha+2}=\frac{4+6}{6+2} \Rightarrow \alpha=14=\beta$$

$\alpha + 2\beta = 42$

22. The distance of the point (2,3) from the line $2x - 3y + 28 = 0$, measured parallel to the line $\sqrt{3}x - y + 1 = 0$, is equal to : **[JEE(Main)Jan.2024]**

बिन्दु (2, 3) की रेखा $2x - 3y + 28 = 0$ से रेखा $\sqrt{3}x - y + 1 = 0$ के समांन्तर मापी दूरी बराबर है : **[JEE(Main)Jan.2024]**

(A) $4+6\sqrt{3}$ (B) $4\sqrt{2}$ (C) $6\sqrt{3}$ (D) $3+4\sqrt{2}$

Ans. A

Sol. Given line $y = \sqrt{3}x + 1$

$$\text{Inclination} = \frac{\pi}{3}$$

Any point on line though (2,3) and parallel to given line.

$$P\left(2+r\cos\frac{\pi}{3},3+r\sin\frac{\pi}{3}\right)\equiv\left(2+\frac{r}{2},3+\frac{r\sqrt{3}}{2}\right)$$

If P lieson $2x - 3y + 28 = 0$

$$2\left(2+\frac{r}{2}\right)-3\left(3+\frac{r\sqrt{3}}{2}\right)+28=0$$

$\Rightarrow |r| = 4+6\sqrt{3}$ = Required distance

23. Let A be the point of intersection of the lines $3x+2y=14$, $5x-y=6$ and B be the point of intersection of the lines $4x+3y=8$, $6x+y=5$. The distance of the point P(5,–2) from the line AB is : **[JEE(Main)Jan.2024]**

माना रेखाओं $3x+2y=14$, $5x-y=6$ का प्रतिच्छेदन बिन्दु A है तथा रेखाओं $4x+3y=8$, $6x+y=5$ का प्रतिच्छेदन बिन्दु B है। तो रेखा AB से बिन्दु P(5,–2) की दूरी है : **[JEE(Main)Jan.2024]**

(A) $\frac{5}{2}$ (B) 8 (C) $\frac{13}{2}$ (D) 6

Ans. D

Sol. Solving lines L_1 $(3x+2y=14)$ and L_2 $(5x-y=6)$

to get A(2, 4) and solving lines L_3 $(4x+3y=8)$

and L_4 $(6x+y=5)$ to get $B\left(\frac{1}{2},2\right)$

Finding Eqn. of AB : $4x-3y+4=0$

Calculate distance PM $\Rightarrow \left|\frac{4(5)-3(-2)+4}{5}\right| = 6$

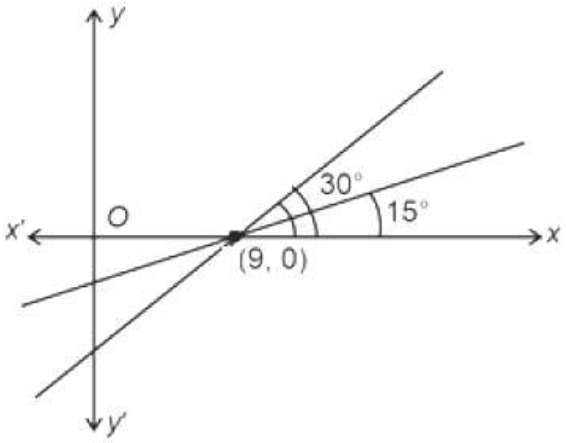
24. A line passing through the point A (9, 0) makes an angle of 30º with the positive direction of x-axis. If this line is rotated about A through an angle of 15º in the clockwise direction, then its equation in the new position is : **[JEE(Main)Jan.2024]**

बिंदु A (9, 0) से होकर जाने वाली एक रेखा x-अक्ष की धनात्मक दिशा से 30º का कोण बानाती है। यदि इस रेखा को घड़ी की दिशा में 15º के कोण तक A के सापेक्ष घुमाया जाता है, तो नयी स्थिति में इसका समीकरण है : **[JEE(Main)Jan.2024]**

(A) $\frac{x}{\sqrt{3}-2}+y=9$ (B) $\frac{x}{\sqrt{3}+2}+y=9$ (C) $\frac{y}{\sqrt{3}-2}+x=9$ (D) $\frac{y}{\sqrt{3}+2}+x=9$

Ans. C

Sol.



Equation : $y-0=\tan 15^\circ (x-9) \Rightarrow y=\left(2-\sqrt{3}\right)(x-9)$

25. If $x^2-y^2+2hxy+2gx+2fy+c=0$ is the locus of a point, which moves such that it is always equidistant from the lines $x+2y+7=0$ and $2x-y+8=0$, then the value of $g+c+h-f$ equals : **[JEE(Main)Jan.2024]**

यदि एक बिन्दु, जो इस प्रकार चलता है कि इसकी रेखाओं $x+2y+7=0$ तथा $2x-y+8=0$ से दूरी बराबर रहती है, का बिन्दुपथ $x^2-y^2+2hxy+2gx+2fy+c=0$ है, तो $g+c+h-f$ का मान बराबर है : **[JEE(Main)Jan.2024]**

(A) 6 (2) 8 (C) 29 (D) 14

Ans. D

Sol. Locus of point P(x, y) whose distance from

Gives

X + 2y + 7 = 0 & 2x – y + 8 = 0 are equal is $\frac{x+2y+7}{\sqrt{5}}=\pm\frac{2x-y+8}{\sqrt{5}}$

$(x+2y+7)^2-(2x-y+8)^2=0$

Combined equation of lines

$(x-3y+1)(3x+y+15)=0$

$3x^2-3y^2-8xy+18x-44y+15=0$

$x^2-y^2-\frac{8}{3}xy+6x-\frac{44}{3}y+5=0$

$x^2-y^2+2h\,xy+2gx+2fy+c=0$

$h=\frac{4}{3},\ g=3,\ f=-\frac{22}{3},\ c=5$

$g+c+h-f=3+5-\frac{4}{3}+\frac{22}{3}=8+6=14$

26. Let $\alpha, \beta, \gamma, \delta \in Z$ and let $A(\alpha, \beta)$, $B(1,0)$, $C(\gamma, \delta)$ and $D(1,2)$ be the vertices of a parallelogram ABCD. If $AB = \sqrt{10}$ and the points A and C lie on the line $3y = 2x + 1$, then $2(\alpha + \beta + \gamma + \delta)$ is equal to :

[JEE(Main)Jan.2024]

माना $\alpha, \beta, \gamma, \delta \in Z$ हैं तथा माना एक समांतर चतुर्भज ABCD के शीर्ष $A(\alpha, \beta)$, $B(1,0)$, $C(\gamma, \delta)$ तथा $D(1,2)$ हैं। यदि $AB = \sqrt{10}$ है तथा बिन्दु A और C रेखा $3y = 2x + 1$ पर हैं, तो $2(\alpha + \beta + \gamma + \delta)$ बराबर है :

[JEE(Main)Jan.2024]

(A) 12 (B) 10 (C) 5 (D) 8

Ans. D

Sol. ABCD is a parallelogram so mid-point of AC = mid-point of BD

$$\Rightarrow \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2}\right) = \left(\frac{1+1}{2}, \frac{0+2}{2}\right)$$

$$\Rightarrow \left(\frac{\alpha+\gamma}{2}, \frac{\beta+\delta}{2}\right) = (1,1)$$

$$\Rightarrow \alpha+\gamma = 2 \text{ and } \beta+\delta = 2$$

So, $2(\alpha+\beta+\gamma+\delta) = 2(2+2) = 8$

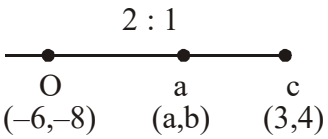
27. Let A(a, b), B(3, 4) and C(–6, –8) respectively denote the centroid, circumcentre and orthocentre of a triangle. Then, the distance of the point P(2a + 3, 7b + 5) from the line $2x + 3y - 4 = 0$ measured parallel to the line $x - 2y - 1 = 0$ is : **[JEE(Main)Jan.2024]**

माना A(a, b), B(3, 4) तथा C(–6, –8) एक त्रिभुज के केन्द्रक, परिकेन्द्र तथा लंबकेन्द्र है। तो बिन्दु P(2a + 3, 7b + 5) की रेखा $2x + 3y - 4 = 0$ से, रेखा $x - 2y - 1 = 0$ के समांतर नापी गई दूरी है : **[JEE(Main)Jan.2024]**

(A) $\frac{17\sqrt{5}}{6}$ (B) $\frac{17\sqrt{5}}{7}$ (C) $\frac{\sqrt{5}}{17}$ (D) $\frac{15\sqrt{5}}{7}$

Ans. B

Sol. Centroid divides orthocentre and circumcentre in 2 : 1.

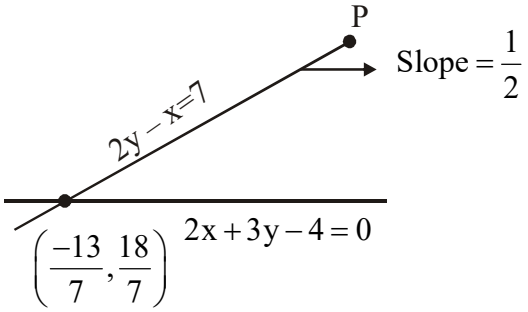


$$a \equiv \frac{2(3)-6}{3} = 0$$

$$b \equiv \frac{2(4)-8}{3} = 0$$

$$\Rightarrow (a, b) \equiv (0, 0)$$

$\Rightarrow P \equiv (3,5)$



$\Rightarrow (y-5)=\frac{1}{2}(x-3)$

$\Rightarrow 2y-10=x-3$

$\Rightarrow 2y-x=7$

$\Rightarrow \sqrt{\left(3+\frac{13}{7}\right)^2+\left(5-\frac{18}{7}\right)^2}=\frac{\sqrt{34^2+17^2}}{7}$

$=\frac{17\sqrt{5}}{7}$

28. If the sum of squares of all real values of α, for which the lines $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ and $\alpha x + 2y - 2 = 0$ do not form a triangle is p, then the greatest integer less than or equal to p is _____.

[JEE(Main)Jan.2024]

माना α के सभी वास्तविक मानों, जिनके लिए रेखाएँ $2x - y + 3 = 0$, $6x + 3y + 1 = 0$ तथा $\alpha x + 2y - 2 = 0$ एक त्रिभुज नहीं बनाती हैं, के वर्गों का योग p है, तो महत्तम पूर्णांक $\leq$ p है _____ **[JEE(Main)Jan.2024]**

Ans. 32

Sol. $2x - y + 3 = 0$

$6x + 3y + 1 = 0$

$\alpha x + 2y - 2 = 0$

will not form a Δ if $\alpha x + 2y - 2 = 0$ is concurrent with $2x - y + 3 = 0$ and $6x + 3y + 1 = 0$ or parallel to either of them so

Case-1: Concurrent lines

$$\begin{vmatrix} 2 & -1 & 3 \\ 6 & 3 & 1 \\ \alpha & 2 & -2 \end{vmatrix} = 0 \Rightarrow \alpha = \frac{4}{5}$$

Case-2 : Parallel lines

$$-\frac{\alpha}{2}=\frac{-6}{3} \text{ or } -\frac{\alpha}{2}=2$$

$\Rightarrow \alpha = 4$ or $\Rightarrow \alpha = -4$

$$P = 16+16+\frac{16}{25}$$

$$[P]=\left[32+\frac{16}{25}\right]=32$$

29. Let for any three distinct consecutive terms a, b, c of an A.P., the lines $ax + by + c = 0$ be concurrent at the point P and Q (α, β) be a point such that the system of equations

$x + y + z = 6$,

$2x + 5y + \alpha z = \beta$ and

$x + 2y + 3z = 4$, has infinitely many solutions. Then $(PQ)^2$ is equal to _____ .

[JEE(Main)Jan.2024]

माना एक A.P. के किसी भी तीन भिन्न क्रमागत पदों a, b, c के लिए रेखाऐं $ax + by + c = 0$ एक बिन्दु P पर संगामी हैं तथा बिन्दु Q (α, β) के लिए समीकरण निकाय

$x + y + z = 6$,

$2x + 5y + \alpha z = \beta$ तथा

$x + 2y + 3z = 4$, के अनंत हल है। तो $(PQ)^2$ बराबर है _____ . **[JEE(Main)Jan.2024]**

Ans. 113

Sol. $b - a = c - b$

$a - 2b + c = 0$

$\Rightarrow$ $ax + by + c = 0$ are concurrent at P $(1, -2)$

$\therefore$ $x + y + z = 6$,

$2x + 5y + \alpha z = \beta$,

$x + 2y + 3z = 4$

$\Delta = 8 - \alpha$,

$\Delta_1 = -8\alpha - \beta + 70$,

$\Delta_2 = 2\alpha + 2\beta - 28$,

$\Delta_3 = 6 - \beta$ for infinitely many solutions

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$\Rightarrow \alpha = 8, \beta = 6$

$\Rightarrow Q(8, 6), P(1, -2)$

$\Rightarrow (PQ)^2 = 113$

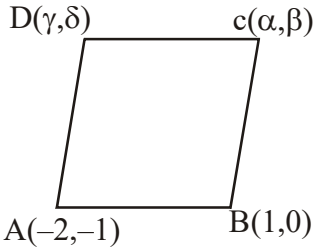
30. Let A(–2, –1), B(1, 0), C(α, β) and D(γ, δ) be the vertices of a parallelogram ABCD. If the point C lies on $2x - y = 5$ and the point D lies on $3x - 2y = 6$, then the value of $|\alpha + \beta + \gamma + \delta|$ is equal to _____.

[JEE(Main)Jan.2024]

माना एक समांतर चतुर्भुज ABCD के शीर्ष A(–2, –1), B(1, 0), C(α, β) तथा D(γ, δ) हैं। यदि बिन्दु C, रेखा $2x - y = 5$ पर है तथा बिन्दु D, रेखा $3x - 2y = 6$ पर है, तो $|\alpha + \beta + \gamma + \delta|$ का मान बराबर है _____ **[JEE(Main)Jan.2024]**

Ans. 32

Sol.



$2\alpha - \beta = 5$ .....(i)

$3\gamma - 2\delta = 6$ .....(ii)

Diagonal bisect each other

$\alpha - 2 = \gamma + 1 \Rightarrow \alpha = \gamma + 3$

$\beta - 1 = \delta \quad \Rightarrow \beta = \delta + 1$

Now form (i)

$2(\gamma + 3) - \delta - 1 = 5 \Rightarrow \delta = 2\gamma$

From (ii) $\gamma = -6, \quad \delta = -12, \quad \alpha = 3, \quad \beta = -11$

$|\alpha + \beta + \gamma + \delta| = 32$

31. Let ABC be an isosceles triangle in which A is at (–1, 0), $\angle A = \frac{2\pi}{3}$, AB = AC and B is on the positive x-axis. If $BC = 4\sqrt{3}$ and the line BC intersects the line $y = x + 3$ at (α, β), then $\frac{\beta^4}{\alpha^2}$ is _____.

[JEE(Main)Jan.2024]

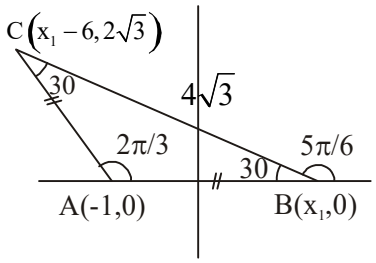
माना एक समद्विबाहु त्रिभुज ABC में A, बिंदु (–1, 0) हैं, $\angle A = \frac{2\pi}{3}$ हैं,, AB = AC है तथा B, धनात्मक x-अक्ष पर है। यदि

$BC = 4\sqrt{3}$ तथा रेखा BC, रेखा $y = x + 3$ को (α, β) पर काटती है, तो $\frac{\beta^4}{\alpha^2}$ बराबर है _____

[JEE(Main)Jan.2024]

Ans. 36

Sol.



Co-ordinates of 'C' are

$$\frac{x - x_1}{\cos\frac{5\pi}{6}} = \frac{y - 0}{\sin\frac{5\pi}{6}} = 4\sqrt{3}$$

$$x = 4\sqrt{3}\cos\frac{5\pi}{6} + x_1 \Rightarrow 4\sqrt{3} \times \left(-\frac{\sqrt{3}}{2}\right) + x_1 \Rightarrow (x_1 - 6)$$

$$y = 4\sqrt{3}\sin\frac{5\pi}{6} \Rightarrow 4\sqrt{3} \times \frac{1}{2} \Rightarrow 2\sqrt{3}$$

$$C\left(x_1 - 6, 2\sqrt{3}\right)$$

$$m_{AC} = -\sqrt{3}$$

$$\frac{0 - 2\sqrt{3}}{-1 - x_1 + 6} = -\sqrt{3} \Rightarrow x_1 = 3$$

$$BC : x + \sqrt{3}y = 3, y = x + 3$$

$$y - 3 + \sqrt{3}y = 3$$

$$y = \frac{6}{\sqrt{3}+1}, x = \frac{6}{\sqrt{3}+1} - 3 \Rightarrow \frac{6 - 3\sqrt{3} - 3}{\sqrt{3}+1}$$

$$x = \frac{3 - 3\sqrt{3}}{\sqrt{3}+1}$$

$$\frac{-6}{(1+\sqrt{3})^2}$$

So, $\frac{\beta^4}{\alpha^2} = 36$

32. The vertices of a triangle are A(–1,3), B(–2,2) and C(3, –1). A new triangle is formed by shifting the sides of the triangle by one unit inwards. Then the equation of the side of the new triangle nearest to origin is :

[JEE(Main)April,2024]

एक त्रिभुज के शीर्ष A(–1,3), B(–2,2) तथा C(3, –1) हैं। इस त्रिभुज की भुजाओं को त्रिभुज के अंदर की तरफ एक इकाई स्थानांतरित कर एक नया त्रिभुज बनाया गया है। तो नए त्रिभुज की मूलबिन्दु के सबसे निकट भुजा का समीकरण है :

[JEE(Main)April,2024]

(A) $x + y - \left(2 - \sqrt{2}\right) = 0$

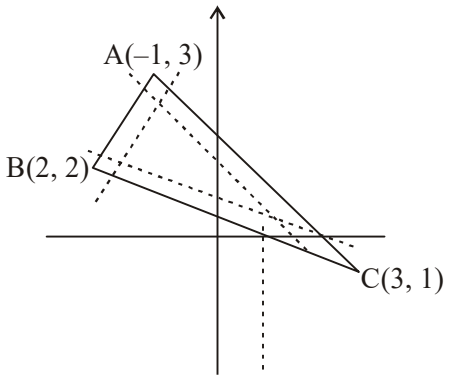
(B) $x - y - \left(2 + \sqrt{2}\right) = 0$

(C) $-x + y - \left(2 - \sqrt{2}\right) = 0$

(D) $x + y + \left(2 - \sqrt{2}\right) = 0$

Ans. A

Sol.



$AB : y - 2 = \frac{1}{1}(x + 2)$

$x - y - 4 = 0$. $x - y + c = 0$

$\left|\frac{c+4}{\sqrt{2}}\right| = 1$

$C = \sqrt{2} - 4, -\sqrt{2} - 4$

$BC : \ y + 1 = \frac{3}{-5}(x - 3)$

$-5y - 5 = 3x - 9$

$3x + 5y = 4$

$\left|\frac{c+4}{\sqrt{34}}\right| = 1$

AC : $y-3=-1(x+1)$

$x+y-2$

$$\left|\frac{c+2}{\sqrt{2}}\right|=1$$

$C=\sqrt{2}-2,-\sqrt{2}-2$

Nearest : $x+y-2+\sqrt{2}=0$

33. Let two straight lines drawn from the origin O intersect the line $3x+4y=12$ at the points P and Q such that ΔOPQ is an isosceles triangle and $\angle POQ=90^{\circ}$. If $l=OP^2+PQ^2+QO^2$, then the greatest integer less than or equal to l is : **[JEE(Main)April,2024]**

माना मूलबिन्दु से खींची गई दो रेखाएँ, रेखा $3x+4y=12$ को बिन्दुओं P तथा Q पर इस प्रकार काटती है कि ΔOPQ एक समद्विबाहु त्रिभुज है तथा $\angle POQ=90^{\circ}$ है। यदि $l=OP^2+PQ^2+QO^2$ है, तो l से कम या l के बराबर महत्तम पूर्णांक है :

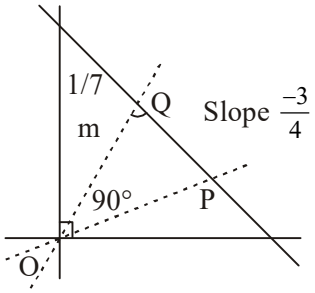
[JEE(Main)April,2024]

(A) 44 (B) 42 (C) 46 (D) 48

Ans. C

Sol.

$$\left|\frac{m+\frac{3}{4}}{1-\frac{3m}{4}}\right|=1$$



$$m+\frac{3}{4}=1-\frac{3}{4}m$$

$$m+\frac{3}{4}m=1-\frac{3}{4}$$

$7m=1$

$$\boxed{m=\frac{1}{7}}$$

$OQ = y=\frac{1}{7}x$ OP $y=7x$

$Q = 3x+4y=12$

$3x+\frac{4}{7}x=12$

$$x=\frac{12\times7}{25}, y=\frac{12\times7}{25\times7}=\frac{12}{25}, Q\left(\frac{84}{25},\frac{12}{25}\right)$$

$P \Rightarrow -25x=12$

$$x = \frac{-12}{25}, y = \frac{84}{25}$$

$$P = \left(\frac{-12}{25}, \frac{84}{25}\right)$$

$$OQ = \sqrt{\frac{84^2}{25^2} + \frac{12^2}{25^2}}$$

$$OP = \sqrt{\frac{84^2}{25^2} + \frac{12^2}{25^2}}$$

$$PQ = \sqrt{2}\sqrt{\frac{84^2}{25^2} + \frac{12^2}{25^2}}$$

$\ell = OQ^2 + OP^2 + PQ^2$

$$= \frac{84^2}{25^2} \times 2 + \frac{12^2}{25^2} \times 2 + 2\left(\frac{84^2}{25^2} + \frac{12^2}{25^2}\right)$$

$$= 4 \times \frac{84^2}{25^2} + 4 \times \frac{12^2}{25^2}$$

$$\ell = 4 \times \frac{12^2}{25^2}\left(7^2 + 1\right) = \frac{144 \times 8}{25} = 46.08$$

$[\ell] = 46$

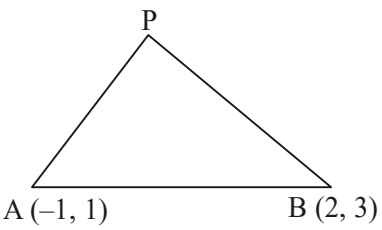
34. Let A (–1, 1) and B(2, 3) be two points and P be a variable point above the line AB such that the area of ΔPAB is 10. If the locus of P is ax + by = 15, then 5a + 2b is : **[JEE(Main)April,2024]**

माना दो बिन्दु A (–1, 1) तथा B(2, 3) हैं और रेखा AB के ऊपर एक चर बिन्दु P इस प्रकार है कि ΔPAB का क्षेत्रफल 10 है। यदि P का बिंदुपथ ax + by = 15 है, तो 5a + 2b बराबर है : **[JEE(Main)April,2024]**

(A) 6 (B) 4 (C) $-\frac{6}{5}$ (D) $-\frac{12}{5}$

Ans. D

Sol.



Locus of P,

$ax + by = 15$

Take point P as (h, k)

Area of Δ ABP

$= \frac{1}{2}|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)| = 10$

$= \frac{1}{2}|-1(3-k) + 2(k-1) + h(1-3)| = 10$

$= |-3 + k + 2k - 2 + h - 3h| = 20$

$= |3k - 2h - 5| = 20$

$= -(3k - 2h - 5) = 20 \Rightarrow 2h - 3k = 15$

$a = 2, b = -3$

$\therefore\ 5a + 2b = 4$

35. Let a variable line of slope $m > 0$ passing through the point $(4, -9)$ intersect the coordinate axes at the points A and B. The minimum value of the sum of the distances of A and B from the origin is : **[JEE(Main)April,2024]**

माना प्रवणता $m > 0$ तथा बिन्दु $(4, -9)$ से गुजरने वाली कोई चर रेखा निर्देशांक अक्षों को बिन्दु A और B पर प्रतिच्छेद करती है। मूल बिन्दु से A और B की दूरियों के योग का न्यूनतम मान है : **[JEE(Main)April,2024]**

(A) 30 (B) 25 (C) 15 (D) 10

Ans. B

Sol. Equation of line is

$y + 9 = m(x - 4)$

$\therefore A = \left(\frac{9 + 4m}{m}, 0\right)$

$B = (0, -9 - 4m)$

$\therefore OA + OB = \frac{9 + 4m}{m} + 9 + 4m$

$\because\ m > 0$

$= 13 + \frac{9}{m} + 4m$

$\because \frac{4m + \frac{9}{m}}{2} \geq \sqrt{36} \qquad \Rightarrow 4m + \frac{9}{m} \geq 12$

$\therefore\ OA + OB \geq 25$

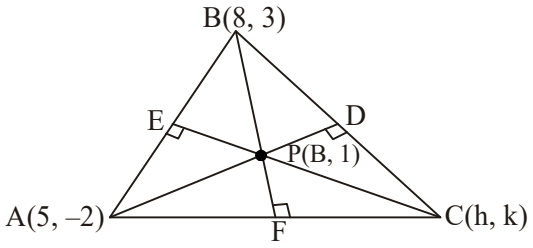
36. If P(6, 1) be the orthocentre of the triangle whose vertices are A(5, –2), B(8, 3) and C(h, k), then the point C lies on the circle : **[JEE(Main)April,2024]**

यदि P(6, 1) उस त्रिभुज का लंबकेन्द्र है, जिसके शीर्ष A(5, –2), B(8, 3) तथा C(h, k) हैं, तो बिन्दु C किस वृत्त पर स्थित है : **[JEE(Main)April,2024]**

(A) $x^2 + y^2 - 52 = 0$ (B) $x^2 + y^2 - 61 = 0$ (C) $x^2 + y^2 - 74 = 0$ (D) $x^2 + y^2 - 65 = 0$

Ans. D

Sol.



Solpe of BF $= (m_{BF}) = \frac{3-1}{8-6} = 1$

$m_{AC} = \frac{k+2}{h-5} - 1 (AC \perp BF)$

$\Rightarrow k + 2 = 5 - h$

$\Rightarrow k = 3 - h$ .....(i)

$m_{AD} = \frac{1+2}{6-5} = 3$

$m_{BC} = \frac{k-3}{h-8} = \frac{1}{3} (AD \perp BC)$

$\Rightarrow 3k - 9 = 8 - h$ .....(ii)

From (i) & (ii)

$h = -4, k = 7$

Now, $h^2 + k^2 = 16 + 49 = 65$

$h^2 + k^2 - 65 = 0$

Locus is $x^2 + y^2 - 65 = 0$

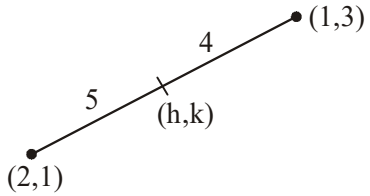
37. If the locus of the point, whose distances from the point (2, 1) and (1, 3) are in the ratio 5 : 4, is $ax^2 + by^2 + cxy + dx + ey + 170 = 0$, then the value of $a^2 + 2b + 3c + 4d + e$ is equal to : **[JEE(Main)April,2024]**

यदि उस बिन्दु, जिसकी बिन्दुओं (2, 1) और (1, 3) से दूरियों का अनुपात 5 : 4 है, का बिन्दुपथ $ax^2 + by^2 + cxy + dx + ey + 170 = 0$ है, तो $a^2 + 2b + 3c + 4d + e$ का मान बराबर है : **[JEE(Main)April,2024]**

(A) –27 (B) 37 (C) 5 (D) 437

Ans. B

Sol.



$$\therefore \frac{(h-2)^2+(k-1)^2}{(h-1)^2+(k-3)^2}=\frac{25}{16}$$

$$\frac{h^2+k^2-4h-2k+5}{h^2+k^2-2h-6k+10}=\frac{25}{16}$$

Replacing $h \to x$ and $k \to y$.

$16x^2 + 16y^2 - 64x - 32y + 80$

$= 25x^2 + 25y^2 - 50x - 150y + 250$

$9x^2 + 9y^2 + 14x - 118y + 170 = 0$

$\Rightarrow a = 9, b = 9, c = 0, d = 14, e = -118$

$a^2 + 2b + 3c + 4d + e$

$81 + 18 + 56 - 118$

$\Rightarrow 37$

38. The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $\left(2, -\frac{4}{3}\right)$ divides the third side BC internally in the ratio 2 : 1. The equation of the side BC is :

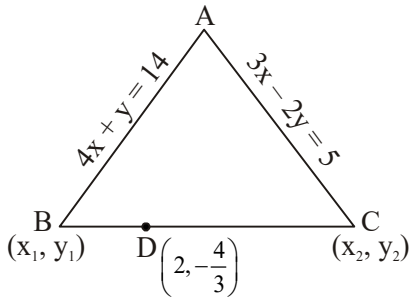
[JEE(Main)April,2024]

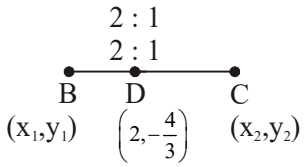
त्रिभुज ABC के दो भुजाओं AB तथा AC का समीकरण क्रमशः $4x + y = 14$ और $3x - 2y = 5$ है। यदि बिन्दु $\left(2, -\frac{4}{3}\right)$ तीसरी भुजा BC को 2 : 1 में अंतः विभाजित करता है, तो BC का समीकरण होगा : **[JEE(Main)April,2024]**

(A) $x + 3y + 2 = 0$ (B) $x + 6y + 6 = 0$ (C) $x - 6y - 10 = 0$ (D) $x - 3y - 6 = 0$

Ans. A

Sol.





$$2 = \frac{2x_2 + x_1}{3}, \frac{-4}{3} = \frac{2y_2 + y_1}{3}$$

$2x_2 + x_1 = 6,\ 2y_2 + y_1 = -4$

$x_1 = 6 - 2x_2$ ...(1)

$y_1 = -4 - 2y_2$ ...(2)

$4x_1 + y_1 = 14$ ...(3)

$3x_2 - 2y_2 = 15$ ...(4)

From here, $x_2 = 1,\ y_2 = -1,\ x_1 = 4,\ y_1 = -2$

B(4, –2) C(1, –1)

$$y + 2 = \frac{-1+2}{1-4}(x-4)$$

$-3y - 6 = x - 4$

$x + 3y + 2 = 0$

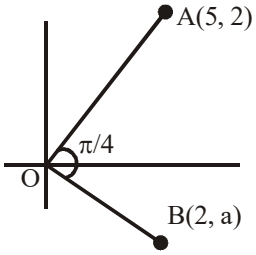
39. If the line segment joining the points (5,2) and (2,a) subtends an angle $\frac{\pi}{4}$ at the origin, then the absolute value of the product of all possible values of a is : **[JEE(Main)April,2024]**

यदि बिन्दुओं (5,2) तथा (2,a) को मिलाने वाला रेखाखंड, मूलबिन्दु पर $\frac{\pi}{4}$ का कोण बनाता है, तो a के सभी संभव मानों के गुणनफल का निरपेक्ष मान है : **[JEE(Main)April,2024]**

(A) 2 (B) 8 (C) 4 (D) 6

Ans. C

Sol.



$M_{OA} = \frac{2}{5}$

$M_{OB} = \frac{a}{2}$

$$= \tan\frac{\pi}{4} = \left|\frac{\frac{2}{5} - \frac{a}{2}}{1 + \frac{2}{5}\cdot\frac{a}{2}}\right|$$

$$\Rightarrow 1 = \left|\frac{4 - 5a}{10 + 2a}\right| \Rightarrow a = \frac{-6}{7}, \frac{14}{3}$$

Product $\Rightarrow -4$, Abs. value = 4

40. A ray of light coming from the point P(1, 2) gets reflected from the point Q on the x-axis and then passes through the point R(4, 3). If the point S(h, k) is such that PQRS is a parallelogram, then hk^2 is equal to :

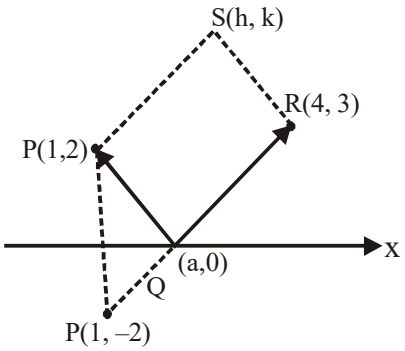
[JEE(Main)April,2024]

बिन्दु P(1, 2) से आने वाली रोशनी की एक किरण x-अक्ष पर बिन्दु Q से परावर्तित होने के पश्चात् बिन्दु R(4, 3) से गुजरती है। यदि बिन्दु S(h, k) इस प्रकार है कि PQRS एक समांतर चतुर्भुज है, तो hk^2 बराबर है : **[JEE(Main)April,2024]**

(A) 80 (B) 90 (C) 60 (D) 70

Ans. D

Sol.



$$P'R : y + 2 = \frac{5}{3}(x - 1)$$

For Point Q $\Rightarrow y = 0$

$$\frac{6}{5} = a - 1 \Rightarrow a = \frac{11}{5}$$

Now, PQRS is parallelogram

$\therefore \frac{h+a}{2}=\frac{4+1}{2} \Rightarrow h=5-\frac{11}{5}=\frac{14}{5}$

and $\frac{2+3}{2}=\frac{k}{2} \Rightarrow K=5$

Now $hk^2=25\times\frac{14}{5}=14\times5=70$

41. Two vertices of a triangle ABC are A(3,–1) and B(–2,3), and its orthocentre is P(1,1). If the coordinates of the point C are (α, β) and the centre of the circle circumscribing the triangle PAB is (h, k), then the value of $(\alpha+\beta)+2(h+k)$ equals : **[JEE(Main)April,2024]**

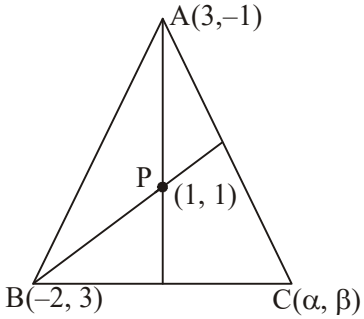
किसी त्रिभुज ABC के दो शीर्ष A(3,–1) और B(–2,3) हैं तथा उसका लंब केन्द्र P(1,1) है। यदि बिन्दु C के निर्देशांक (α, β) हैं तथा त्रिभुज PAB के परिवृत्त का केन्द्र (h, k) है, तब $(\alpha+\beta)+2(h+k)$ का मान बराबर है :

[JEE(Main)April,2024]

(A) 5 (B) 51 (C) 15 (D) 81

Ans. A

Sol.



Equation of PC : $y-1=\frac{5}{4}(x-1)\ \left(\because m_{AB}=-\frac{4}{5}\right)$

$4y-4=5x-5$

$5x-4y=1$ _____(1)

Equation of BC : $y-3=1(x+2)\ (\because m_{AP}=-1)$

$x-y=-5$ _____(2)

Solving equation (1) and (2), $x=21, y=26$

$\therefore \alpha=21, \beta=26 \Rightarrow (\alpha+\beta)=47$

Now, equation of $\perp$ bisector of AP : $y-0=1(x-2)$

$x-y=2$ _____(3)

Equation of $\perp$ bisector of AB : $y-1=\frac{5}{4}\left(x-\frac{1}{2}\right)$ _____(4)

Solving equation (3) and (4),

$x = -\frac{19}{2}, y = -\frac{23}{2}$

i.e $2h = -19, 2k = -23$

$\therefore\ 2(h + k) = -42$

So, $(\alpha + \beta) + 2(h + k) = 47 - 42 = 5$

42. Consider a triangle ABC having the vertices A(1,2), B(α, β) and C(γ, δ) and angles $\angle ABC = \frac{\pi}{6}$ and $\angle BAC = \frac{2\pi}{3}$. If the points B and C lie on the line $y = x + 4$, then $\alpha^2 + \gamma^2$ is equal to _____.

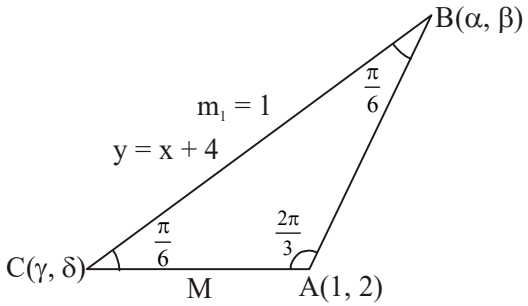
[JEE(Main)April,2024]

एक त्रिभुज ABC जिसके शीर्ष A(1,2), B(α, β) तथा C(γ, δ) और कोण $\angle ABC = \frac{\pi}{6}$ तथा $\angle BAC = \frac{2\pi}{3}$ हैं पर विचार कीजिए। यदि बिन्दु B तथा C रेखा $y = x + 4$ पर स्थित हैं, तो $\alpha^2 + \gamma^2$ बराबर है _____

[JEE(Main)April,2024]

Ans. 14

Sol.



Equation of line passing through P(1, 2) and making angle $\frac{\pi}{6}$ with $y = x + 4$ _____(1)

$y - 2 = m(x - 1)$

$y - 2 = \left(2 \pm \sqrt{3}\right)(x - 1)$ _____(2)

$\tan\frac{\pi}{6} = \left|\frac{m - m_1}{1 + mm_1}\right| \pm \frac{1}{\sqrt{3}} = \frac{m - 1}{1 + m}$

$m = 2 \pm \sqrt{3}$

Solve (1) and (2)

$x + 4 - 2 = \left(2 \pm \sqrt{3}\right)(x - 1)$

$x = \frac{4 + \sqrt{3}}{1 + \sqrt{3}} = \alpha$ and $x = \frac{4 - \sqrt{3}}{1 - \sqrt{3}} = \gamma$

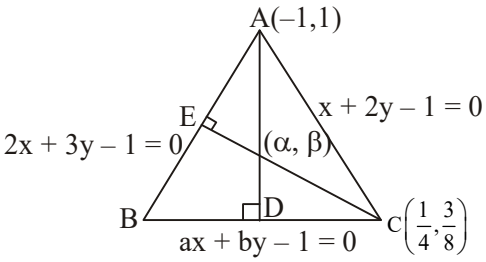
$\therefore\ \alpha^2 + \gamma^2 = 14$

43. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are (3, 4) and (–6, –8), then the value of $|a - b|$ is _____. **[JEE(Main)April,2024]**

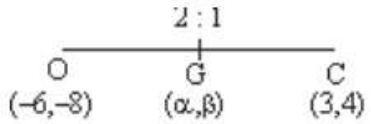
यदि रेखाओं $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ और $ax + by - 1 = 0$ द्वारा बनाया गया त्रिभुज, जिसका लंब केन्द्र किसी अन्य त्रिभुज के केन्द्रक के बराबर है एवं जिसका परिकेन्द्र और लंब केन्द्र क्रमशः (3, 4) तथा (–6, –8) है, तो $|a - b|$ का मान होगा _____ **[JEE(Main)April,2024]**

Ans. 16

Sol.



For another Δ



$$\alpha = \frac{2\times 3 - 6}{3} = 0 \qquad \beta = \frac{2\times 4 - 8}{3} = 0$$

$$(\alpha, \beta) = (0, 0)$$

$$x + 2y - 1 = 0$$
$$2x + 3y - 1 = 0$$
$$\underline{- \; - \quad +}$$
$$y - 1 = 0$$

$y = 1$

$x = -1$

A(–1, 1)

$$\text{Slope of AD} = \frac{0-1}{0+1} = -1$$

and

$M_{AD} \times M_{BC} = -1$

$(-1)\, M_{BC} = -1$

$M_{BC} = 1$

Now for point C

find equation of CE

$M_{CE} = \frac{2}{3}$

$y = \frac{2}{3}x$

and equation of AC

$$\begin{array}{r} x + 2y - 1 = 0 \\ -3x + 2y = 0 \\ \hline 4x - 1 = 0 \end{array}$$

$x = \frac{1}{4}$

$y = \frac{3}{2} \times \frac{1}{4} = \frac{3}{8}$

Eq. of BC

$C\left(\frac{1}{4}, \frac{3}{8}\right)$

$m_{BC} = 1$

$y - \frac{3}{8} = 1\left(x - \frac{1}{4}\right)$

$\frac{8y-3}{8} = \frac{4x-1}{4}$

$8y - 8x + 2 - 3 = 0$

$-8x + 8y - 1 = 0$

$ax + by - 1 = 0$

$a = -8$

$b = 8$

Value of $|a - b| = |-8 - 8| = 16$

44. Let a ray of light passing through the point (3,10) reflects on the line $2x + y = 6$ and the reflected ray passes through the point (7,2). If the equation of the incident ray is $ax + by + 1 = 0$, then $a^2 + b^2 + 3ab$ is equal to _____.

[JEE(Main)April,2024]

माना बिन्दु (3,10) से गुजरने वाली प्रकाश की कोई किरण रेखा $2x + y = 6$ पर परावर्तित होती है। तथा परावर्तित होती है तो परावर्तित किरण बिन्दु (7,2) से गुजरती है। यदि आपतित किरण समीकरण $ax + by + 1 = 0$ है, तो $a^2 + b^2 + 3ab$ बराबर है।

[JEE(Main)April,2024]

Ans. 1

Sol. $\frac{x-3}{2}=\frac{y-10}{1}=\frac{-2(6+10-6)}{4+1}=-4$

$x=-5,\ y=6$

$A'(-5,6),\ B(7,2)$

equation A'B $\quad y-6=\frac{2-6}{7+5}(x+5)$

$3y-18=-x-5$

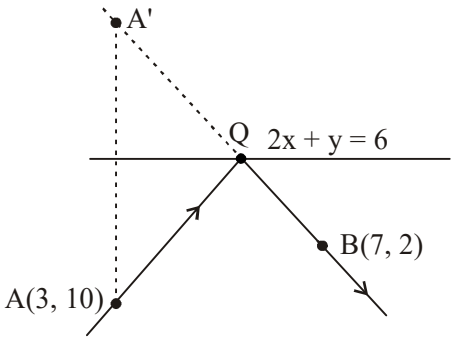
$x+3y=13$

$2x+6y=26$

$2x+y=6$ (given line)

$5y=20, \quad y=4,\ x=1$

$Q=(1,4)$



Required line $\quad y-10=\frac{4-10}{1-3}(x-3)$

$y-10=3x-9$

$3x-y+1=0$

$ax+by+1=0$

$a=3,\ b=-1$

$a^2+b^2+3ab=9+1-9=1$

Exercise J - A

1. For $a > b > c > 0$, the distance between (1, 1) and the point pf intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then : **[JEE(Adv.)2013]**

$a > b > c > 0$ के लिए, (1, 1) तथा रेखाओं $ax + by + c = 0$ व $bx + ay + c = 0$ के प्रतिच्छेद बिन्दु के बीच की दूरी $2\sqrt{2}$ से कम है, तब : **[JEE(Adv.)2013]**

(A) $a + b - c > 0$ (B) $a - b + c < 0$ (C) $a - b + c > 0$ (D) $a + b - c < 0$

Ans. AC

Sol. Point of intersection of both lines is

$$\left(-\frac{c}{(a+b)}, -\frac{c}{(a+b)}\right)$$

Distance between $\left(-\frac{c}{(a+b)}, -\frac{c}{(a+b)}\right)$ & (1,1) is

$$\text{Distance} = \sqrt{\frac{(a+b+c)^2}{(a+b)^2} \times 2} < 2\sqrt{2}$$

$a + b + c < 2(a + b)$

$a + b - c > 0$

According to given condition option (C) also correct.

2. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4$, is _____. **[JEE(Adv.)2014]**

समतल में स्थित किसी बिन्दु P से रेखाओं $x - y = 0$ तथा $x + y = 0$ की दूरी क्रमशः $d_1(P)$ तथा $d_2(P)$ है। यदि क्षेत्र R उन सभी बिन्दुओं P से बना है जो प्रथम चतुर्थांश (quadrant) में स्थित हैं तथा $2 \le d_1(P) + d_2(P) \le 4$ को सन्तुष्ट करते हैं, तब क्षेत्र R का क्षेत्रफल है _____ **[JEE(Adv.)2014]**

Ans. 6

Sol. let p(h, k)

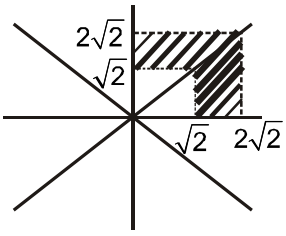
$$2 \le \left|\frac{h-k}{\sqrt{2}}\right| + \left|\frac{h+k}{\sqrt{2}}\right| \le 4$$

$\Rightarrow \quad 2\sqrt{2} \le |h - k| + |h + k| \le 4\sqrt{2}$

If $h \ge k$

$\Rightarrow \quad 2\sqrt{2} \le x - y + x + y \le 4\sqrt{2}$ or $\sqrt{2} \le x \le 2\sqrt{2}$

similarly when $k > h$



we have $\sqrt{2} \le y \le 2\sqrt{2}$

The required area = $(2\sqrt{2})^2 - (\sqrt{2})^2 = 6$.

3. Consider the lines L_1 and L_2

defined by $L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S. where the distance between R and S is $\sqrt{270}$. Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'. The value of λ^2 is_____. **[JEE(Adv.)2021]**

माना रेखाऐं L_1 तथा L_2

$L_1 : x\sqrt{2} + y - 1 = 0$ व $L_2 : x\sqrt{2} - y + 1 = 0$ द्वारा परिभाषित हैं। एक निश्चित अचर λ के लिए, माना C एक बिन्दु P का बिन्दुपथ है जबकि P की L_1 से दूरी तथा P की L_2 से दूरी का गुणनफल λ^2 है। रेखा $y = 2x + 1$, C से दो बिन्दुओं R व S पर मिलती है, जहाँ R व S के मध्य दूरी $\sqrt{270}$ है। माना RS का लम्ब समद्विभाजक C से दो भिन्न बिन्दुओं R' व S' पर मिलता है। माना R' and S' के मध्य दूरी का वर्ग D है। λ^2 मा मान _____है। **[JEE(Adv.)2021]**

Ans. 9

Sol. Locus C,

$$\left|\frac{x\sqrt{2} + y - 1}{\sqrt{3}}\right|\left|\frac{x\sqrt{2} - y + 1}{\sqrt{3}}\right| = \lambda^2$$

$\Rightarrow |2x^2 - (y - 1)^2| = 3\lambda^2$

C cuts $y - 1 = 2x$ at $R(x_1, y_1)$ and $S(x_2, y_2)$

So, $|2x^2 - (2x)^2| = 3\lambda^2$

$$\Rightarrow x = \pm\lambda\sqrt{\frac{3}{2}}$$

$$\therefore R\left(\lambda\sqrt{\frac{3}{2}}, 1 + \lambda\sqrt{6}\right) \text{ and } S\left(-\lambda\sqrt{\frac{3}{2}}, 1 - \lambda\sqrt{6}\right)$$

Distance between R and S is $\sqrt{270}$

$$\Rightarrow \sqrt{\left(\lambda\sqrt{6}\right)^2 + \left(2\lambda\sqrt{6}\right)^2} = \sqrt{270}$$

$$\therefore \lambda^2 = 9$$

4. Consider the lines L_1 and L_2 defined by

$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S. where the distance between R and S is $\sqrt{270}$. Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'. The value of D is_____. **[JEE(Adv.)2021]**

माना रेखाऐं L_1 तथा L_2

$L_1 : x\sqrt{2} + y - 1 = 0$ व $L_2 : x\sqrt{2} - y + 1 = 0$ द्वारा परिभाषित हैं।

एक निश्चित अचर λ के लिए, माना C एक बिन्दु P का बिन्दुपथ है जबकि P की L_1 से दूरी तथा P की L_2 से दूरी का गुणनफल λ^2 है। रेखा $y = 2x + 1$, C से दो बिन्दुओं R व S पर मिलती है, जहाँ R व S के मध्य दूरी $\sqrt{270}$ है। माना RS का लम्ब समद्विभाजक C से दो भिन्न बिन्दुओं R' व S' पर मिलता है। माना R' तथा S' के मध्य दूरी का वर्ग D है। D का मान ____है। **[JEE(Adv.)2021]**

Ans. $\frac{540}{7}$

Sol. Locus C,

$$\left|\frac{x\sqrt{2}+y-1}{\sqrt{3}}\right|\left|\frac{x\sqrt{2}-y+1}{\sqrt{3}}\right| = \lambda^2$$

$\Rightarrow |2x^2 - (y-1)^2| = 3\lambda^2$

C cuts $y - 1 = 2x$ at $R(x_1, y_1)$ and $S(x_2, y_2)$

So, $|2x^2 - (2x)^2| = 3\lambda^2$

$$\Rightarrow x = \pm\lambda\sqrt{\frac{3}{2}}$$

$$\therefore R\left(\lambda\sqrt{\frac{3}{2}}, 1+\lambda\sqrt{6}\right) \text{ and } S\left(-\lambda\sqrt{\frac{3}{2}}, 1-\lambda\sqrt{6}\right)$$

Distance between R and S is $\sqrt{270}$

$$\Rightarrow \sqrt{\left(\lambda\sqrt{6}\right)^2 + \left(2\lambda\sqrt{6}\right)^2} = \sqrt{270}$$

$$\therefore \lambda^2 = 9$$

Now, the curve C is

$$\Rightarrow \left|2x^2-(y-1)^2\right|=27 \qquad ...(1)$$

Mid-point of RS is (0,1) and perpendicular bisector is

$$y=-\frac{1}{2}x+1 \qquad ...(2)$$

From (1) and (2), we have

$$\frac{7x^2}{2}=27 \Rightarrow x=\pm 6\sqrt{\frac{3}{7}}$$

$$\therefore R\left(6\sqrt{\frac{3}{7}},1-3\sqrt{\frac{3}{7}}\right) \text{ and } S\left(-6\sqrt{\frac{3}{7}},1+3\sqrt{\frac{3}{7}}\right)$$

$$D=(R'S')^2=\left(12\sqrt{\frac{3}{7}}\right)^2+\left(6\sqrt{\frac{3}{7}}\right)^2$$

$$\Rightarrow D=\frac{3}{7}\times 180=\frac{540}{7}$$

$$\therefore D\approx 77.14$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. The orthocentre of the triangle ABC is 'B' and the circumcentre is 'S' (a, b). If A is the origin then the co–ordinates of C are :

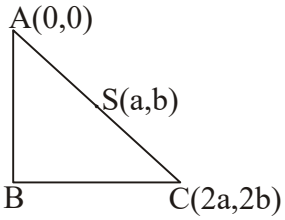
(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) none

त्रिभुज ABC का लम्बकेन्द्र 'B' है तथा परिकेन्द्र 'S' (a, b) है। यदि A मूल बिन्दु हो, तो शीर्ष C के निर्देशांक हैं :

(A) (2a, 2b) (B) $\left(\frac{a}{2}, \frac{b}{2}\right)$ (C) $\left(\sqrt{a^2 + b^2}, 0\right)$ (D) इनमें से कोई नहीं

Ans. A

Sol.



B is right angle

2. A triangle ABC with vertices A $(-1, 0)$, B $(-2, 3/4)$ & C $(-3, -7/6)$ has its orthocentre H. Then the orthocentre of triangle BCH will be :

(A) $(-3, -2)$ (B) $(1, 3)$ (C) $(-1, 2)$ (D) none of these

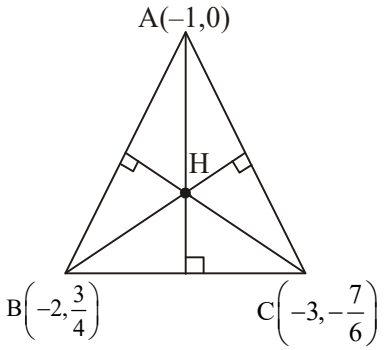
किसी त्रिभुज ABC के शीर्ष A $(-1, 0)$, B $(-2, 3/4)$ एवं C $(-3, -7/6)$ है तथा इसका लम्बकेन्द्र H है, तो त्रिभुज BCH का लम्बकेन्द्र होगा :

(A) $(-3, -2)$ (B) $(1, 3)$ (C) $(-1, 2)$ (D) इनमें से कोई नहीं

Ans. D

Sol. **Alter :**

If H is orthocentre of triangle ABC, then orthocentre of triangle BCH is point A



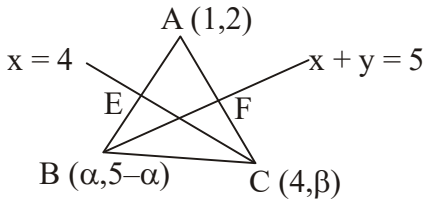
3. In a triangle ABC, co–ordinates of A are (1, 2) and the equations to the medians through B and C are $x + y = 5$ and $x = 4$ respectively. Then the co–ordinates of B and C will be :

त्रिभुज ABC में A के निर्देशांक (1, 2) है तथा B तथा C से गुजरने वाली माध्यिकाओं की समीकरण क्रमशः $x + y = 5$ एंव $x = 4$ है, तो B एवं C के निर्देशांक क्रमशः होगें :

(A) (–2, 7), (4, 3) (B) (7, –2), (4, 3) (C) (2, 7), (–4, 3) (D) (2, –7), (3, –4)

Ans. B

Sol.



$F \equiv F\left(\frac{5}{2}, \frac{\beta+2}{2}\right)$ $E \equiv \left(\frac{\alpha+1}{2}, \frac{5-\alpha+2}{2}\right)$

$\frac{5}{2} + \frac{\beta+2}{2} = 5$ $\frac{\alpha+1}{2} = 4$

$\beta + 7 = 10$ $\alpha = 7$

$\beta = 3$ $\Rightarrow E\left(\frac{5}{2}, \frac{5}{2}\right)$ & C (4,3) B(7, –2)

Alter :

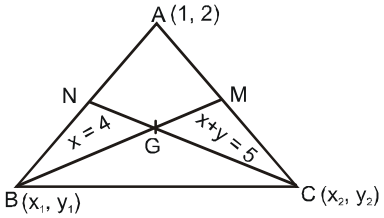
$x_1 + y_1 = 5$ _____(i)

$x_2 = 4$ _____(ii)

co - ordinates of G are ≡ (4, 1)

$\Rightarrow \quad \frac{1 + x_1 + x_2}{3} = 4$ _____(iii)

and $\frac{y_1 + y_2 + 2}{3} = 1$ _____(iv)



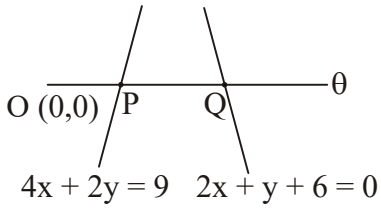
4. A straight line through the origin O meets the parallel lines $4x + 2y = 9$ and $2x + y + 6 = 0$ at points P and Q respectively. Then the point O divides the segment PQ in the ratio :

मूलबिन्दु O से जाने वाली एक सरल रेखा, दो समान्तर रेखाओं $4x + 2y = 9$ और $2x + y + 6 = 0$ से क्रमशः P और Q पर मिलती है, तो बिन्दु O, रेखाखण्ड PQ को किस अनुपात में विभाजित करेगा :

(A) 1 : 2 (B) 3 : 4 (C) 2 : 1 (D) 4 : 3

Ans. B

Sol.



Let $\frac{OP}{OQ} = \frac{\lambda}{1}$

$OP = \lambda k$

$OQ = k$

$P(\lambda k \cos\theta, \lambda k \sin\theta) \Rightarrow 4x + 2y = 9$

$Q(k\cos\theta, k\sin\theta) \Rightarrow 2x + y + 6 = 0$

$$2\lambda k = \frac{9}{2\cos\theta + \sin\theta}$$

$$k = \frac{-6}{2\cos\theta + \sin\theta}$$

$$2\lambda = -\frac{9}{6} = -\frac{3}{2}$$

$\lambda = -\frac{3}{4}$ O divides PQ in 3 : 4, internally

5. The area bounded by the curves $y = |x| - 1$ and $y = -|x| + 1$ is :

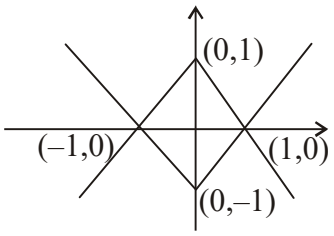
वक्रो $y = |x| - 1$ एवं $y = -|x| + 1$ से परिबद्ध क्षेत्रफल है :

(A) 1 (B) 2 (C) $2\sqrt{2}$ (D) 4

Ans. B

Sol. $y = |x| - 1$ & $y = -|x| + 1$

$$y = |x| - 1 = \begin{cases} x - 1 & x \geq 0 \\ -x - 1 & x < 0 \end{cases}$$



$$\text{Area} = \frac{1}{2}d_1d_2 = \frac{1}{2}\times 2\times 2 = 2$$

6. The set of values of 'b' for which the origin and the point (1, 1) lie on the same side of the straight line, $a^2x + aby + 1 = 0 \ \forall\ a \in R, b > 0$ are :

मूल बिन्दु एवं बिन्दु (1,1) सरल रेखा $a^2x + aby + 1 = 0 \ \forall\ a \in R, b > 0$ के एक ही ओर स्थित होने के लिए b के मानों का समुच्चय है :

(A) $b \in (2, 4)$ (B) $b \in (0, 2)$ (C) $b \in [0, 2]$ (D) $(2, \infty)$

Ans. B

Sol. $a^2x + aby + 1 = 0$ origin and (1, 1) lies on same side. $a^2 + ab + 1 > 0 \ \forall\ a \in R$

$D < 0 \Rightarrow b^2 - 4 < 0 \Rightarrow b \in (-2, 2)$ but $b > 0 \Rightarrow b \in (0, 2)$

7. The point $(a^2, a + 1)$ is a point in the angle between the lines $3x - y + 1 = 0$ and $x + 2y - 5 = 0$ containing the origin if:

(A) $a \geq 1$ or $a \leq -3$ (B) $a \in (-3, 0) \cup (1/3, 1)$

(C) $a \in (0, 1)$ (D) none of these

बिन्दु $(a^2, a + 1)$ रेखाओं $3x - y + 1 = 0$ एवं $x + 2y - 5 = 0$ के मध्य उस कोण में स्थित होगा जिसमें मूल बिन्दु स्थित है यदि :

(A) $a \geq 1$ या $a \leq -3$ (B) $a \in (-3, 0) \cup (1/3, 1)$

(C) $a \in (0, 1)$ (D) इनमें से कोई नहीं

Ans. B

Sol. Origin & $(a^2, a + 1)$ lies same side w.r.t. to given lines

$a^2 + 2a + 2 - 5 < 0 \Rightarrow a^2 + 2a - 3 < 0$

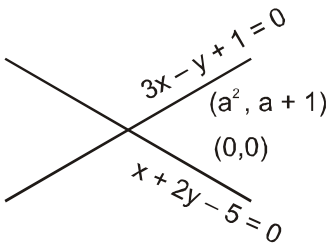
$\Rightarrow (a + 3)(a - 1) < 0$

$\Rightarrow a \in (-3, 1)$

$3a^2 - (a + 1) + 1 > 0 \Rightarrow 3a^2 - a > 0$

$\Rightarrow a(3a - 1) > 0$

$\Rightarrow a \in (-\infty, 0) \cup \left(\frac{1}{3}, \infty\right)$



take intersection we get $a \in (-3, 0) \cup \left(\frac{1}{3}, 1\right)$

8. The image of the point A (1, 2) by the line mirror $y = x$ is the point B and the image of B by the line mirror $y = 0$ is the point (α, β) then :

(A) $\alpha = 1, \beta = -2$ (B) $\alpha = 0, \beta = 0$ (C) $\alpha = 2, \beta = -1$ (D) none of these

रेखा दर्पण $y = x$ में बिन्दु A (1, 2) का प्रतिबिम्ब B है तथा रेखा दर्पण $y = 0$ में B के प्रतिबिम्ब के निर्देशांक (α, β) है, तो :

(A) $\alpha = 1, \beta = -2$ (B) $\alpha = 0, \beta = 0$ (C) $\alpha = 2, \beta = -1$ (D) इनमें से कोई नहीं

Ans. C

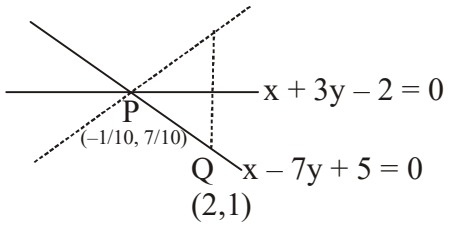
Sol. Image of A(1, 2) in line mirror $y = x$ is (2, 1); Image of B(2, 1) in $y = 0$ (x – axes) is 2, – 1)
Hence, $\alpha = 2, \beta = -1$

9. The line $x + 3y - 2 = 0$ bisects the angle between a pair of straight lines of which one has equation $x - 7y + 5 = 0$. The equation of the other line is :
(A) $3x + 3y - 1 = 0$ (B) $x - 3y + 2 = 0$ (C) $5x + 5y - 3 = 0$ (D) none

रेखा $x + 3y - 2 = 0$ एक सरल रेखा युग्म के मध्य कोण को समद्विभाजित करती हैं। यदि रेखा युग्म की एक समीकरण $x - 7y + 5 = 0$ हो, तो अन्य रेखा की समीकरण है :
(A) $3x + 3y - 1 = 0$ (B) $x - 3y + 2 = 0$ (C) $5x + 5y - 3 = 0$ (D) इनमें से कोई नहीं

Ans. C

Sol.



$$P \equiv \left(\frac{-1}{10}, \frac{7}{10}\right)$$

$$Q' \to \frac{x-2}{1} = \frac{y-1}{3} = -2\left(\frac{2+3-2}{1+9}\right) = -\frac{3}{5} \qquad (Q' \equiv \text{image of } Q)$$

$$x = \frac{7}{5}, y = -\frac{4}{5}$$

Equation of line PQ′

$$\left(y - \frac{7}{10}\right) = \left(\frac{\frac{-4}{5} - \frac{7}{10}}{\frac{7}{5} + \frac{1}{10}}\right)\left(x + \frac{1}{10}\right)$$

$5x + 5y - 3 = 0$

10. On the portion of the straight line, $x + 2y = 4$ intercepted between the axes, a square is constructed on the side of the line away from the origin. Then the point of intersection of its diagonals has co–ordinates :
(A) (2, 3) (B) (3, 2) (C) (3, 3) (D) none

सरल रेखा $x + 2y = 4$ द्वारा अक्षों के मध्य काटे गये अन्तःखण्ड को एक भुजा लेकर एक वर्ग, मूल बिन्दु के विपरीत ओर, बनाया जाता है, तो इसके विकर्णों के प्रतिच्छेद बिन्दु के निर्देशांक होगें :

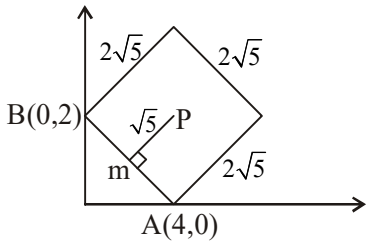
(A) (2, 3) (B) (3, 2) (C) (3, 3) (D) इनमें से कोई नहीं

Ans. C

Sol. $AB = \sqrt{16+4} = 2\sqrt{5}$

M(2,1)

Slope of PM = $\tan\theta = 2$



$\sin\theta = \frac{2}{\sqrt{5}}$

$\cos\theta = \frac{1}{\sqrt{5}}$

$r = \sqrt{5}$

$P(2 + r\cos\theta, 1 + r\sin\theta)$

$= P(3,3)$

11. AB is a variable line sliding between the coordinate axes is such a way that a lies on x–axis and B lies on y–axis. If P is a variable point on AB such that PA = b, PB = a and AB = a + b, then equation of locus of P is :

(A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $x^2 + y^2 = a^2 + b^2$ (D) none of these

AB एक चर रेखा है जो निर्देशी अक्षों के बीच इस प्रकार फिसल रही है कि A, x–अक्ष पर स्थित है और B, y–अक्ष पर स्थित है। यदि AB पर एक चर बिन्दु P इस प्रकार है कि PA = b, PB = a और AB = a + b, तो P के बिन्दुपथ का समीकरण है :

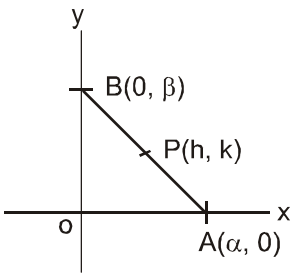
(A) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ (B) $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ (C) $x^2 + y^2 = a^2 + b^2$ (D) इनमें से कोई नही

Ans. A

Sol. **M : 1 :-**By geometry

$\alpha^2 + \beta^2 = (a + b)^2$ ....(i)

By section formula



$h = \frac{\alpha a}{a+b} \Rightarrow \alpha = \frac{h(a+b)}{a}$

$k = \frac{\beta b}{a+b} \Rightarrow \beta = \frac{k(a+b)}{b}$

Put value of α and β in (i)

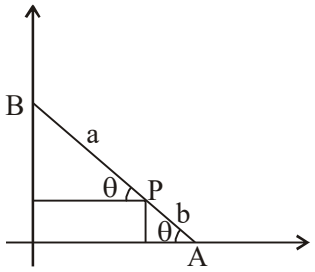
$$\frac{h^2(a+b)^2}{a^2}+\frac{k^2(a+b)^2}{b^2}=(a+b)^2 \quad \Rightarrow \quad \frac{h^2}{a^2}+\frac{k^2}{b^2}=1$$

Locus of P is $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$

M : 2 :- $P(h,k)=(a\cos\theta, b\sin\theta)$

$$\frac{h}{a}=\cos\theta$$

$$\frac{k}{b}=\sin\theta$$



$$\frac{h^2}{a^2}+\frac{k^2}{b^2}=1$$

$$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$$

12. Equations of the line pair through the origin and perpendicular to the line pair $xy-3y^2+y-2x+10=0$ is :

रेखा युग्म $xy-3y^2+y-2x+10=0$ के लम्बवत् एवं मूल बिन्दु से गुजरने वाले रेखा युग्म का समीकरण है :

(A) $xy-3y^2=0$ (B) $xy+3x^2=0$ (C) $xy+3y^2=0$ (D) $x^2-y^2=0$

Ans. B

Sol. Let equations of lines represented by the line pair $xy-3y^2+y-2x+10=0$ are

$y+c_1=0,\ x-3y+c_2=0$

lines $\perp$ to these lines and passing through origin are

$x=0,\ y=-3x$

Joint equation

$x(3x+y)=0 \ \Rightarrow \ xy+3x^2=0$

13. If the straight lines joining the origin and the points of intersection of the curve $5x^2+12xy-6y^2+4x-2y+3=0$ and $x+ky-1=0$ are equally inclined to the x–axis then the value of k is equal to :

वक्र $5x^2+12xy-6y^2+4x-2y+3=0$ एवं रेखा $x+ky-1=0$ के प्रतिच्छेद बिन्दुओं को मूल बिन्दु से मिलाने वाली सरल रेखाएँ x–अक्ष से समान कोण पर झुकी हुई हो, तो k का मान है :

(A) 1 (B) -1 (C) 2 (D) 3

Ans. B

Sol. Homogenize $5x^2+12xy-6y^2+4x-2y+3=0$ by $x+ky=1$

$$5x^2 + 12xy - 6y^2 + 4x(x + ky) - 2y(x + ky) + 3(x + ky)^2 = 0$$

it is equally indined with x-axes hence coeff. $xy = 0 \Rightarrow 12 + 4k - 2 + 6k = 0 \Rightarrow k = -1$

14. Area of the parallelogram formed by the lines $y = mx$, $y = mx + 1$, $y = nx$ and $y = nx + 1$ equals :

रेखाओं $y = mx$, $y = mx + 1$, $y = nx$ और $y = nx + 1$ से बनने वाले समान्तर चतुर्भुज का क्षेत्रफल है :

(A) $\frac{|m+n|}{(m-n)^2}$ (B) $\frac{2}{|m+n|}$ (C) $\frac{1}{|m+n|}$ (D) $\frac{1}{|m-n|}$

Ans. D

Sol. Area of ||gram $= \left|\frac{1}{m-n}\right|$

15. The area bounded by the angle bisectors of the lines $x^2 - y^2 + 2y = 1$ and the line $x + y = 3$, is :

रेखा युग्म $x^2 - y^2 + 2y = 1$ के कोण समद्विभाजको एवं रेखा $x + y = 3$ से निर्मित त्रिभुज का क्षेत्रफल है :

(A) 2 (B) 3 (C) 4 (D) 6

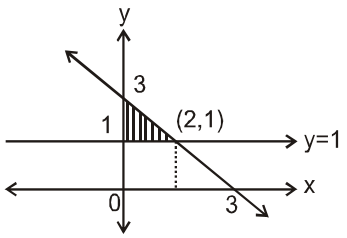
Ans. A

Sol. $x^2 - y^2 + 2y = 1 \quad \Rightarrow \quad x = \pm (y - 1)$

Bisector of above lines are $x = 0$, $y = 1$

so Area between $x = 0$, $y = 1$ and $x + y = 3$

$= \frac{1}{2} \times 2 \times 2 = 2$ sq. units



16. If L is the line whose equation is $ax + by = c$. Let M be the reflection of L through the y–axis, and let N be the reflection of L through the x–axis. Which of the following must be true about M and N for all choices of a, b and c :

(A) The x–intercept of M and N are equal (B) The y–intercepts of M and N are equal

(C) The slopes of M and N are equal (D) The slopes of M and N are reciprocal

यदि रेखा L जिसका समीकरण $ax + by = c$ है। माना L का y–अक्ष में प्रतिबिम्ब M तथा माना L का x–अक्ष में प्रतिबिम्ब N है, सभी a, b तथा c के लिए M तथा N के लिए निम्न में से कौनसा सत्य है?

(A) M तथा N का x–काट बराबर है (B) M तथा N का y–काट बराबर है

(C) M तथा N की प्रवणता बराबर है (D) M तथा N की प्रवणता व्युत्क्रमानुपाती है

Ans. C

Sol. L : $ax + by = c$

M : $-ax + by = c$

N : $ax - by = c$

M is parallel to N.

17. The complete set of real values of 'a' such that the point P(a, sin a) lies inside the triangle formed by the line $x - 2y + 2 = 0$, $x + y = 0$ and $x - y - \pi = 0$, is :

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$ (C) $(0, \pi)$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

'a' के सभी वास्तविक मानों का सम्पूर्ण समुच्चय जिसके लिए बिन्दु P(a, sin a), रेखाओं $x - 2y + 2 = 0$, $x + y = 0$ तथा $x - y - \pi = 0$ से बनने वाले त्रिभुज के अन्दर स्थित है, होगा :

(A) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ (B) $\left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{2\pi}{2}, 2\pi\right)$

(C) $(0, \pi)$ (D) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right)$

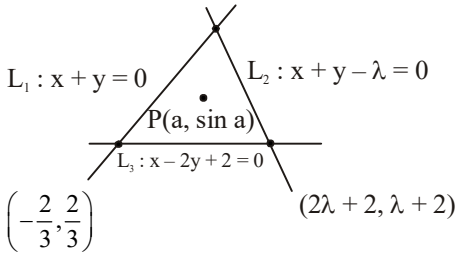
Ans. C

Sol. P(a, sin a)

$L_1 : a + \sin a > 0$ ...(1)

$L_2 : a - \sin a - \pi < 0$ ...(2)

$L_3 : a - 2 \sin a + 2 > 0$ ...(3)



checking by option

Correct answer is C.

18. A piece of cheese is located at (12, 10) in a coordinate plane. A mouse is at (4, –2) and is running up the line $y = -5x + 18$. at the point (a, b) the mouse starts getting farther from the cheese rather than closer to it. The value of $(a + b)$ is :

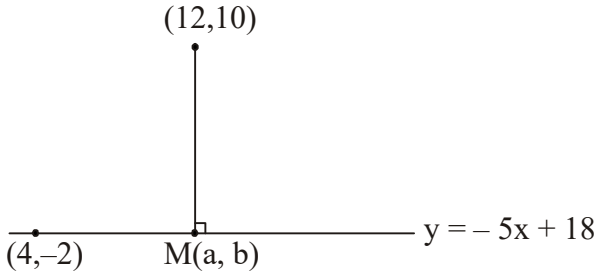
(A) 6 (B) 10 (C) 18 (D) 14

एक पनीर का टुकड़ा निर्देशांक समतल में बिन्दु (12, 10) पर स्थित है। एक चुहा बिन्दु (4, –2) पर है तथा रेखा $y = -5x + 18$ के अनुदिश दौड़ रहा है। बिन्दु (a, b) पर चुहा पनीर के नजदीक जाने की अपेक्षा दूर जाना शुरू हो जाता है, तो $(a + b)$ का मान होगा :

(A) 6 (B) 10 (C) 18 (D) 14

Ans. B

Sol.



$$\frac{a-12}{5}=\frac{b-10}{1}=-\frac{(60+10-18)}{26}=-\frac{52}{26}=-2$$

$a = 2, b = 8 \Rightarrow a + b = 10$

19. If the equation $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ represent three straight lines, two of them are perpendicular then sum of all possible values of a is equal to :

(A) $\frac{3}{8}$ (B) $\frac{-3}{4}$ (C) $\frac{1}{4}$ (D) –2

यदि समीकरण $4y^3 - 8a^2yx^2 - 3ay^2x + 8x^3 = 0$ तीन सरल रेखाओं को निरूपित करता है, जिसमें दो लम्बवत् रेखायें है, तो a के सभी सम्भावित मानों का योग होगा :

(A) $\frac{3}{8}$ (B) $\frac{-3}{4}$ (C) $\frac{1}{4}$ (D) –2

Ans. B

Sol. $4\left(\frac{y}{x}\right)^3 - a^2\left(\frac{y}{x}\right) - 3a\left(\frac{y}{a}\right)^2 + 8 = 0$

$4m^3 - 3am^2 - 8a^2m + 8 = 0$

$m_1m_2m_3 = -\frac{8}{4} = -2$ $\qquad$ $m_1m_2 = -1$

$m_3 = 2$

$\Rightarrow 32 - 12a - 16a^2 + 8 = 0$

$\Rightarrow 8 - 3a - 4a^2 + 2 = 0$

$\Rightarrow 4a^2 + 3a + 10 = 0$

$\text{sum of roots} = -\frac{3}{4}$

20. The graph of the function, $y = \cos x \cos(x+2) - \cos^2(x+1)$ is :

(A) a straight line passing through $(0, -\sin^2 1)$ with slope 2

(B) a straight line passing through $(0, 0)$

(C) a parabola with vertex $(1, -\sin^2 1)$

(D) a straight line passing through the point $\left(\frac{\pi}{2}, -\sin^2 1\right)$ and parallel to the x–axis

फलन $y = \cos x \cos(x+2) - \cos^2(x+1)$ का आरेख है :

(A) एक सरल रेखा है जो $(0, -\sin^2 1)$ से गुजरती है जिसकी प्रवणता 2 है

(B) एक सरल रेखा जो $(0, 0)$ से गुजरती है

(C) एक परवलय जिसका शीर्ष $(1, -\sin^2 1)$ है

(D) एक सरल रेखा जो $\left(\frac{\pi}{2}, -\sin^2 1\right)$ से गुजरती है तथा x–अक्ष के समान्तर है

Ans. D

Sol. $y = \cos x \cdot \cos(x+2) - \cos^2(x+1)$

$$y = \cos x \cdot \cos(x+2) - \left(\frac{1+\cos(2x+2)}{2}\right)$$

$2y = 2\cos x \cdot \cos(x+2) - 1 - \cos(2x+2)$

$2y = \cos(2x+2) + \cos 2 - 1 - \cos(2x+2)$

$2y = \cos 2 - 1$

$$y = \frac{\cos 2 - 1}{2} = -\sin^2(1)$$

Slope $= 0 \Rightarrow$ parallel to x-axis

21. P is a point inside the triangle ABC. Lines are drawn through P, parallel to the sides of the triangle. The three resulting triangles with the vertex at P have areas 4, 9 and 49 sq. units. The area of the triangle ABC is :

(A) 121 (B) 100 (C) 81 (D) 144

त्रिभुज ABC के अन्दर एक बिन्दु P है, बिन्दु P से तीनों भुजाओं के समान्तर रेखा खींची जाती है इस प्रकार से एक उभयनिष्ट शीर्ष P से तीन त्रिभुज बनती है। जिनका क्षेत्रफल 4, 9 एवं 49 है, तो ABC का क्षेत्रफल होगा :

(A) 121 (B) 100 (C) 81 (D) 144

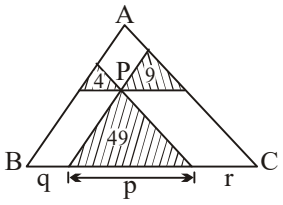
Ans. D

Sol. $$\frac{\text{ar}(PQR)}{QR^2} = \frac{\text{ar}(ABC)}{BC^2}$$

$\frac{49}{\Delta} = \frac{p^2}{a^2}$ or $\frac{7}{\sqrt{\Delta}} = \frac{p}{a}$ _____(1)

|||ly $\frac{2}{\sqrt{\Delta}} = \frac{q}{a}$ _____(2)

and $\frac{3}{\sqrt{\Delta}} = \frac{r}{a}$ _____(3)



(1) + (2) + (3) gives $\frac{12}{\sqrt{\Delta}} = \frac{p+q+r}{a} = 1 \Rightarrow \Delta = 144$

22. The complete set of values of the parameter α so that the point $P\left(\alpha, (1+\alpha^2)^{-1}\right)$ does not lie outside the triangle formed by the lines $L_1 : 15y = x + 1$, $L_2 : 78y = 118 - 23x$ and $L_3 : y + 2 = 0$, is :

यदि बिन्दु $P\left(\alpha, (1+\alpha^2)^{-1}\right)$ भुजाओं $L_1 : 15y = x + 1$, $L_2 : 78y = 118 - 23x$ एवं $L_3 : y + 2 = 0$ से बनने वाली त्रिभुज के भीतर या त्रिभुज पर स्थित है, तो प्राचल α का मान होगा :

(A) (0, 5) (B) [2, 5] (C) [1, 5] (D) [0, 2] (E) (2, 5]

Ans. B

Sol. As $P(\alpha, (1+\alpha^2)^{-1})$ lie on $y = \frac{1}{1+x^2}$

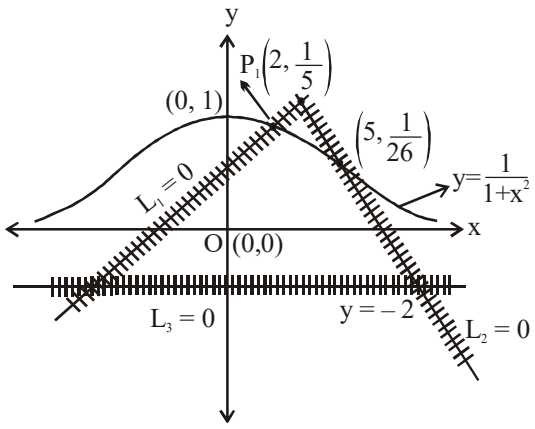
$\therefore$ On solving $y = \frac{1}{1+x^2}$ with L_1,

$\frac{15}{1+x^2} = x + 1 \Rightarrow x = 2,\ y = \frac{1}{5}$

we get $P_1\left(2, \frac{1}{5}\right)$ _____(1)

and with L_2 $\frac{78}{1+x^2} = 118 - 23x$, we get $P_2\left(5, \frac{1}{26}\right)$ _____(2)

$\therefore$ From (1) and (2), we get $2 \le \alpha \le 5$



ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

23. Line $\frac{x}{a}+\frac{y}{b}=1$ cuts the co–ordinate axes at A(a, 0) and B (0, b) and the line $\frac{x}{a'}+\frac{y}{b'}=-1$ at A′ (–a′, 0) and B′ (0, –b′) . If the points A, B, A′, B′ are concyclic then the orthocentre of the triangle ABA′ is :

रेखा $\frac{x}{a}+\frac{y}{b}=1$ तथा $\frac{x}{a'}+\frac{y}{b'}=-1$ निर्देशांक अक्ष को क्रमशः A(a, 0), B (0, b) तथा A′ (–a′, 0) एवं B′ (0, –b′) पर काटती है यदि बिन्दु A, B, A′, B′ चक्रीय हो, तो त्रिभुज ABA′ का लम्ब केन्द्र होगा :

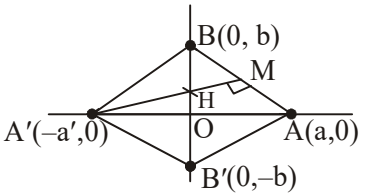
(A) (0, 0) (B) (0, b') (C) $\left(0, \frac{aa'}{b}\right)$ (D) $\left(0, \frac{bb'}{a}\right)$

Ans. BC

Sol. $\frac{x}{a}+\frac{y}{b}=1, \quad \frac{x}{a'}+\frac{y}{b'}=1$

$m_{AB}=-\frac{b}{a}$

$m_{A'M}=\frac{a}{b}$



$\frac{\beta-0}{0+a'}=\frac{a}{b}\Rightarrow\beta=\frac{aa'}{b}$

$H\left(0,\frac{aa'}{b}\right)\Rightarrow$ option (C)

ABA′B′ is concyclic then

Intersecting chord theorem

$a\times(a')=b\times(+b')$

$\Rightarrow b'=\frac{aa'}{b}$

$H\left(0,\frac{aa'}{b}\right)\equiv H(0,b')$

Aliter :

$m_{AB}=\frac{-b}{a}$

$$m_{A'H} = \frac{a}{b}$$

$$y - 0 = \frac{a}{b}(x + a')$$

$$y = \frac{a}{b}x + a\frac{a'}{b}$$

$$y - 0 = \frac{a'}{b}(x - a)$$

$$y = \frac{a'}{b}x - a\frac{a'}{b}$$

$$\text{P.O.I.} \left(0, \frac{aa'}{b}\right)$$

24. Consider the equation $y - y_1 = m(x - x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1, then :

(A) the lines will pass through a fixed point (B) there will be a set of parallel lines

(C) all the lines intersect the line $x = x_1$ (D) all the lines will be parallel to the line $y = x_1$

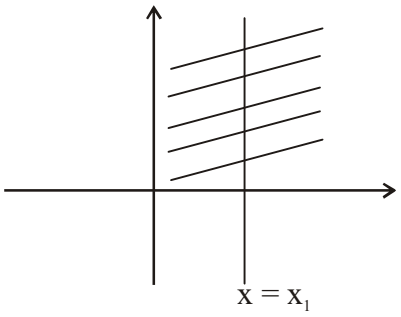
दिया हुआ रेखा का समीकरण $y - y_1 = m(x - x_1)$ है। यदि m एवं x_1 स्थिर है। तथा y_1 के अलग–अलग मान के लिये अलग–अलग रेखा खींची जाती हो, तो :

(A) सारी रेखाऐं स्थिर बिन्दु से जायेगी (B) सारी रेखाऐं समान्तर होगी

(C) सारी रेखाऐं रेखा $x = x_1$ को काटेगी (D) सारी रेखाऐं $y = x_1$ के समान्तर होगी

Ans. BC

Sol. $(y - y_1) = m(x - x_1)$



(B) All lines are parallel.

(C) All the lines intersect the line $x = x_1$.

25. If $\frac{x}{c}+\frac{y}{d}=1$ is a line through the intersection of $\frac{x}{a}+\frac{y}{b}=1$ and $\frac{x}{b}+\frac{y}{a}=1$ and the lengths of the perpendiculars drawn from the origin to these lines are equal in lengths then :

(A) $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}+\frac{1}{d^2}$ (B) $\frac{1}{a^2}-\frac{1}{b^2}=\frac{1}{c^2}-\frac{1}{d^2}$

(C) $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}+\frac{1}{d}$ (D) none

यदि रेखाओं $\frac{x}{a}+\frac{y}{b}=1$ एवं $\frac{x}{b}+\frac{y}{a}=1$ के प्रतिच्छेद बिन्दु से गुजरने वाली रेखा का समीकरण $\frac{x}{c}+\frac{y}{d}=1$ है तथा (0, 0) से इन रेखाओं पर डाले गये लम्ब की लम्बाई समान हो, तो :

(A) $\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}+\frac{1}{d^2}$ (B) $\frac{1}{a^2}-\frac{1}{b^2}=\frac{1}{c^2}-\frac{1}{d^2}$

(C) $\frac{1}{a}+\frac{1}{b}=\frac{1}{c}+\frac{1}{d}$ (D) इनमें से कोई नहीं

Ans. AC

Sol. $\left.\begin{matrix} bx+ay=ab \\ ax+by=ab \end{matrix}\right] \Rightarrow y=\frac{ab}{a+b}$

$P\left(\frac{ab}{a+b},\frac{ab}{a+b}\right)$

$x=\frac{ab}{a+b}$

$\frac{x}{c}+\frac{y}{a}=1 \Rightarrow \frac{ab}{c(a+b)}+\frac{ab}{d(a+b)}=1$

$\left(\frac{1}{c}+\frac{1}{d}\right)=\left(\frac{1}{a}+\frac{1}{b}\right) \rightarrow \text{option (C)}$

$P=\frac{1}{\sqrt{\frac{1}{c^2}+\frac{1}{d^2}}}=\frac{1}{\sqrt{\frac{1}{a^2}+\frac{1}{b^2}}}=\frac{1}{a^2}+\frac{1}{b^2}=\frac{1}{c^2}+\frac{1}{d^2} \rightarrow \text{option (A)}$

26. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2-3|x|+2=0$ and the y-coordinates of the vertices are the roots of the equation $y^2-3y+2=0$ then the possible vertices of the square is/are :

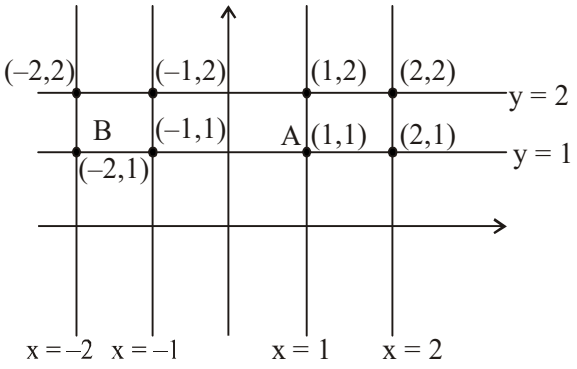
इकाई क्षेत्रफल वाले वर्ग के शीर्षो का भुज तथा कोटि समीकरण $x^2 - 3|x| + 2 = 0$ तथा $y^2 - 3y + 2 = 0$ का मूल है तो वर्ग के शीर्षो के सम्भावित निर्देशांक होगे।

(A) (1, 1), (2, 1), (2, 2), (1, 2) (B) (–1, 1), (–2, 1), (–2, 2), (–1, 2)

(C) (2, 1), (1, –1), (1, 2), (2, 2) (D) (–2, 1), (–1, –1), (–1, 2), (–2, 2)

Ans. AB

Sol. $x = \pm 1, \pm 2, y = 1, 2$



27. P(x, y) moves such that the area of the triangle formed by P, Q (a , 2a) and R (–a, – 2a) is equal to the area of the triangle formed by P, S (a, 2a) and T (2a, 3a). The locus of 'P' is a straight line given by :

यदि P (x, y) इस प्रकार से गति करता है कि ΔPQR का क्षेत्रफल ΔPST के क्षेत्रफल के बराबर है जहां Q (a , 2a) तथा R (– a, – 2a) एवं S (a, 2a) तथा T (2a, 3a) है तो P का बिन्दुपथ सरल रेखा द्वारा निरूपित होगा।

(A) $3x - y = a$ (B) $5x - 3y + a = 0$ (C) $5x - 5y + a = 0$ (D) $2y = ax$

Ans. AB

Sol. P (x, y) Q (a, 2a) & R (–a, –2a)

S (a, 2a) T (2a, 3a)

$$\begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ -a & -2a & 1 \end{vmatrix} = \pm \begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ 2a & 3a & 1 \end{vmatrix}$$

–ve +ve

$$\Rightarrow \begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ 3a & 5a & 0 \end{vmatrix} = 0 \qquad \begin{vmatrix} x & y & 1 \\ a & 2a & 1 \\ a & a & 2 \end{vmatrix} = 2$$

$\Rightarrow -5ax + 3ay - a^2 = 0$ $3ax - ay - a^2 = 0$

$\Rightarrow 5x - 3y + a = 0$ $3x - y - a = 0$

28. Let $u \equiv ax + by + a\sqrt[3]{b} = 0$ $v \equiv bx - ay + b\sqrt[3]{a} = 0$ $a, b \in R$ be two straight lines. The equation of the bisectors of the angle formed by $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ for non zero real k_1 and k_2 are :

माना कि $u \equiv ax + by + a\sqrt[3]{b} = 0$, $v \equiv bx - ay + b\sqrt[3]{a} = 0$ जहाँ $a, b \in R$ रेखाओ $k_1u - k_2v = 0$ एवं $k_1u + k_2v = 0$ के कोणों की अर्द्धक रेखा का समीकरण होगा (जहाँ $k_1, k_2 \neq 0$) :

(A) $u = 0$ (B) $k_2u + k_1v = 0$ (C) $k_2u - k_1v = 0$ (D) $v = 0$

Ans. AD

Sol. Note that the lines are perpendicular.

Assume the co-ordinate axes to be directed along $u = 0$ and $v = 0$.

Now the lines $k_1u - k_2v = 0$ and $k_1u + k_2v = 0$ are equally inclined with uv axes.

Hence the bisectors are $u = 0$ and $v = 0$. i.e. $\dfrac{k_1u - k_2u}{\sqrt{k_1^2 + k_2^2}} = \pm \dfrac{k_1u + k_2u}{\sqrt{k_1^2 + k_2^2}}$

29. The bisectors of angle between the st. lines, $y - b = \dfrac{2m}{1-m^2}(x-a)$ and $y - b = \dfrac{2m'}{1-m'^2}(x-a)$ are :

रेखाओं $y - b = \dfrac{2m}{1-m^2}(x-a)$ तथा $y - b = \dfrac{2m'}{1-m'^2}(x-a)$ के कोणों की अर्द्धक रेखा होगी।

(A) $(y-b)(m+m') + (x-a)(1-mm') = 0$ (B) $(y-b)(m+m') - (x-a)(1-mm') = 0$

(C) $(y-b)(1-mm') + (x-a)(m+m') = 0$ (D) $(y-b)(1-mm') - (x-a)(m+m') = 0$

Ans. AD

Sol. $M_1 = \tan 2\theta_1$ $M_2 = \tan 2\theta_2$ $\tan\theta_1 = M_1 \tan\theta_2 = M$

$\phi_1 = \dfrac{2\theta_1 + 2\theta_2}{2} = \theta_1 + \theta_2$ $\phi_2 = \dfrac{\pi}{2} + \theta_1 + \theta_2$

$\tan\phi_1 = \dfrac{m+m'}{1-mm'}$ $\tan\phi_2 = -\cos(\theta_1 + \theta_2)$

$= -\left(\dfrac{mm'-1}{m+m'}\right)$

$(y-b) = \dfrac{m+m'}{1-mm'}(x-a) \rightarrow (D)$

$(y-b) = \left(\dfrac{mm'-1}{m+m'}\right)(x-a) \rightarrow (A)$

30. If the vertices P, Q, R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s)?

(A) centriod (B) incentre (C) circumcentre (D) orthocentre

यदि ΔPQR के शीर्ष यदि परिमेय है, तो इनमें से कौन सा निर्देशांक हमेशा परिमेय होगा

(A) केन्द्रक (B) अन्तः केन्द्र (C) परिकेन्द्र (D) लम्बकेन्द्र

Ans. ACD

Sol. put $m = \tan\alpha$ and $m' = \tan\beta$

$(x_1, y_1), (x_2, y_2), (x_3, y_3)$ all are rational.

Hence

(A) Centroid $\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$ is rational.

(B) In centre $\left(\frac{\Sigma a x_1}{\Sigma a}, \frac{\Sigma a y_1}{\Sigma a}\right)$ can be irrational as side length can be irrational.

(C) Circumcentre will be rational as coffiecient in equations of $\perp$ bisector will be rational.

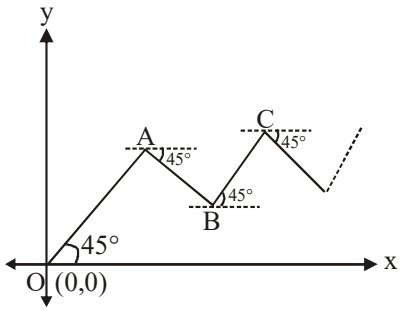
Hence coordinate of there Point of Intersection will also be rational.

(D) Similarly, Orthocentre is also rational.

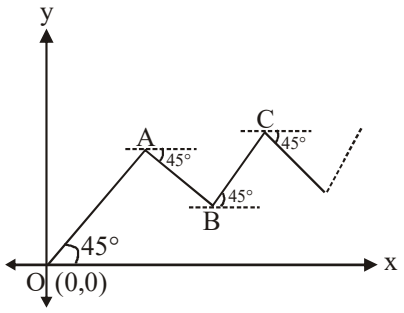
Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. Points O, A, B, C.....are shown in figure where OA = 2AB = 4BC = ... so on. Let A is the centroid of a triangle whose orthocentre and circumcenter are (2, 4) and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point O(0, 0) along the straight line in zig–zag fashion and terminates ultimately at point P(α, β) then find the value of ($\alpha + \beta$)



आरेख में प्रदर्शित बिन्दुओं O, A, B, C ..... जहाँ OA = 2AB = 4BC = ... यदि बिन्दु A, किसी त्रिभुज का केन्द्रक है एवं जिसका लम्बकेन्द्र (2, 4) एवं परिकेन्द्र $\left(\frac{7}{2}, \frac{5}{2}\right)$ पर स्थित है, यदि कोई कीट मूल बिन्दु O(0, 0) से रेखा के साथ उतार–चढ़ाव (zig–zag) में बिन्दु P तक चलता है। यदि बिन्दु P के निर्देशांक (α, β) हो तो ($\alpha + \beta$) का मान ज्ञात करो।



Ans. 8

Sol. $H(2,4) \bullet \underset{G}{\overset{2 \quad 1}{\longrightarrow}} \bullet 0\left(\frac{7}{2}, \frac{5}{2}\right)$

$$\left(\frac{7+2}{+3}, \frac{5+4}{3}\right) \equiv G(3,3)$$

$$\alpha = \frac{3\sqrt{2}}{\sqrt{2}} + \frac{3\sqrt{2}}{2\sqrt{2}} + \frac{3\sqrt{2}}{4\sqrt{2}} + ...\infty = \frac{\frac{3\sqrt{2}}{\sqrt{2}}}{1-\frac{1}{2}} = 6$$

$$\beta = \frac{3\sqrt{2}}{\sqrt{2}} - \frac{3\sqrt{2}}{2\sqrt{2}} + \frac{3\sqrt{2}}{4\sqrt{2}} .....\infty = \frac{\frac{3\sqrt{2}}{\sqrt{2}}}{\frac{3}{2}} = 2$$

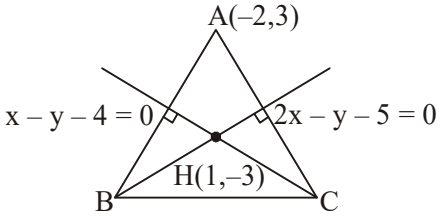
$\alpha + \beta = 6 + 2 = 8$

2. The equations of perpendiculars of the sides AB & AC of triangle ABC are $x - y - 4 = 0$ and $2x - y - 5 = 0$ respectively. If the vertex A is $(-2, 3)$ and point of intersection of perpendiculars bisectors is $\left(\frac{3}{2}, \frac{5}{2}\right)$, find the equation of medians to the sides AB & AC respectively.

त्रिभुज ABC समें भुजाओं AB एवं AC के लम्बो का समीकरण क्रमशः $x - y - 4 = 0$ एवं $2x - y - 5 = 0$ हो, यदि शीर्ष बिन्दु A के निर्देशांक $(-2, 3)$ हो, एवं $\left(\frac{3}{2}, \frac{5}{2}\right)$ भुजाओं के लम्ब अर्द्धको का प्रतिच्छेद बिन्दु हो, तो क्रमशः AB एवं AC की मध्यिकाओं के समीकरण ज्ञात करो।

Ans. $x + 4y = 4$; $5x + 2y = 8$

Sol.



$$\begin{array}{rrrrrr} x & - & y & - & 4 & = & 0 \\ 2x & - & y & - & 5 & = & 0 \\ - & & + & & + & & \\ \hline & - & x & + & 1 & = & 0 \\ \hline \end{array}$$

$x = 1 \qquad y = -3$

Eqn of $\perp$ bisector of AB and AC $\rightarrow$ AB $\perp^r$ Bisector $\rightarrow x - y + 1 = 0$

$x - y + 1 = 0$

AC $\perp^r$ Bisector

$$2x - y - \frac{1}{2} = 0$$

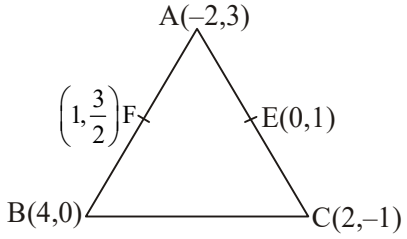
$$4x - 2y - 1 = 0$$

$B \rightarrow$ Image of $A(-2, 3)$ in $4x - 2y - 1 = 0$

$$\frac{x+2}{4} = \frac{y-3}{-2} = -2\frac{(-8-6-1)}{20} = \frac{3}{2} \Rightarrow B(4,0)$$

$C \rightarrow$ Image of $A(-2, 3)$ in $x - y + 1 = 0$

$$\frac{x+2}{1} = \frac{y-3}{(-1)} = -2\frac{(-2-3+1)}{2} = 4 \Rightarrow C(2,-1)$$



CF : $\rightarrow$ $\quad (y+1) = \left(\frac{\frac{3}{2}+1}{1-2}\right)(x-2)$

$$2y + 2 = -5x + 10$$

$$5x + 2y = 8 \text{ Ans.}$$

BE : $\rightarrow$ $\quad (y-0) = \frac{1}{-4}(x-4)$

$$-4y = x - 4$$

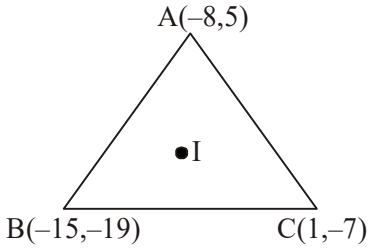
$$x + 4y = 4 \text{ Ans.}$$

3. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are A(–8, 5); B(–15, –19) and C(1, – 7) has the equation $ax + 2y + c = 0$. Find 'a' and 'c'.

यदि त्रिभुज ABC के शीर्ष बिन्दु के निर्देशांक A(–8, 5); B(–15, –19) एवं C(1, – 7) है, एवं कोण के आंतरिक कोण अर्द्धक का समीकरण $ax + 2y + c = 0$ है तो 'a' एवं 'c' का मान ज्ञात करो।

Ans. $a = 11$, $c = 78$

Sol.



$I \equiv$ Incentre

$a = BC = \sqrt{16^2 + 12^2}$

$a = 20$

$b = AC = \sqrt{9^2 + 12^2}$

$b = 15$

$C = AB = \sqrt{7^2 + 24^2}$

$C = 25$

$$I = \left(\frac{-160 - 225 + 25}{60}, \frac{100 - 285 - 175}{60} \right) \qquad \left(\because I \equiv \frac{ax_1 + by_1 + cz_1}{a+b+c}, \frac{ax_2 + by_2 + cz_2}{a+b+c} \right)$$

$I \equiv (-6, -6)$

AI eqn. : $\rightarrow \quad (y+6) = \frac{11}{-2}(x+6)$

$-2y - 12 = 11x + 66$

$11x + 2y + 78 = 0$

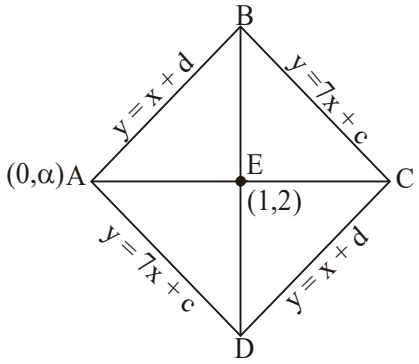
$a = 11,\ c = 78$

4. Two sides of a rhombous ABCD are parallel to the lines $y = x + 2$ & $y = 7x + 3$. If the diagonals of the rhombous intersect at the point (1, 2) & the vertex A is on the y-axis, find the possible coordinates of A.

किसी समचतुर्भुज ABCD की दो भुजाऐं, रेखाओं $y = x + 2$ एवं $y = 7x + 3$ के समान्तर हो, एवं समचतुर्भुज के विकर्णो का प्रतिच्छेद बिन्दु (1, 2) हो तथा बिन्दु A, y-अक्ष पर स्थित हो, तो बिन्दु A के सम्भावित निर्देशांक ज्ञात करो।

Ans. $(0, 0)$ or $\left(0, \frac{5}{2}\right)$

Sol.



Diagonals will be parallel to angle bisectors of line

$y = x + 2$

$y = 7x + 3$

$$\frac{x - y + 2}{\sqrt{2}} = \pm\frac{(7x - y + 3)}{5\sqrt{2}}$$

(+)	(–)
$5x - 5y + 10 = 7x - y + 3$	$5x - 5y + 10 = -7x + y - 3$
$B_1 : 2x + 4y - 7 = 0$	$B_2 : 12x - 6y + 13 = 0$
$x + 2y - \frac{7}{2} = 0$	$2x - y + \frac{13}{6} = 0$

$$m_{AE} = \frac{-2}{4} \text{ or } \frac{12}{6}$$

$$\Rightarrow \frac{2-\alpha}{1-0} = -\frac{1}{2} \text{ or } \frac{2-\alpha}{1-0} = 2$$

$$\alpha = \frac{5}{2} \text{ or } \alpha = 0$$

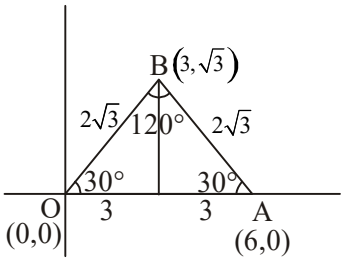
Coordinates $\left(0, \frac{5}{2}\right)$ or $(0,0)$

5. Let O(0, 0), A(6, 0) and B(3, $\sqrt{3}$) be the vertices of ΔOAB. Let R be the region consisting of all those points P inside ΔOAB which satisfy d(P, OA) ≤ minimum $\left(d(P, OB), d(P, AB)\right)$ where d(P, OA), d(P, OB) and d(P, AB) represent the distance of P from the sides OA, OB and AB respectively. If the area of region R is 9 ($a - \sqrt{b}$) where a and b are coprime, then find the value of (a + b).

यदि O(0, 0), A (6, 0) एवं B(3, $\sqrt{3}$) त्रिभुज OAB के शीर्ष बिन्दु है तथा R त्रिभुज OAB के अंदर स्थित सभी बिन्दुओं के क्षेत्र को प्रदर्शित करता है जो कि संबंध d(P, OA) ≤ न्यूनतम मान $\left(d(P,OB), d(P,AB)\right)$ को संतुष्ट करता है जहाँ d(P, OA), d(P, OB) तथा d(P, AB) बिन्दु क्रमशः भुजाओं OA, OB एवं AB दूरी को दर्शाता है। यदि क्षेत्र R का क्षेत्रफल 9 (a – $\sqrt{b}$) है जहाँ a एवं b परस्पर अभाज्य (coprime) है, तो (a + b) का मान ज्ञात कीजिये।

Ans. 5

Sol.



Let P(h, k)

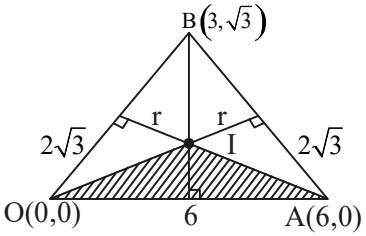
$OB \equiv \sqrt{3}y = x$

$AB = \sqrt{3}y = -x + 6$

$d(P,OA) \le \min\left\{d(P,OB), d(P,AB)\right\}$

$d(P,OA) \le d(P,OB)$

$d(P,OA) \le d(P,AB)$



$$I\left(\frac{18+0+12\sqrt{3}}{6+4\sqrt{3}}, \frac{6\sqrt{3}+0+0}{6+4\sqrt{3}}\right)$$

$$\text{Area} = R = \frac{1}{2}\times(6)\times\left(\frac{6\sqrt{3}}{6+4\sqrt{3}}\right)$$

$$= 9\left(\frac{\sqrt{3}}{3+2\sqrt{3}}\right) = 9\left(\frac{\sqrt{3}\left(2\sqrt{3}-3\right)}{3}\right)$$

$$= 9\left(2-\sqrt{3}\right)$$

a = 2, b = 3, a + b = 5

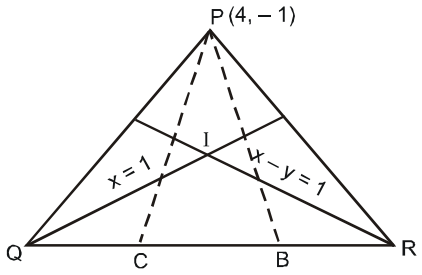
6. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines x – 1 = 0 and x – y –1 = 0 are the equations of two internal bisectors of its angles.

त्रिभुज की भुजाओं के समीकरण ज्ञात करो यदि एक शीर्ष बिन्दु के निर्देशांक (4, –1) एवं दो आन्तरिक कोर्णाद्धको के समीकरण x – 1 = 0 एवं x – y –1 = 0 है।

Ans. 2x – y + 3 = 0, 2x + y – 7 = 0, x – 2y – 6 = 0

Sol. **Method I** :

Alter :



Mirror image of a vertex of ΔABC w.r.t. it's angle bisector always lie on the side opposite to the vertex.

Reflection of (4, – 1) about x = 1 is B and about x – y = 1 is C

$B \equiv (x_1, y_1)$

$C \equiv (x_2, y_2)$

$\left(\frac{x_1+4}{2}, \frac{y_1-1}{2}\right)$ will lie on x = 1

$\Rightarrow \quad \frac{x_1+4}{2} = 1 \quad \Rightarrow \quad x_1 = -2 \qquad$ and $\qquad \frac{y_1+1}{x_1-4} = 0 \quad \Rightarrow \quad y_1 = -1$

$\therefore \quad B \equiv (-2, -1) \Rightarrow \left(\frac{x_2+4}{2}, \frac{y_2-1}{2}\right)$ will lie on x – y – 1 = 0

$x_2 + 4 - y_2 + 1 - 2 = 0 \quad \Rightarrow \quad x_2 - y_2 + 3 = 0 \quad$ _____(i)

and $\left(\frac{y_2+1}{x_2-4}\right).1 = -1 \quad \Rightarrow \quad x_2 + y_2 - 3 = 0 \quad$ _____(ii)

from (i) & (ii) $x_2 = 0, y_2 = 3$

$\therefore$ C(0, 3) $\therefore$ Equation of BC

$y + 1 = \frac{4}{2}(x + 2) \quad \Rightarrow \quad 2x - y + 3 = 0$

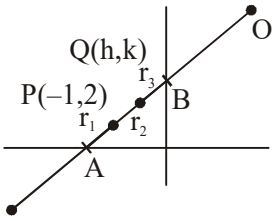
To find PQ & PR we have to find Q and R ;
solving QI & BC, we get Q (1, 5)
equation of PQ is $2x + y - 7 = 0$
For R, we solve RI & BC
$\therefore$ R (–4, – 5)
equation of PR is $x - 2y - 6 = 0$

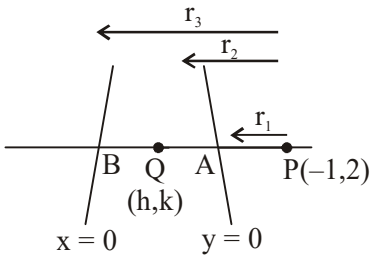
7. P is the point (–1, 2), a variable line through P cuts the x & y axes at A & B respectively Q is the point on AB such that PA, PQ, PB are H.P. Find the locus of Q.

यदि P के निर्देशांक (–1, 2) हो, एवं एक चर रेखा जो बिन्दु P से गुजरती है, तथा x एवं y अक्षों को क्रमशः A एवं B पर काटती है, एवं Q, AB पर स्थित कोई बिन्दु है, तथा PA, PQ, PB हरात्मक श्रेणी में हो तो, बिन्दु Q का बिन्दुपथ ज्ञात करो।

Ans. $y = 2x$

Sol.





$$Q(h,k) = Q(-1 + r_2\cos\theta, 2 + r_2\sin\theta)$$

$$A(-1 + r_1\cos\theta, 2 + r_1\sin\theta) \Rightarrow x - \text{axis}$$

$$r_1\sin\theta = -2$$

$$r_1 = \frac{-2}{\sin\theta}$$

$$B(-1 + r_3\cos\theta, 2 + r_3\sin\theta) \Rightarrow y - \text{axis}$$

$$-1 + r_3\cos\theta = 0$$

$$r_3 = \frac{1}{\cos\theta}$$

$$\frac{2}{r_2} = \frac{1}{r_1} + \frac{1}{r_3} = \frac{-\sin\theta}{2} + \frac{\cos\theta}{1}$$

$$2 = \frac{-r_2 \sin\theta}{2} - r_2 \cos\theta$$

$$4 = -(k-2) + 2(h+1)$$

$2h - k = 0$

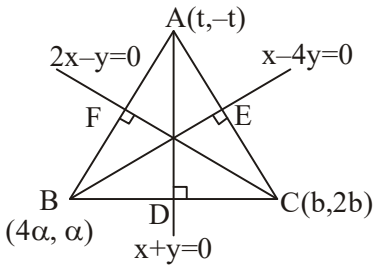
$y = 2x$

8. The equations of the altitudes AD, BE, CF of a triangle ABC are $x+y=0$, $x-4y=0$ and $2x-y=0$ respectively. The coordinates of A are $(t, -t)$. Find coordinates of B & C. Prove that if t varies the locus of the centroid of the triangle ABC is $x+5y=0$.

त्रिभुज ABC के लम्बो AD, BE, CF के समीकरण के क्रमशः $x+y=0$, $x-4y=0$ एवं $2x-y=0$ है एवं बिन्दु A के निर्देशांक $(t, -t)$ है। तो बिन्दु B एवं C के निर्देशांक ज्ञात कीजिये। दर्शाओं यदि t प्राचल हो तो त्रिभुज ABC के केन्द्रक का बिन्दुपथ $x+5y=0$ होगा।

Ans. $B\left(-\frac{2t}{3}, -\frac{t}{6}\right)$, $C\left(\frac{t}{2}, t\right)$

Sol.



$$\frac{-t-\alpha}{t-4\alpha} = \frac{-1}{2}$$

$2t + 2\alpha = t - 4\alpha$

$t = -6\alpha$

$$\alpha = \frac{t}{-6}$$

$$\frac{2b+t}{b-t} = -4$$

$2b + t = 4t - 4b$

$6b = 3t$

$t = 2b$

$$b = \frac{t}{2}$$

$$B\left(\frac{-2t}{3}, \frac{-t}{6}\right) \qquad C\left(\frac{t}{2}, t\right)$$

$$G\left(\frac{t - \frac{2t}{3} + \frac{t}{2}}{3}, \frac{-t - \frac{t}{6} + t}{3}\right) \equiv (h, k)$$

$$\left.\begin{array}{l} \frac{5t}{18} = h \Rightarrow t = \frac{18h}{5} \\ t = -18k \end{array}\right\rangle \quad h + 5k = 0$$

$$\Rightarrow x + 5y = 0$$

hence, locus of G is st. line $x + 5y = 0$

9. The distance of a point (x_1, y_1) from each of two straight lines which passes through the origin of co-ordinates is δ; find the combined equation of these straight lines.

बिन्दु P (x_1, y_1) की मूल बिन्दु से गुजरने वाली दो रेखाओं में प्रत्येक दूरी δ है। तो दोनो रेखाओं का सम्मिलित समीकरण ज्ञात करो।

Ans. $(y_1^2 - \delta^2)\, x^2 - 2\, x_1 y_1\, xy + (x_1^2 - \delta^2)\, y^2 = 0$

Sol. $$P_1\, P_2 = \frac{\left|ax_1^2 + 2hx_1y_1 + by_1^2\right|}{\sqrt{(a-b)^2 + 4h^2}}$$

$$\delta^2 = \frac{\left|ax_1^2 + 2hx_1y_1 + by_1^2\right|}{\sqrt{(a-b)^2 + 4h^2}}$$

$y = m_1x, \qquad y = m_2x$

$$\delta = \left|\frac{m_1x_1 - y_1}{\sqrt{1 + m_1^2}}\right| \qquad \delta = \left|\frac{m_2x_2 - y_2}{\sqrt{1 + m_2^2}}\right|$$

$$\delta^2 = \frac{(mx_1 - y_1)^2}{1 + m_1^2} \Rightarrow \delta^2 = \frac{m^2x_1^2 - 2x_1y_1m + y_1^2}{1 + m_1^2}$$

$$\left(x_1^2 - \delta^2\right)m^2 - 2x_1y_1m + \left(y_1^2 - \delta^2\right) = 0$$

Put $m = \frac{y}{x}$

$\left(y_1^2 - \delta^2\right)x^2 - 2x_1y_1xy + \left(x_1^2 - \delta^2\right)xy^2 = 0$

10. The sides of a triangle have the combined equation $x^2 - 3y^2 - 2xy + 8y - 4 = 0$. The third side, which is variable always passes through the point (–5, –1). If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a, b) then find $\left(a + \frac{1}{b^2}\right)$.

त्रिभुज की दो भुजाओं का संयुक्त रूप का समीकरण $x^2 - 3y^2 - 2xy + 8y - 4 = 0$ तथा तीसरी चर भुजा जो कि हमेशा बिन्दु (–5, –1) से गुजरती है। तीसरी भुजा की प्रवणता का अन्तराल (a, b) इस तरह है, कि मूल बिन्दु त्रिभुज का आंतरिक बिन्दु है तो $\left(a + \frac{1}{b^2}\right)$ का मान ज्ञात करो।

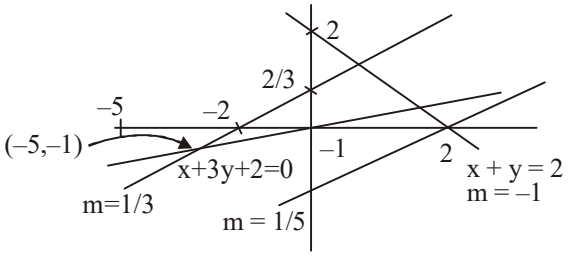
Ans. 24

Sol. $x^2 - 2xy - 3y^2 + 8y - 4 = 0$

$x^2 - 2xy - 3y^2 = x^2 - 3xy + xy - 3y^2 = (x + y)(x - 3y) = 0$

$(x + y + c_1)(x - 3y + c_2) = 0$

$$\begin{array}{rrrrr} & C_1 & + & C_2 & = 0 \\ - & 3C_1 & + & C_2 & = 8 \\ + & & - & & - \\ \hline & & 4C_1 & = & -8 \\ \hline \end{array}$$



$C_1\, C_2 = -4$

$C_1 = -2 \qquad C_2 = 2$

$L_1 : x + y - 2 = 0 \qquad L_2 : x - 3y + 2 = 0$

Let equation of variable line be

$y + 1 = m(x + 5)$

For origin to lie inside the required Δ, y intercept for the variable line should be –ve.

$\therefore\ 5m - 1 = 0 \Rightarrow m < \frac{1}{5}$

$(-5, -1)$ also lie on $x - 3y + 2 = 0$

Now, Δ can form when $m > -1$.

So, $m \in \left(-1, \frac{1}{5}\right)$, $a + b = 4$

11. Consider a line pair $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ and another line L passing through origin with gradient 3. The line pair and line L form a triangle whose vertices are A, B and C.

(a) Find the sum of the contangents of the interior angles of the triangle ABC.

(b) Find the area of triangle ABC

(c) Find the radius of the circle touching all the 3 sides of the triangle.

एक रेखा युग्म $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ तथा रेखा L जिसकी प्रवणता 3 एवं मूलबिन्दु से गुजरती है। तो रेखा युग्म एवं रेखा L से बनने वाले त्रिभुज के शीर्ष बिन्दु A, B एवं C है।

(a) CotA + CotB + CotC का मान ज्ञात कीजिये। (जहाँ A,B,C त्रिभुज ABC के आन्तरिक कोण हैं)

(b) त्रिभुज ABC का क्षेत्रफल ज्ञात करो

(c) त्रिभुज की तीनो भुजाओं को स्पर्श करने वाले वृत की त्रिज्या ज्ञात करे।

Ans. (a) $\frac{50}{7}$; (b) $\frac{63}{10}$; (c) $\frac{3}{10}\left(8\sqrt{5} - 5\sqrt{10}\right)$

Sol. $2x^2 + 3xy - 2y^2 - 2y^2 - 10x + 15y - 28 = 0$

$y = 3x$

$2x^2 + 3xy - 2y^2 = 0$

$2 + 3m - 2m^2 = 0 \qquad \left(\because \frac{y}{x} = m\right)$

$2m^2 - 3m - 2 = 0$

$2m^2 - 4m + m - 2 = 0$

$(2m + 1)(m - 2) = 0$

$m = -\frac{1}{2}, \qquad m = 2$

$m_1 = 3$

$m_2 = 2$

$$m_3 = -\frac{1}{2}$$

$$\tan I = \frac{m_1 - m_2}{1 + m_1 m_2} = \frac{3-2}{1+6} = \frac{1}{7}$$

$$\tan II = \frac{m_2 - m_3}{1 + m_2 m_3} = \frac{2 + \frac{1}{2}}{1-1} = \infty$$

$$\tan III = \frac{-\frac{1}{2} - 3}{1 - \frac{3}{2}} = \frac{-17}{1+1}$$

(a) $\sum \cot I = 7 + \frac{1}{7} + 0 = \frac{50}{7}$

(d) $2x^2 + 3xy - 2y^2 = 0$

$2x^2 + 4xy - xy - 2y^2 = 0$

$2x(2x + 2y) - y(x + 2y) = 0$

$(x + 2y)(2x - y) = 0$

$2C_1 + C_2 = -10$ $\qquad C_1 C_2 = -28$

$-C_1 + 2C_2 = 15$

$C_1 = -7 \qquad C_2 = 4$

$x + 2y + C_1 = 0 \qquad \Rightarrow x + 2y - 7 = 0$ and $y = 3x$

$2x - y + C_2 = 0 \qquad \Rightarrow 2x - y + 4 = 0$

P.O.I $\quad A\left(-\frac{1}{5}, \frac{18}{5}\right)$

B (4, 12)

C (1, 3)

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} -\frac{1}{5} & \frac{18}{5} & 1 \\ 4 & 12 & 1 \\ 1 & 3 & 1 \end{vmatrix} = \frac{1}{10}\begin{vmatrix} -1 & 18 & 5 \\ 4 & 12 & 1 \\ 1 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{10}\begin{vmatrix} -6 & 3 & 5 \\ 3 & 9 & 1 \\ 0 & 0 & 1 \end{vmatrix} \Rightarrow \frac{1}{10}[-54-9] = \frac{63}{10}$$

12. The two line pairs $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ enclose a 4 sided convex polygon find
(i) area of the polygon (ii) length of its diagonals.

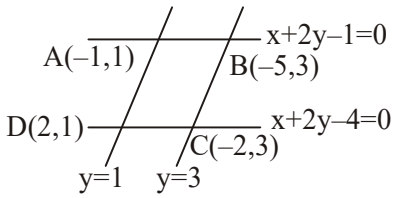
दो रेखा युग्मों $y^2 - 4y + 3 = 0$ एवं $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ से चार भुजाओं वाला उत्तल बहुभुज निर्मित होता है, तो
(i) बहुभुज का क्षेत्रफल (ii) विकर्णो की लम्बाई ज्ञात करो

Ans. (i) area = 6 sq. units, (ii) diagonals are $\sqrt{5}$ & $\sqrt{53}$

Sol. $y^2 - 4y + 3 = 0$ $\qquad x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$

$y = 3, 1$ $\qquad (x + 2y)^2 - 5(x + 2y) + 4 = 0$

$(x + 2y - 4)(x + 2y - 1) = 0$



(i) Area of ploygon

$$\text{Area} = \frac{1}{2}\begin{vmatrix} -1 & 2 & -2 & -5 & -1 \\ 1 & 1 & 3 & 3 & 1 \end{vmatrix}$$

$$= \frac{1}{2}[-1 - 2 + 6 + 2 - 6 + 15 - 5 + 3]$$

$= 6$

(ii) Length of diagonals

$AC = \sqrt{1+4} = \sqrt{5}$

$BD = \sqrt{49+4} = \sqrt{53}$

13. Find the equation of the two straight lines which together with those given by the equation $6x^2 - xy - y^2 + x + 12y - 35 = 0$ will make a parallelogram whose diagonals intersect in the origin.

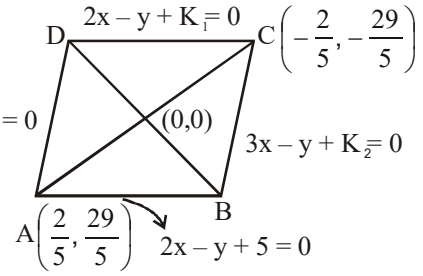
उन दोनों रेखाओं का समीकरण ज्ञात करो जो कि समीकरण $6x^2 - xy - y^2 + x + 12y - 35 = 0$ से प्रदर्शित होने वाली अन्य रेखाओं के साथ मिलकर एक सामान्तर चतुर्भुज इस प्रकार निर्मित करती है कि समान्तर चतुर्भुज के विकर्ण मूलबिन्दु पर प्रतिच्छेद करते है।

Ans. $6x^2 - xy - y^2 - x - 12y - 35 = 0$

Sol. $6x^2 - xy - y^2 + x + 12y - 35 = 0 \quad \Rightarrow \quad (2x - y + 5)(3x + y - 7) = 0$

Now point A is intersection point of $2x - y + 5 = 0$ and $3x + y - 7$

$\therefore \quad A\left(\frac{2}{5}, \frac{29}{5}\right)$



Now C $\left(-\frac{2}{5}, -\frac{29}{5}\right)$ as midpoint of diagonals is (0, 0)

$2x - y + K_1 = 0$ and $3x + y + K_2 = 0$ are passing through C

$\therefore \quad K_1 = -5$, and $K_2 = 7$

Now combined equation $(2x - y - 5)(3x + y + 7) = 0$

$6x^2 - xy - y^2 - x - 12y - 35 = 0$

14. A straight line passing through O(0, 0) cuts the lines $x = a$, $y = b$ and $x + y = 8$ at A, B and C respectively such that $OA \cdot OB \cdot OC = 48\sqrt{2}$ and $f(a, b) \le 0$

where $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$.

(i) Find the point of intersection of lines $x = a$ and $y = b$.

(ii) Find the value of $(OA + OB + OC)$.

(iii) Find the equation of line OA.

एक सरल रेखा जो कि मूल बिन्दु O(0, 0) से गुजरती है, तथा रेखाओं $x = a$, $y = b$ एवं $x + y = 8$ को क्रमशः A, B एवं C पर इस प्रकार प्रतिच्छेद करती है, कि $OA \cdot OB \cdot OC = 48\sqrt{2}$ एवं $f(a, b) \le 0$

जहाँ $f(x, y) = \left|\frac{y}{x} - \frac{3}{2}\right| + (3x - 2y)^6 + \sqrt{ex + 2y - 2e - 6}$

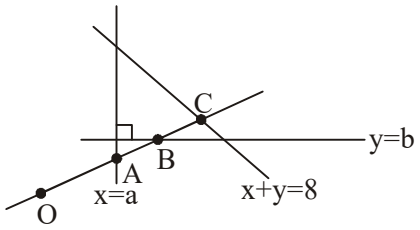
(i) रेखाओं $x = a$ एवं $y = b$ का प्रतिच्छेद बिन्दु ज्ञात करो

(ii) $(OA + OB + OC)$ का मान ज्ञात करो।

(iii) OA का समीकरण ज्ञात करो।

Ans. (i) a = 2 and b = 3; (ii) $9\sqrt{2}$; (iii) x – y = 0

Sol.



(i) $f(a, b) \le 0$

$\Rightarrow f(a, b) = 0$ (only possible)

$\frac{b}{a} = \frac{3}{2}$

$2b = 3a$

$ea + 2b - ea - 6 = 0$

$(a, b) = (2, 3)$ $\quad b = 3$

$\quad a = 2$

(ii) $A\left(r_1 \cos\theta, r_1 \sin\theta\right)$ $\quad \Rightarrow r_1 \cos\theta = 2$

$B\left(r_2 \cos\theta, r_2 \sin\theta\right)$ $\quad \Rightarrow r_2 \sin\theta = 3$

$C\left(r_3 \cos\theta, r_3 \sin\theta\right)$ $\quad \Rightarrow r_3 = \frac{8}{\cos\theta + \sin\theta}$

$r_1 r_2 r_3 = 48\sqrt{2}$

$\frac{2 \times 3 \times 8}{\cos\theta.\sin\theta.\left(\cos\theta + \sin\theta\right)} = 48\sqrt{2}$ $\quad \Rightarrow \sin\theta\cos\theta\left(\sin\theta + \cos\theta\right) = \frac{1}{\sqrt{2}}$

$\Rightarrow \sin 2\theta.\left(\sin\theta + \cos\theta\right) = \sqrt{2}$

$\sin 2\theta = 1,$ $\quad \sin\theta + \cos\theta = \sqrt{2}$

$\theta = 45°$

$r_1 = 2\sqrt{2}$, $r_2 = 3\sqrt{2}$, $r_3 = \frac{8}{2}\sqrt{2} = 4\sqrt{2}$

$r_1 + r_2 + r_3 = 9\sqrt{2}$

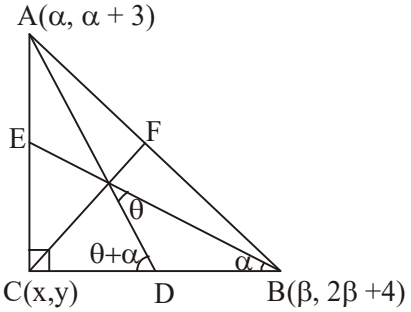
(iii) $y = x$ $\quad m = \tan 45° = 1,$ $\quad O(0,0)$

15. The triangle ABC, right angled at C, has median AD, BE and CF. AD lies along the line $y = x + 3$, BE lies along the line $y = 2x + 4$. If the length of the hypotenuse is 60, find the area of the triangle ABC.

त्रिभुज ABC, जिसक कोण C समकोण है, तथा माध्यिकाये AD, BE एवं CF है जहाँ AD एवं BE क्रमशः रेखाओं $y = x + 3$ एवं $y = 2x + 4$ के अनुसरित है यदि कर्ण की लम्बाई 60 है तो त्रिभुज ABC का क्षेत्रफल ज्ञात करो।

Ans. 400 sq. units

Sol.



$AC^2 + BC^2 = 60^2$

$b^2 + a^2 = 60^2$

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right| = \left|\frac{1-2}{1+2}\right| = \frac{1}{3}$$

In ΔCBE

$$\tan\theta = \frac{CE}{BC} = \frac{\frac{b}{2}}{a} = \frac{b}{2a}$$

In ΔADC

$$\tan(\theta + \alpha) = \frac{AC}{CD} = \frac{b}{\frac{a}{2}} = \frac{2b}{a}$$

$$\frac{\tan\theta + \tan\alpha}{1 - \tan\theta\tan\alpha} = \frac{2b}{a} \Rightarrow \frac{\frac{1}{3} + \frac{b}{2a}}{1 - \frac{1}{3}\cdot\frac{b}{2a}} = \frac{2b}{a}$$

$$\Rightarrow \frac{2a + 3b}{6a - b} = \frac{2b}{a}$$

$\Rightarrow ab = 800$

$ar.(\Delta ABC) = 400$

16. A straight line L through the origin meets the line $x+y=1$ and $x+y=3$ at P and Q respectively. Through P and Q two straight lines L_1 and L_2 are drawn, parallel to $2x-y=5$ and $3x+y=5$ respectively. Lines L_1 and L_2 intersect at R. Show that the locus of R, as L varies, is a straight line.

मूलबिन्दु से होकर जाने वाली एक सरल रेखा L रेखाओं $x+y=1$ और $x+y=3$ से क्रमशः P और Q पर मिलती है। P और Q से दो सरल रेखाऐं L_1 एवं L_2 क्रमशः $2x-y=5$ और $3x+y=5$ के समान्तर खींची जाती है। रेखाऐं L_1 व L_2 बिन्दु R पर प्रतिच्छेद करती है। दिखाइए कि बिन्दु R का बिन्दुपथ एक सरलरेखा है।

Ans. $x-3y+5=0$;

Sol. The line $y=mx$ meets the given lines in

$$P\left(\frac{1}{m+1},\frac{m}{m+1}\right) \text{ and } Q\left(\frac{3}{m+1},\frac{3m}{m+1}\right)$$

Hence equation of L_1 is

$$y-\frac{m}{m+1}=2\left(x-\frac{1}{m+1}\right)$$

$$\Rightarrow y-2x-1=-\frac{3}{m+1} \qquad \text{_____(1)}$$

and that of L_2 is

$$y-\frac{3m}{m+1}=-3\left(x-\frac{3}{m+1}\right)$$

$$\Rightarrow y+3x-3=\frac{6}{m+1} \qquad \text{_____(2)}$$

From (1) and (2) $\frac{y-2x-1}{y+3x-3}=-\frac{1}{2}$

$\Rightarrow x-3y+5=0$ which is a straight line

17. A straight line L with negative slope passes through the point (8,2) and cuts the positive coordinates axes at points P and Q. Find the absolute minimum value of OP + OQ, as L varies, where O is the origin.

ऋणात्मक प्रवणता वाली सरलरेखा L बिन्दु (8,2) से गुजरती है तथा निर्देश अक्षों का धनात्मक दिशा में P तथा Q बिन्दुओं पर काटती है। L के परिवर्तित होने के अनुसार OP + OQ का न्यूनतम निरपेक्ष मान ज्ञात कीजिये, जहाँ O मूल बिन्दु है।

Ans. 18

Sol. Equation of line

(A) $y-2=m(x-8)$ and $m<0 \quad \Rightarrow \quad P\left(8-\frac{2}{m},\ 0\right)$ and $Q(0,\ 2-8m)$

$$OP+OQ=\left|8-\frac{2}{m}\right|+|2-8m|=10+\frac{2}{(-m)}+8(-m) \geq 10+2\sqrt{\frac{2}{(-m)}\times 8(-m)} \geq 18.$$

Area of ΔOPQ is minimum when (8,2) is midpoint of line. So, P(16, 0), Q(0,4) $\Delta OPQ = \frac{1}{2}(16)(4) = 32$.

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 18 to 20

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that AE = AF. Let P be a point inside the square ABCD.

माना ABCD एक वर्ग है जिसकी भुजाओं की लम्बाई एक है। बिन्दु E तथा F क्रमशः भुजाओं AB तथा AD पर इस प्रकार लिये गये है कि AE = AF है। माना बिन्दु P वर्ग ABCD के अन्दर है।

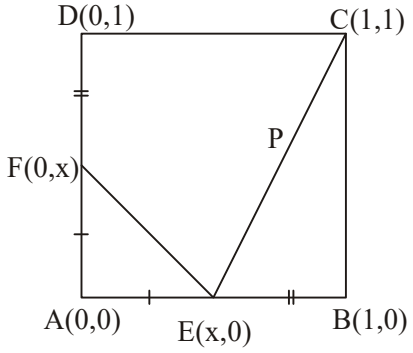
18. The maximum possible area of quadrilateral CDFE is :

चतुर्भुज CDFE का अधिकतम क्षेत्रफल है :

(A) $\frac{1}{8}$ (B) $\frac{1}{4}$ (C) $\frac{5}{8}$ (D) $\frac{3}{8}$

Ans. C

Sol.



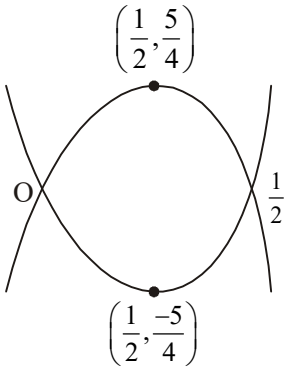
$$\text{Area} = \frac{1}{2}\begin{vmatrix} x & 0 & 0 & 1 & x \\ 0 & x & 1 & 1 & 0 \end{vmatrix}$$

$$A = \frac{1}{2}\left|x^2 - 1 - x\right|$$

$$A = \frac{1}{2}\left|x^2 - x - 1\right|$$

$$x \in (0,1)$$

$$\frac{-b}{2a} = \frac{1}{2}$$



$$\frac{-D}{4a}=\frac{-(1+4)}{4}=\frac{-5}{4}$$

$$A_{max}=\frac{5}{8}$$

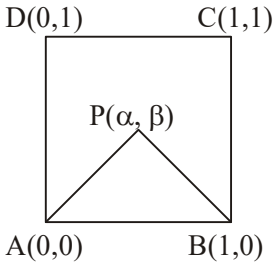
19. The value of $(PA)^2-(PB)^2+(PC)^2-(PD)^2$ is equal to :

$(PA)^2-(PB)^2+(PC)^2-(PD)^2$ का मान है :

(A) 3 (B) 2 (C) 1 (D) 0

Ans. D

Sol.



$$Z=\alpha^2+\beta^2-(\alpha-1)^2-\beta^2+(\alpha-1)^2+(\beta+2)^2-\alpha^2-(\beta\alpha)^2$$

$Z=0$

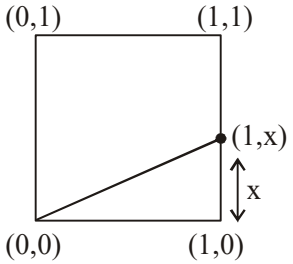
20. Let a line passing through point A divides the square ABCD into two parts so that area of one portion is double the other, then the length of portion of line inside the square is :

माना एक रेखा बिन्दु A से गुजरती है जो वर्ग ABCD को दो भागों में विभाजित करती है जिनमें एक का क्षेत्रफल अन्य से दोगुना है तो रेखा के वर्ग के भीतर हिस्से की लम्बाई है :

(A) $\frac{\sqrt{10}}{3}$ (B) $\frac{\sqrt{13}}{3}$ (C) $\frac{\sqrt{11}}{3}$ (D) $\frac{2}{\sqrt{3}}$

Ans. B

Sol.



$3A=1$

$$A = \frac{1}{3}$$

$$\frac{1}{2} \times 1 \times x = \frac{1}{3}$$

$$x = \frac{2}{3}$$

$$\text{Length} = \sqrt{1 + x^2}$$

$$= \sqrt{1 + \frac{4}{9}} = \frac{\sqrt{13}}{3}$$

Paragraph for Question Nos. 21 to 23

The base of an isosceles triangle is equal to 4, the base angle is equal to 45°. A straight line cuts the extension of the base at a point M at the angle θ and bisects the lateral side of the triangle which is nearest to M.

एक समद्विबाहु त्रिभुज का आधार 4 है तथा आधार कोण 45° है एक रेखा इस बढ़े हुये आधार को बिन्दु M पर θ कोण पर काटती है तथा M से करीब भुजा को समद्विभाजित करती है।

21. The area 'A' of the quadrilateral which the straight line cuts off from given triangle is :

यह रेखा त्रिभुज के साथ जो चतुर्भुज बनाती है। उसका क्षेत्रफल 'A' होगा :

(A) $\frac{3 + \tan\theta}{1 + \tan\theta}$ (B) $\frac{3 + 2\tan\theta}{1 + \tan\theta}$ (C) $\frac{3 + \tan\theta}{1 - \tan\theta}$ (D) $\frac{3 + 5\tan\theta}{1 + \tan\theta}$

Ans. D

22. The range of values of 'A' for different values of θ, lie in the interval :

θ के भिन्न–भिन्न मान के लिय 'A' का परिसर होगा :

(A) $\left(\frac{5}{2}, \frac{7}{2}\right)$ (B) (4, 5) (C) $\left(4, \frac{9}{2}\right)$ (D) (3, 4)

Ans. D

23. The length of portion of straight line inside the triangle may lie in the range :
इस रेखा का वह हिस्सा जो कि त्रिभुज के भीतर है, उसकी लम्बाई का परिसर होगा :

(A) (2, 4) (B) $\left(\frac{3}{2}, \sqrt{3}\right)$ (C) $(\sqrt{2}, 2)$ (D) $(\sqrt{2}, \sqrt{3})$

Ans. C

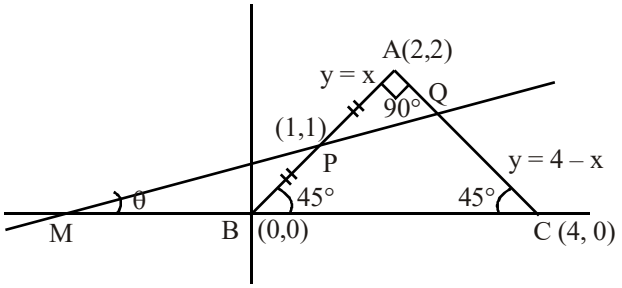
Sol. Equation of line PM :

$y - 1 = \tan\theta\,(x - 1)$

Intersection point 'Q' of AC and MP

$4 - x - 1 = \tan\theta\,(x - 1)$

$$Q \equiv \left(\frac{3+\tan\theta}{1+\tan\theta}, \frac{1+3\tan\theta}{1+\tan\theta}\right)$$



$$\text{Area of } \Delta APQ = \text{modulus of } \frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ 2 & 2 & 1 \\ \frac{3+\tan\theta}{1+\tan\theta} & \frac{1+3\tan\theta}{1+\tan\theta} & 1 \end{vmatrix} = \frac{1-\tan\theta}{1+\tan\theta}$$

MATCH THE COLUMN TYPE QUESTION(S)

24. Match the following column :

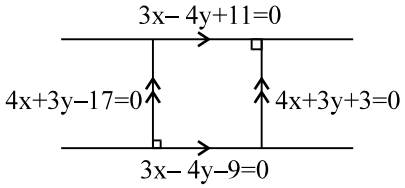
Column–I	**Column–II**
(A) The four lines $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ and $4x + 3y - 17 = 0$ enclose a figure which is	(P) A quadrilateral which is neither a parallelogram nor a trapezium nor a kite.
(B) The lines $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ and $x + 2y = 3$ from a figure which is	(Q) A parallelogram which is neither a rectangle nor a rhombus
(C) If 'O' is the origin, P is the intersection of the lines $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$, A and B are the points in which these lines are cut by the line $x + 2y - 5 = 0$, then the points O, A, P, B (in some order) are the vertices of	(R) A rhombus which is not a square
	(S) A square

निम्न कॉलम को सुमेलित कीजिए :

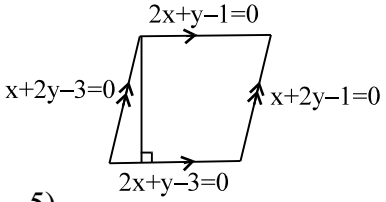
Column–I	**Column–II**
(A) चार भुजाऐं $3x - 4y + 11 = 0$; $3x - 4y - 9 = 0$; $4x + 3y + 3 = 0$ एवं $4x + 3y - 17 = 0$ किस आकृति को बनाएगी।	(P) एक चतुर्भुज जो न तो समान्तर चतुर्भुज एवं नही काइट है
(B) रेखाऐं $2x + y = 1$, $x + 2y = 1$, $2x + y = 3$ तथा $x + 2y = 3$ किस आकृति को बनाएगी	(Q) एक समान्तर चतुर्भुज जो न तो आयत एवं न ही समचतुर्भुज है
(C) यदि रेखा $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0$ का प्रतिच्छेद बिन्दु P हो तथा इस रेखा युग्म को दूसरी रेखा $x + 2y - 5 = 0$ यदि A एवं B पर काटती है तथा O मूलबिन्दु है। तो O, A, P, B किसके शीर्ष होगे	(R) एक समचतुर्भुज जो वर्ग नहीं है
	(S) वर्ग

Ans. (A) S; (B) R; (C) Q

Sol. (A) $\left.\begin{array}{l} d_1 = \dfrac{20}{5} = 4 \\ d_2 = \dfrac{20}{5} = 4 \end{array}\right] \Rightarrow$ square

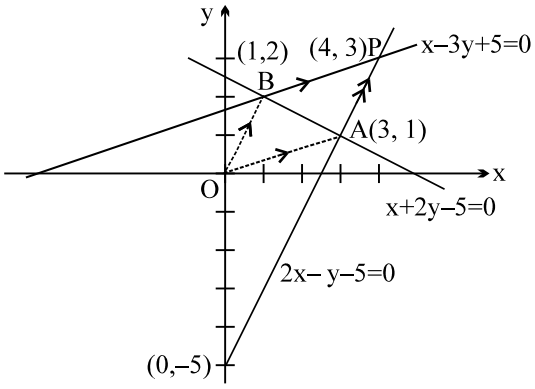


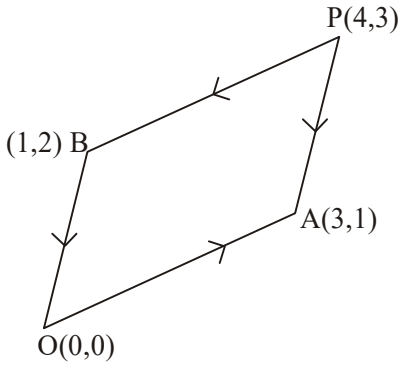
(B) $\left.\begin{array}{l} d_1 = \dfrac{2}{\sqrt{5}} \\ d_2 = \dfrac{2}{\sqrt{5}} \end{array}\right] \Rightarrow$ interior not 90° $\Rightarrow$ rhombus



(C) $2x^2 - 7xy + 3y^2 + 5x + 10y - 25 = 0 \equiv (x - 3y + 5)(2x - y - 5)$

the point of intersection is (4, 3)





$m_{OA} = \dfrac{1}{3}$ $\qquad$ $m_{BP} = \dfrac{1}{3}$

$m_{AP} = \dfrac{2}{1}$ $\qquad$ $m_{OB} = \dfrac{2}{1}$

$OA \parallel AP$ & $AP \parallel OB$

$OA \not\perp AP$

Hence, parallelogram.

25. Consider the 3 linear equations

$ax + by + c = 0$

$bx + cy + a = 0$

$cx + ay + b = 0$

where $a, b, c \in R$.

Column–I	**Column–II**
(A) If $a + b + c = 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$ then	(P) Entire xy plane
(B) If $a + b + c = 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$ then	(Q) The lines are concurrent
(C) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 \neq ab + bc + ca$ then	(R) Line are concident
(D) If $a + b + c \neq 0$ and $a^2 + b^2 + c^2 = ab + bc + ca$ then	(S) Lines are neither coincident nor concurrent

तीन रेखाओं का समीकरण दिया है

$ax + by + c = 0$

$bx + cy + a = 0$

$cx + ay + b = 0$

जहाँ $a, b, c \in R$.

Match the following column :

Column – I	**Column – II**
(A) यदि $a + b + c = 0$ एवं $a^2 + b^2 + c^2 \neq ab + bc + ca$ तो	(P) xy तल होगा
(B) यदि $a + b + c = 0$ एवं $a^2 + b^2 + c^2 = ab + bc + ca$ तो	(Q) रेखाऐं संगामी होगी
(C) यदि $a + b + c \neq 0$ एवं $a^2 + b^2 + c^2 \neq ab + bc + ca$ तो	(R) रेखाऐं सम्पाती होगी
(D) यदि $a + b + c \neq 0$ एवं $a^2 + b^2 + c^2 = ab + bc + ca$ तो	(S) न तो रेखाऐं सम्पाती होगी एवं न ही संगामी होगी

Ans. (A) Q; (B) P; (C) S; (D) R

Sol. $D = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ and $D = 0$ is condition of concurrency

$D = -(a^3 + b^3 + c^3 - 3abc) = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$

(A) if $a + b + c = 0$ but $\sum a^2 \neq \sum ab$ i.e. a, b, c are not all equal, then $D = 0$

hence lines are concurrent $\Rightarrow$ (Q)

(B) if $a + b + c = 0$ and $\sum a^2 = \sum ab$ $\Rightarrow a = b = c$

$\therefore a = 0; \quad b = 0; \quad c = 0$

$\Rightarrow$ lines becomes identical and of the form $0x + 0y + 0 = 0$

any ordered pair (x, y) will satisfy $\Rightarrow$ complete xy plane $\Rightarrow$ (P)

(C) if $a + b + c \neq 0$ and $\sum a^2 \neq \sum ab$ $\Rightarrow$ a, b, c are not all equal $\Rightarrow D \neq 0$

In this case equations represents set of lines which are neither concident nor concurrent $\Rightarrow$ (S)

(D) if $a + b + c \neq 0$ and $\sum a^2 = \sum ab$ $\Rightarrow a = b = c$

hence lines becomes identical or concident $\Rightarrow$ (R)

26. Match the following column :

Column–I	**Column–II**
(A) The lines $y = 0$; $y = 1$; $x - 6y + 4 = 0$ and $x + 6y - 9 = 0$ constitute a figure which is	(P) a cyclic quadraliteral
(B) The points A(a, 0), B(0, b), C(c, 0) and D(0, d) are such that $ac = bd$ and a, b, c, d are all non-zero. The points A, B, C and D always constitute	(Q) a rhombus
(C) The figure formed by the four lines $ax \pm by \pm c = 0$ $(a \neq b)$, is	(R) a square
(D) The line pairs $x^2 - 8x + 12 = 0$ and $y^2 - 14y + 45 = 0$ constitute a figure which is	(S) a trapezium

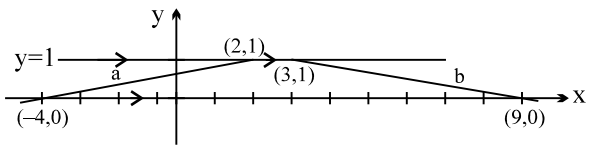
निम्न कॉलम को सुमेलित कीजिए :

Column–I	**Column–II**
(A) रेखा $y = 0$; $y = 1$; $x - 6y + 4 = 0$ एवं $x + 6y - 9 = 0$ से बनने वाली आकृति होगी	(P) चक्रीय चतुर्भुज
(B) यदि $a, b, c, d \neq 0$ तथा A(a, 0), B(0, b), C(c, 0) एवं D(0, d) इस प्रकार से है कि $ac = bd$ है तो आकृति A B C D इनमें से क्या होगा	(Q) समचतुर्भुज
(C) चार रेखाओं $ax \pm by \pm c = 0$ $(a \neq b)$ से बनने वाली आकृति होगी	(R) वर्ग
(D) रेखा युग्म $x^2 - 8x + 12 = 0$ एवं $y^2 - 14y + 45 = 0$ कौनसी आकृति बनाएगें।	(S) समलम्ब चतुर्भुज

Ans. (A) P, S; (B) P; (C) Q; (D) P, Q, R

Sol. (A) Obviously trapezium

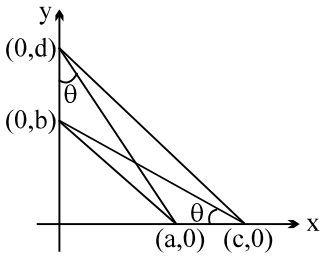
$$\left.\begin{matrix} a = \sqrt{37} \\ b = \sqrt{37} \end{matrix}\right] \quad \Rightarrow \quad a = b$$



hence isosceles trapezium

$\Rightarrow$ a cyclic quadrilateral also $\Rightarrow$ P, S

(B) $ac = bd$ $\Rightarrow$ $\dfrac{b}{c} = \dfrac{a}{d}$



$$\left.\begin{matrix} \tan\theta = \dfrac{b}{c} \\ \tan\phi = \dfrac{a}{d} \end{matrix}\right] \quad \Rightarrow \quad \theta = \phi$$

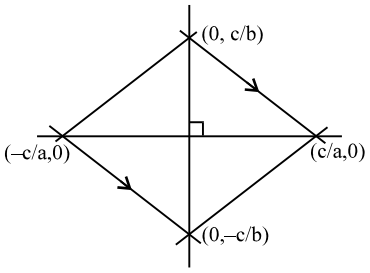
hence cyclic quadrilateral $\Rightarrow$ P

(C) $ax \pm by \pm c = 0$

if $y = 0$, $x = \pm \frac{c}{a}$

if $x = 0$, $y = \pm \frac{c}{b}$

$\Rightarrow$ rhombus $\Rightarrow$ Q



(D) $(x - 6)(x - 2) = 0$

$x = 6$ and $x = 2$

$y^2 - 14y + 45 = 0$

$(y - 9)(y - 5) = 0$

$\Rightarrow$ a square $\Rightarrow$ P, Q, R

